A Bayesian Hierarchical Hurdle Beta-Binomial Model for Survey-Weighted Bounded Counts and Its Application to Childcare Enrollment

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Abstract. Bounded discrete proportions—counts out of known totals—present modeling challenges when data exhibit structural zeros, overdispersion, and hierarchical clustering. We develop a Bayesian hierarchical hurdle beta-binomial model with state-varying coefficients that addresses all four features. The framework makes three methodological contributions: (i) it studies cross-margin dependence via a cross-block covariance component and clarifies when and how this parameter is identified through the hierarchical layer rather than the conditional likelihood; (ii) it proposes a Cholesky-based sandwich variance calibration for pseudo-posterior inference under survey weights, guided by a parameter-specific design effect ratio diagnostic; and (iii) it introduces a log-scale marginal effect decomposition for hurdle models that translates regression coefficients into policy-relevant quantities. Applied to 6,785 childcare providers across 51 states from the 2019 National Survey of Early Care and Education, the model reveals a "poverty reversal": poverty reduces enrollment participation yet increases intensity among participants, with the extensive margin accounting for two-thirds of the total effect. Design-calibrated simulation shows that sandwich-corrected intervals substantially improve coverage, reaching 82–88.5% at the 90% nominal level for fixed effects. The R package hurdlebb implements all methods.

Keywords: Bayesian hierarchical model; beta-binomial distribution; bounded discrete proportions; hurdle model; sandwich variance correction; survey weighting

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1. **Introduction**

Bounded discrete proportions—counts out of known totals—arise whenever a finite capacity is partially utilized: hospital beds occupied, insurance plans enrolled, dental surfaces decayed, survey items endorsed, or enrollment slots filled at a childcare center. When the outcome can take the value zero for structural rather than sampling reasons and displays variance well beyond the binomial, standard generalized linear models are inadequate. Infant and toddler childcare enrollment in the United States offers a canonical instance of all these complications simultaneously. Data from the 2019 National Survey of Early Care and Education (National Survey of Early Care and Education Project Team, 2022) record, for each center-based provider, the number of children under age three enrolled out of total capacity. Over one-third of centers report zero infant and toddler enrollment—not because demand is absent, but because stringent staff-to-child ratios and elevated per-child costs lead many providers to decline serving this age group entirely (Adams & Rohacek, 2010). The resulting access gaps fall disproportionately on high-poverty communities (Dobbins et al., 2016; Malik et al., 2018), yet the relationship between poverty and enrollment turns out to be considerably more complex than a simple deficit story.

A preliminary examination of the data reveals a distinctive pattern that we term the *poverty reversal*: community poverty reduces the probability that a center serves infants and toddlers at all, yet among centers that *do* participate, those in higher-poverty areas devote a *larger* share of capacity to this age group. The reversal poses a methodological challenge: it *cannot be detected* by any model that conflates participation and intensity into a single equation, nor by one that fails to accommodate bounded support, overdispersion, structural zeros, and survey weighting simultaneously. These four features jointly motivate the hierarchical hurdle beta-binomial framework developed in this paper.

Two-part models for count data with excess zeros have a substantial history. Cragg (1971) introduced the hurdle framework for continuous expenditures and Mullahy (1986) adapted it for counts, separating a binary participation stage from a truncated count stage. Bayesian extensions have expanded their scope considerably: hierarchical zero-inflated or hurdle models with Poisson and negative-binomial kernels now accommodate spatial, longitudinal, and multilevel structures (Altinisik, 2023; B. Neelon, 2019; B. Neelon et al., 2013, 2016; B. H. Neelon et al., 2010). Most relevant to our work, Ghosal et al. (2020) propose a hierarchical hurdle model for spatiotemporal count data that captures cross-margin dependence through a single shared scalar.

However, all of these models assume Poisson or negative-binomial kernels, which treat the count as unbounded. When the response has bounded support, the beta-binomial distribution is the natural alternative, introducing overdispersion while respecting the $\{0, 1, \ldots, n\}$ support (Wagner et al., 2015). Bandyopadhyay et al. (2011) embed the beta-binomial in a spatial framework for clustered bounded counts, and Wen et al. (2025) propose a zero-inflated beta-binomial with spatially varying coefficients and cross-component covariance. Yet they adopt a zero-inflation formulation in which zeros arise from a latent mixture, whereas in the childcare setting a center reporting zero infant enrollment has made a deliberate operational decision (cf. Hall, 2000; Lambert, 1992), making the hurdle specification the natural choice.

A separate methodological strand addresses design-based inference in Bayesian models for survey data. Savitsky and Toth (2016) develop a pseudo-posterior framework in which survey weights enter the likelihood exponent, and Williams and Savitsky (2021) provide calibrated sandwich variance corrections that restore nominal credible-interval coverage (building on Binder, 1983; Pfeffermann, 1993). These contributions target standard single-equation regressions; their extension to two-part hierarchical specifications has not been attempted. In sum, the literature provides the individual building blocks—hurdle, beta-binomial, hierarchical structure, survey correction—but not their synthesis.

This paper develops that synthesis. The model extends the framework of Ghosal et al. (2020) in two directions, each addressing a limitation of existing approaches.

First, we introduce a Bayesian hierarchical hurdle beta-binomial model with state-varying coefficients and cross-margin covariance. The hurdle component separates participation from intensity; the beta-binomial kernel respects the bounded support and accommodates the observed overdispersion, which exceeds the binomial variance by a factor of approximately twelve; and the hierarchical prior on state-level coefficients identifies a cross-margin covariance structure whose effective sample size is $S = 51$ states rather than $N = 6{,}785$ providers, with the LKJ prior providing finite-sample regularization for the high-dimensional covariance ($2q = 10$ parameters). The model nests the scalar cross-margin parameter of Ghosal et al. (2020) as a special case and extends the beta-binomial models of Bandyopadhyay et al. (2011) and Wen et al. (2025) by incorporating a hurdle for structural zeros. Applied to the NSECE data, this specification reveals the reversal—and its geographic heterogeneity across 51 states—with full uncertainty quantification.

Second, we extend the pseudo-posterior framework of Savitsky and Toth (2016) and Williams and Savitsky (2021) to the two-part hierarchical setting, implementing a Cholesky-based sandwich variance correction (Binder, 1983; Pfeffermann, 1993) and introducing the design effect ratio as a parameter-specific diagnostic. A companion simulation study (Section 4) provides candid diagnostics: sandwich-corrected intervals substantially improve coverage for fixed effects under realistic design effects (82–88.5% at the 90% nominal level), while variance components require unweighted estimation—a finding we report transparently rather than treating the sandwich as a universal remedy.

These two innovations jointly enable a marginal effect decomposition (Proposition 5) that separates the reversal into extensive-margin (access) and intensive-margin (intensity) components. A single-equation specification conflates these offsetting forces into a muted net effect; the two-part decomposition quantifies each channel and their relative magnitudes, with the extensive margin accounting for approximately two-thirds of the total effect. The companion R package hurdlebb (Lee, 2026) implements the full methodology.

The remainder of the paper is organized as follows. Section 2 introduces the NSECE data and documents the empirical patterns motivating the model. Section 3 develops the hierarchical hurdle beta-binomial specification, its analytic properties, and the sandwich correction. Section 4 reports the simulation study. Section 5 applies the model to the 2019 NSECE data. Section 6 discusses implications and limitations. Supplementary Materials A–E provide proofs, identification conditions, extended tables, and computational details.

2. Motivating Data and Preliminary Analysis

This section introduces the data and documents the empirical features that jointly determine the model developed in Section 3. We organize the presentation around four distributional characteristics—structural zeros, bounded support, overdispersion, and complex survey design—each of which maps directly to a component of the hierarchical hurdle beta-binomial specification. A preliminary analysis then establishes the reversal phenomenon and the cross-state heterogeneity that motivate the hierarchical structure and cross-margin covariance.

2.1. **The NSECE survey.** The 2019 National Survey of Early Care and Education (NSECE; National Survey of Early Care and Education Project Team, 2022) is a nationally representative survey of early care and education providers conducted

by NORC at the University of Chicago under contract with the Office of Planning, Research, and Evaluation (OPRE) within the Administration for Children and Families. Data collection took place from November 2018 through July 2019, with 6,917 center-based providers completing the survey. The center-based provider component employs a stratified multistage cluster design: 30 strata defined by state groupings, 415 primary sampling units, and 6,809 centers with nonmissing survey responses. Of these, 24 report a total enrollment of $n_i = 0$ and are excluded because the beta-binomial kernel requires a positive denominator, yielding a final analytic sample of $N = 6{,}785$ providers across all $S = 51$ states (50 states plus the District of Columbia). Full details of the sampling frame, weight construction, and informativeness testing appear in Supplementary Material D.

For each provider i, the survey records total enrollment n_i (ages 0–5) and infant/toddler (IT) enrollment $Y_i \in \{0, 1, \ldots, n_i\}$ (ages 0–2). The outcome pair (Y_i, n_i) is the natural unit of analysis. Provider-level covariates include the community poverty rate, urbanicity, and racial/ethnic composition of the provider's Census tract (Dobbins et al., 2016; Malik et al., 2018). State-level policy variables capture three dimensions of Child Care and Development Fund (CCDF) subsidy policy—market rate percentile, tiered reimbursement, and infant/toddler rate add-on—and enter the model as cross-level moderators in Section 5. Table 1 summarizes the key descriptive statistics.

2.2. **Four data features and their modeling implications.** We now examine four distributional characteristics of the NSECE data that individually and jointly determine the model specification of Section 3. Figure 1 displays the marginal distribution of the infant/toddler enrollment share, which exhibits all four features simultaneously.

Structural zeros and the hurdle specification. Of the 6,785 providers, 2,393 (35.3%) report zero infant/toddler enrollment—a rate far too large for sampling variability and reflecting a deliberate operational decision not to serve children under age three (Adams & Rohacek, 2010). The hurdle specification (Mullahy, 1986) treats participation and intensity as distinct processes governed by separate parameters, as developed in Section 3.2.

Bounded support and the beta-binomial kernel. Among servers ($z_i = 1$), infant/toddler enrollment Y_i is bounded above by total enrollment n_i, so the share Y_i/n_i lies in $(0, 1]$. The beta-binomial distribution respects this $\{0, 1, \ldots, n_i\}$ support and incorporates extra-binomial variation through the concentration parameter κ (Prentice, 1986; Wagner et al., 2015); see Section 3.2.

TABLE 1. Data summary: NSECE 2019 center-based providers. Panel A describes the outcome variables; Panel B the provider-level covariates (continuous variables shown before standardization, as means with standard deviations and ranges); Panel C the survey design features, including the Kish design effect $\mathrm{DEFF}_{\mathrm{Kish}} = 1 + \mathrm{CV}^2(w)$.

Variable	Value
Panel A: Outcome	
Total providers (N)	6,785
IT-serving providers ($z_i = 1$)	4,392 (64.7%)
Non-servers ($z_i = 0$)	2,393 (35.3%)
Total enrollment (n_i): Mean / Median / Range	60.5 / 48 / 1–378
IT enrollment ($Y_i \mid z_i = 1$): Mean / Median	29.4 / 24
IT share ($Y_i/n_i \mid z_i = 1$): Mean (SD)	0.478 (0.214)
Panel B: Covariates — Mean (SD) [Range]	
Community poverty rate (%)	17.3 (8.3) [1.9, 54.2]
Urban (%)	93.3
Community % Black	17.2 (20.5) [0.0, 96.5]
Community % Hispanic	24.2 (22.6) [0.0, 98.4]
Panel C: Survey design	
States (S)	51 (50 + DC)
Strata / PSUs	30 / 415
Sampling weight range	1–462
Weight CV	1.66
Kish effective sample size	1,803
Kish DEFF	3.76

Overdispersion and the concentration parameter κ. The observed variance of Y_i/n_i among participants exceeds the binomial prediction by a factor of approximately 12 at the pooled level (marginal overdispersion, before accounting for covariates), and all 51 states individually exhibit overdispersion. After conditioning on the model covariates and hierarchical structure, the estimated concentration $\hat{\kappa} \approx 7$ implies a conditional overdispersion factor of approximately 7. The beta-binomial nests the binomial as $\kappa \to \infty$, modeling this overdispersion structurally rather than correcting for it post hoc (Prentice, 1986).

Complex survey design and the pseudo-posterior. Sampling weights range from 1 to 462, yielding a Kish effective sample size of $\mathrm{ESS}_{\mathrm{Kish}} = 1{,}803$ and a design effect of $\mathrm{DEFF}_{\mathrm{Kish}} = 3.76$; a formal test confirms that the design is informative for the intensive margin (Pfeffermann, 1993). Ignoring the design would yield credible

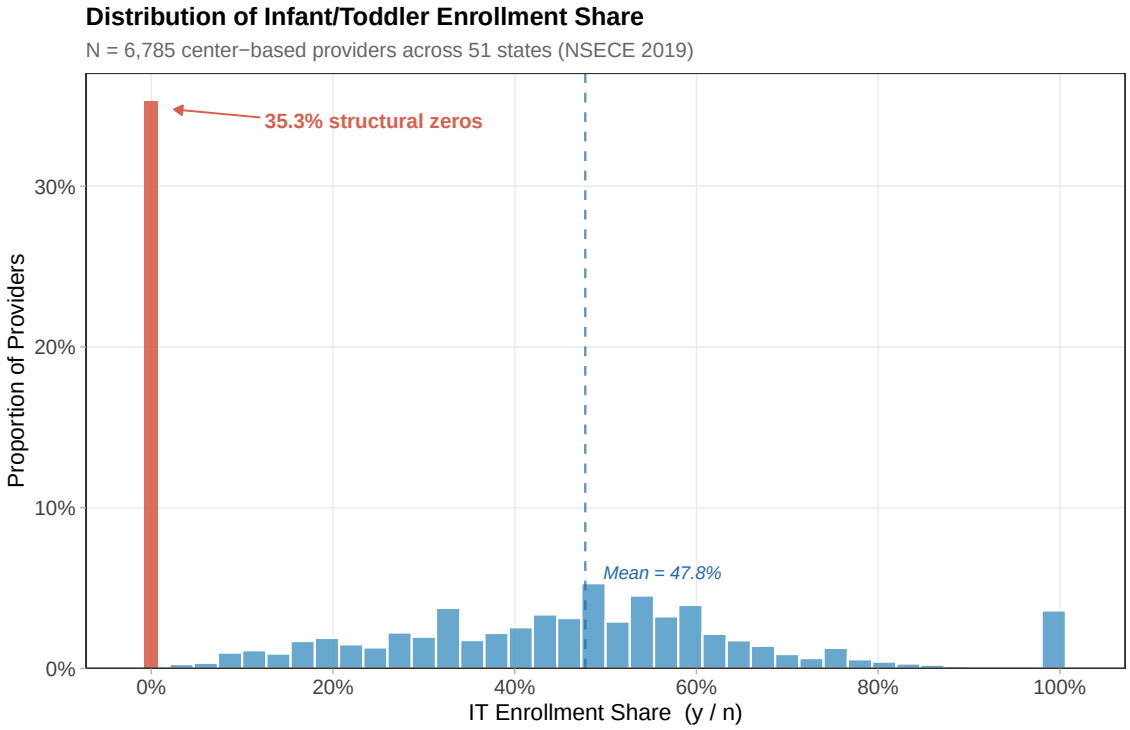

FIGURE 1. Distribution of the infant/toddler enrollment share Y_i/n_i across $N = 6{,}785$ center-based providers (NSECE 2019). The spike at zero (35.3% of centers) represents providers that do not serve any children under age 3. Among IT-serving providers ($z_i = 1$, $N_{\text{pos}} = 4{,}392$), the distribution is roughly unimodal with a mean of 0.478 and substantial dispersion (SD = 0.214), consistent with overdispersion approximately twelve times beyond the binomial—three features that jointly motivate the hurdle beta-binomial model.

intervals with empirical coverage well below nominal; Section 3.5 develops the pseudo-posterior and sandwich correction.

Each feature alone eliminates a class of candidate models: structural zeros eliminate single-equation specifications, bounded support eliminates Poisson and negative-binomial kernels, universal overdispersion eliminates the binomial, and the complex survey design eliminates naive Bayesian posteriors. The *intersection* of all four constraints narrows the viable specification to a hurdle model with a beta-binomial kernel, estimated via a pseudo-posterior with sandwich correction.

2.3. **Preliminary evidence for the poverty reversal.** The four data features determine the model's distributional specification. The model's *hierarchical* structure—state-varying coefficients with cross-margin covariance—is motivated by a fifth empirical pattern: the reversal varies systematically across states.

Pooled estimates. A pooled logistic regression of infant/toddler participation (z_i) on the standardized community poverty rate yields an extensive-margin coefficient of $\hat{\alpha}_{\text{pov}} = -0.020$ ($p < 10^{-10}$), confirming that providers in higher-poverty communities are significantly less likely to serve infants. A pooled OLS regression of the enrollment share (Y_i/n_i among servers) on poverty yields an intensive-margin coefficient of $\hat{\beta}_{\text{pov}} = +0.003$ ($p < 10^{-14}$), establishing that, conditional on serving, providers in poorer communities devote a larger share of enrollment to infants. The opposing signs constitute the reversal in its simplest form: poverty *reduces* participation but *increases* intensity.

The substantive interpretation is noteworthy. Poverty erects a barrier to entry: the elevated costs of infant care discourage participation precisely where demand is greatest (Dobbins et al., 2016; Malik et al., 2018). Yet among centers that clear this barrier, those in disadvantaged communities tilt their enrollment mix toward the youngest age group, reflecting either market demand in areas with fewer alternatives or mission-driven service provision. Any model that collapses participation and intensity into a single equation would estimate a muted net effect, obscuring the two countervailing forces at work.

State heterogeneity. When these regressions are estimated separately by state, 23 of 51 states (45%) display the classic reversal pattern ($\hat{\alpha}_{\text{pov},s} < 0$ and $\hat{\beta}_{\text{pov},s} > 0$). The remaining states show varying combinations: some exhibit negative effects on both margins, while others show positive effects on both. The cross-state correlation between the two margin-specific poverty coefficients is $r = 0.021$—near zero—indicating that the extensive and intensive mechanisms operate largely independently. Figure 2 displays the aggregate reversal in the raw data; the state-level bivariate scatter appears in Figure 5. The near-zero cross-margin correlation implies that knowing a state has a strong participation barrier provides essentially no information about its intensity response, a finding that motivates the full covariance structure developed in Section 3 (Sipple et al., 2020).

Policy correlations. Among the state-level policy variables, tiered reimbursement shows the strongest association with the poverty–participation relationship: states

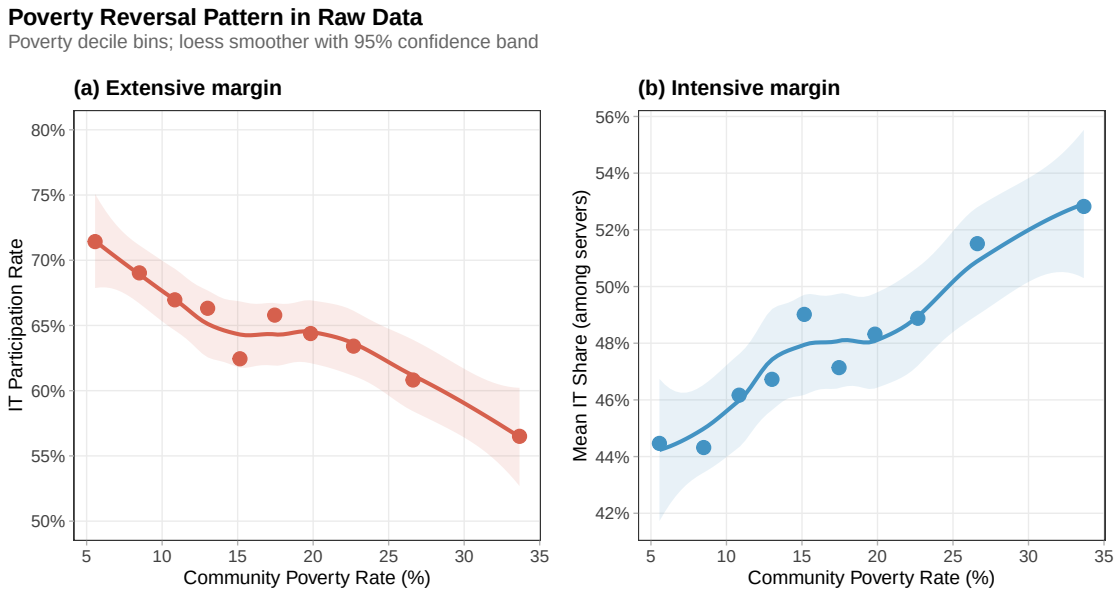

FIGURE 2. Preliminary evidence for the poverty reversal in the raw NSECE 2019 data. *Left panel:* IT participation rate (fraction of centers serving any infants) by community poverty decile; a loess smoother with 95% confidence band confirms the declining trend. *Right panel:* mean IT enrollment share (Y_i/n_i) among servers by poverty decile, showing the opposing positive trend. The two panels together display the reversal in its simplest form: participation falls with poverty, but intensity rises.

with tiered reimbursement tend to have less negative extensive-margin poverty coefficients, consistent with the hypothesis that quality-linked payment incentives offset the cost disadvantage of serving infants in disadvantaged communities. These preliminary correlations motivate the cross-level moderation analysis in Section 5.

In sum, the data require a model that simultaneously (i) separates structural zeros from positive counts via a hurdle, (ii) respects the bounded support via a beta-binomial kernel, (iii) accommodates overdispersion through the concentration parameter, (iv) incorporates survey weights through a pseudo-posterior with sandwich correction, and (v) captures cross-state heterogeneity and cross-margin dependence through a hierarchical prior with full covariance. Section 3 develops this model.

3. The Hierarchical Hurdle Beta-Binomial Model

This section develops the hierarchical hurdle beta-binomial (HBB) model whose distributional specification was motivated by the four data features documented in

Section 2.2. The model combines three building blocks—hurdle, beta-binomial kernel, and hierarchical prior—into a unified framework and adds two methodological contributions that are new to this class of models: a cross-margin covariance whose identification operates entirely through the hierarchical prior rather than the conditional likelihood, and a Cholesky-based sandwich correction that extends pseudo-posterior inference to the two-part hierarchical setting.

In plain terms, the model works as follows. A *hurdle* first separates providers that serve infants and toddlers from those that do not, treating participation as a logistic regression with state-specific intercepts and slopes. For participants, a *beta-binomial* kernel models the share of enrollment devoted to infants and toddlers, accommodating the bounded count and within-state overdispersion. The two margins share a joint hierarchical prior on their state-level deviations, so that a state's participation barrier and its conditional enrollment intensity can be positively or negatively correlated—the cross-margin covariance that is the paper's central structural novelty. Sampling weights enter through a pseudo-likelihood, and a sandwich correction restores design-consistent coverage for population-average parameters.

3.1. **Notation and data structure.** We observe $N = 6{,}785$ center-based childcare providers drawn from $S = 51$ states (including the District of Columbia). Each provider i is nested in a state $s[i]$. The outcome pair (Y_i, n_i) records infant/toddler enrollment $Y_i \in \{0, 1, \ldots, n_i\}$ out of total enrollment n_i; the participation indicator is $z_i = \mathbf{1}(Y_i > 0)$. Table 2 collects the symbols used throughout the paper. All continuous covariates are standardized prior to analysis.

3.2. **The hurdle beta-binomial likelihood. The beta-binomial distribution.** The beta-binomial distribution in the mean–precision parameterization (Griffiths, 1973; Prentice, 1986) (μ, κ) sets $a = \mu\kappa$ and $b = (1-\mu)\kappa$, where $\mu \in (0, 1)$ is the conditional mean parameter and $\kappa > 0$ is the concentration parameter. Larger κ implies less overdispersion, with the $\mathrm{Bin}(n, \mu)$ distribution recovered as $\kappa \to \infty$. The probability mass function (PMF) is

$$\mathbb{P}(Y = y \mid n, \mu, \kappa) = \binom{n}{y} \frac{B\big(y + \mu\kappa,\ n - y + (1-\mu)\kappa\big)}{B\big(\mu\kappa,\ (1-\mu)\kappa\big)}, \quad y = 0, 1, \ldots, n, \tag{3.1}$$

where $B(\cdot, \cdot)$ is the beta function. The first two moments are

$$\mathbb{E}[Y] = n\mu, \qquad \mathrm{Var}(Y) = n\mu(1-\mu)\,\frac{n+\kappa}{1+\kappa}. \tag{3.2}$$

TABLE 2. Notation summary. Dimensions are shown in parentheses.

Symbol	Dimension	Domain	Description
Indices and dimensions			
i	—	$\{1, \ldots, N\}$	Provider index ($N = 6{,}785$)
s	—	$\{1, \ldots, S\}$	State index ($S = 51$, incl. DC)
$s[i]$	—	$\{1, \ldots, S\}$	State membership of provider i
N_s	—	$\mathbb{Z}_{>0}$	Number of providers in state s; $\sum_s N_s = N$
Outcome variables			
n_i	(1)	$\mathbb{Z}_{>0}$	Total enrollment, ages 0–5 (treated as fixed)
Y_i	(1)	$\{0, 1, \ldots, n_i\}$	Infant/toddler (IT) enrollment count
z_i	(1)	$\{0, 1\}$	$z_i = \mathbf{1}(Y_i > 0)$: IT participation indicator
Covariates			
$\mathbf{x}_i$	$(P \times 1)$	$\mathbb{R}^P$	Provider covariates, $P = 5$: intercept, poverty, urban, Black, Hispanic
$\mathbf{x}_i^{(r)}$	$(q \times 1)$	$\mathbb{R}^q$	State-varying subvector of $\mathbf{x}_i$ ($q = P$ in M2/M3)
q_i	(1)	$(0, 1)$	Participation probability (extensive margin); cf. (3.10)
$\mathbf{v}_s$	$(Q \times 1)$	$\mathbb{R}^Q$	State policy vector, $Q = 4$: intercept + $Q - 1$ policy instruments
Parameters			
$\boldsymbol{\alpha}$	$(P \times 1)$	$\mathbb{R}^P$	Population-average extensive-margin coefficients
$\boldsymbol{\beta}$	$(P \times 1)$	$\mathbb{R}^P$	Population-average intensive-margin coefficients
κ	(1)	$\mathbb{R}_{>0}$	Beta-binomial concentration
$\boldsymbol{\delta}_{k,s}$	$(q \times 1)$	$\mathbb{R}^q$	State deviations, margin $k \in \{1, 2\}$, state s
$\boldsymbol{\Gamma}_k$	$(q \times Q)$	$\mathbb{R}^{q \times Q}$	Cross-level policy moderator matrix, margin k
$\boldsymbol{\Sigma}_\delta$	$(2q \times 2q)$	$\mathbb{S}^{2q}_{++}$	Cross-margin state-effect covariance
Survey design			
w_i	(1)	$\mathbb{R}_{>0}$	Sampling weight
$\tilde{w}_i$	(1)	$\mathbb{R}_{>0}$	Normalized weight: $\tilde{w}_i = w_i N / \sum_j w_j$

The variance inflation factor $(n + \kappa)/(1 + \kappa)$ exceeds unity for all $n \geq 2$, capturing overdispersion relative to the binomial. In the childcare data, with a typical enrollment of $n_i \approx 50$ and an estimated $\kappa \approx 7$, this factor is approximately $57/8 \approx 7$, far beyond what a binomial model can accommodate.

Computational remark. In practice, we compute $\log f_{\text{BetaBin}}$ using the `lbeta()` function to avoid numerical overflow when n_i exceeds 50. Specifically, the log-PMF is evaluated as $\log \binom{n}{y} + \texttt{lbeta}(y + a,\, n - y + b) - \texttt{lbeta}(a, b)$, which remains stable for n_i up to several hundred.

Zero probability. At $y = 0$ the PMF simplifies to a product form that is both interpretable and numerically stable:

$$p_0(n, \mu, \kappa) = \prod_{j=0}^{n-1} \frac{(1-\mu)\kappa + j}{\kappa + j}. \tag{3.3}$$

Each factor in the product is the conditional probability that the j-th "trial" yields a non-IT outcome, given all preceding trials did so. The product form shows that p_0 is strictly decreasing in both μ and κ (for $n \geq 2$): higher mean intensity and lower overdispersion each reduce the probability of observing zero IT enrollment.

Hurdle construction. As documented in Section 2.2, the zeros in the childcare data are structural: they reflect a deliberate decision not to serve infants rather than a random draw from a count distribution. This motivates a hurdle specification (Cragg, 1971; Mullahy, 1986) that separates the participation decision from the enrollment intensity.

Definition 1 (Hurdle beta-binomial model)**.** For provider i with enrollment capacity n_i, the HBB model specifies

$$f(y_i \mid q_i, \mu_i, \kappa, n_i) = \begin{cases} 1 - q_i, & \text{if } y_i = 0, \\ q_i \cdot f_{\text{ZT-BetaBin}}(y_i \mid n_i, \mu_i, \kappa), & \text{if } y_i \in \{1, \ldots, n_i\}, \end{cases} \tag{3.4}$$

where $q_i \in (0, 1)$ is the participation probability (extensive margin), $\mu_i \in (0, 1)$ is the conditional mean IT share (intensive margin), and the zero-truncated beta-binomial PMF is

$$f_{\text{ZT-BetaBin}}(y \mid n, \mu, \kappa) = \frac{f_{\text{BetaBin}}(y \mid n, \mu, \kappa)}{1 - p_0(n, \mu, \kappa)}, \quad y = 1, \ldots, n. \tag{3.5}$$

The hurdle separates two distinct decisions that childcare centers face. The *extensive margin* captures whether a center serves any infants at all—the access decision—governed by q_i. The *intensive margin* captures how many infant slots a center offers relative to its total capacity—the intensity decision—governed by μ_i. This separation reflects a genuine two-stage process: a center must first decide to accept the costs and regulatory burden of infant care (clearing the hurdle), and only then does it determine how many infant slots to offer.

The log-likelihood under conditional independence decomposes additively into extensive and intensive components,

$$\ell(\boldsymbol{\theta}) = \underbrace{\sum_{i=1}^{N} \big[z_i \log q_i + (1 - z_i) \log(1 - q_i) \big]}_{\ell_{\text{ext}}(\boldsymbol{\alpha})} + \underbrace{\sum_{i:\, z_i = 1} \big[\log f_{\text{BetaBin}}(y_i \mid n_i, \mu_i, \kappa) - \log\big(1 - p_{0,i}\big) \big]}_{\ell_{\text{int}}(\boldsymbol{\beta}, \kappa)}, \quad (3.6)$$

where $p_{0,i} = p_0(n_i, \mu_i, \kappa)$. This separation is a structural property of hurdle models: the binary indicator z_i is a sufficient statistic partition that renders the extensive parameters $\boldsymbol{\alpha}$ and the intensive parameters $(\boldsymbol{\beta}, \kappa)$ variation-independent in the likelihood. In contrast, zero-inflated models introduce latent at-risk indicators that couple the two components, complicating both estimation and interpretation.

Unconditional expectation and the intensity function. The unconditional mean enrollment count is

$$\mathbb{E}[Y_i] = q_i \cdot n_i \cdot h(\mu_i, n_i, \kappa), \qquad h(\mu, n, \kappa) \equiv \frac{\mu}{1 - p_0(n, \mu, \kappa)}, \quad (3.7)$$

where h is the *intensity function* mapping the latent mean proportion μ to the expected proportion among participants. Since $h(\mu) \geq \mu$ (truncation shifts the conditional mean upward), participating centers are predicted to have a higher IT share than the latent parameter μ alone would suggest. The function h is central to the reversal interpretation: the key question is whether the opposing signs of the extensive and intensive coefficients can be "absorbed" by a nonlinear response of h, or whether the reversal is genuinely a coefficient-level phenomenon. The following theorem resolves this.

Theorem 1 (Monotonicity of h)**.** *For all $n \geq 2$, $\kappa > 0$, and $\mu \in (0, 1)$,*

$$\frac{\partial h}{\partial \mu} > 0. \quad (3.8)$$

For $n = 1$, $h(\mu) \equiv 1$ for all $\mu \in (0, 1)$.

Proof sketch. Define $\Phi(\mu) = (1 - p_0) - \mu\, p_0\, \Lambda$, where $\Lambda = \kappa \sum_{j=0}^{n-1} [(1-\mu)\kappa + j]^{-1} > 0$. Since $\partial h / \partial \mu = \Phi(\mu)/(1 - p_0)^2$, the sign of the derivative equals the sign of Φ. One verifies that $\Phi(0) = 0$ and $\Phi'(\mu) = \mu\, p_0 (\Lambda^2 - \Lambda_2)$, where $\Lambda_2 = \kappa^2 \sum_{j=0}^{n-1} [(1-\mu)\kappa + j]^{-2}$. For $n \geq 2$, the Cauchy–Schwarz inequality gives $\Lambda^2 > \Lambda_2$, so $\Phi'(\mu) > 0$ on $(0, 1)$. Since

Φ starts at zero and is strictly increasing, $\Phi(\mu) > 0$ for all $\mu \in (0, 1)$. The full proof appears in Section A. □

Interpretation for the poverty reversal. Theorem 1 establishes that the intensity function h is strictly increasing in μ: a center with a higher underlying IT intensity μ will always have a higher expected IT share among participants, regardless of how much overdispersion is present. Consequently, the poverty reversal—whereby higher poverty reduces participation ($\alpha_{\text{pov}} < 0$) yet increases conditional intensity ($\beta_{\text{pov}} > 0$)—is purely a *coefficient-level* phenomenon, not an artifact of the nonlinear response function. Researchers can examine the posterior distributions of α_{pov} and β_{pov} directly to assess the reversal, without needing to account for nonlinear link function effects.

Proposition 1 (Elasticity of h)**.** *Define the elasticity $\varepsilon_h = (\partial h/\partial \mu)(\mu/h)$. Then*

$$\varepsilon_h(\mu) = 1 - \omega(\mu), \qquad \omega(\mu) = \frac{\mu\, p_0\, \Lambda}{1 - p_0} \in (0, 1) \quad \text{for } n \geq 2. \tag{3.9}$$

Hence $\varepsilon_h \in (0, 1)$: the intensity function responds inelastically to changes in μ.

The inelastic response arises because increasing μ simultaneously increases the probability that a center would have had a positive draw even without the hurdle (raising $1 - p_0$, the denominator of h), partially offsetting the numerator effect. In practical terms, a one-percent increase in a center's underlying IT propensity μ translates into *less* than a one-percent increase in its expected IT share among participants. At our estimated parameters ($\mu \approx 0.30$, $\kappa \approx 7$, $n \approx 50$), $\omega \approx 0.07$, so the elasticity is approximately 0.93—relatively modest damping in this parameter region.

3.3. **State-varying coefficients (SVC) with cross-margin covariance.** The reversal is not a single national pattern. Preliminary estimates show that fewer than half of states exhibit the classic reversal—a negative extensive-margin poverty coefficient paired with a positive intensive-margin one (Section 2.3). This geographic variation raises a fundamental question: *Does the poverty–enrollment relationship vary systematically across states, and if so, are the extensive and intensive margins linked?*

Linear predictors. Following the spatially varying coefficient framework of Gelfand et al. (2003), the extensive and intensive margins are linked to provider characteristics

through logit linear predictors:

$$\operatorname{logit}(q_i) = \mathbf{x}_i \intercal \boldsymbol{\alpha} + \mathbf{x}_i^{(r)\intercal} \boldsymbol{\delta}_{1,s[i]}, \tag{3.10}$$

$$\operatorname{logit}(\mu_i) = \mathbf{x}_i \intercal \boldsymbol{\beta} + \mathbf{x}_i^{(r)\intercal} \boldsymbol{\delta}_{2,s[i]}, \tag{3.11}$$

where $\boldsymbol{\alpha}, \boldsymbol{\beta} \in \mathbb{R}^P$ are population-average fixed effects—the "national story"—and $\boldsymbol{\delta}_{k,s} \in \mathbb{R}^q$ ($k = 1$ for extensive, $k = 2$ for intensive) are zero-mean state-specific deviations that capture how each state's poverty–enrollment relationship departs from the national average. The subvector $\mathbf{x}_i^{(r)}$ consists of the first q components of $\mathbf{x}_i$ whose coefficients are permitted to vary across states. The *total state-specific coefficient* for covariate j on margin k is

$$\tilde{\alpha}_{j,s} = \alpha_j + \delta_{1,s,j}, \qquad \tilde{\beta}_{j,s} = \beta_j + \delta_{2,s,j}. \tag{3.12}$$

In the childcare application, a state with $\tilde{\alpha}_{\text{pov},s} < 0$ and $\tilde{\beta}_{\text{pov},s} > 0$ exhibits the reversal: higher community poverty is associated with lower participation but higher conditional intensity.

Cross-margin covariance: the key novelty. Standard hurdle models estimate the two margins independently. We depart from this convention by stacking the state deviations into a single $2q$-dimensional vector:

$$\boldsymbol{\delta}_s = \begin{pmatrix} \boldsymbol{\delta}_{1,s} \\ \boldsymbol{\delta}_{2,s} \end{pmatrix} \sim \mathsf{N}_{2q}(\mathbf{0},\ \boldsymbol{\Sigma}_\delta), \qquad s = 1, \ldots, S, \tag{3.13}$$

with the $2q \times 2q$ positive-definite covariance

$$\boldsymbol{\Sigma}_\delta = \begin{pmatrix} \boldsymbol{\Sigma}_{11} & \boldsymbol{\Sigma}_{12} \\ \boldsymbol{\Sigma}_{21} & \boldsymbol{\Sigma}_{22} \end{pmatrix}, \tag{3.14}$$

where $\boldsymbol{\Sigma}_{11}$ governs between-state variation in extensive-margin coefficients, $\boldsymbol{\Sigma}_{22}$ governs the intensive margin, and the off-diagonal block $\boldsymbol{\Sigma}_{12} = \boldsymbol{\Sigma}_{21}\intercal$ captures *cross-margin dependence* (in the policy-moderated model M3b, $\boldsymbol{\Sigma}_{12}$ refers to the *residual* cross-margin covariance after removing policy effects; see Remark B.19). This cross-margin covariance is the central structural novelty of the model: it allows states with stronger participation barriers (larger $|\delta_{1,\text{pov},s}|$) to simultaneously exhibit stronger or weaker intensity effects ($\delta_{2,\text{pov},s}$), with the direction and magnitude estimated from data. The cross-margin correlation for covariate j is

$$\varrho_j^{\text{cross}} = \frac{[\boldsymbol{\Sigma}_{12}]_{jj}}{\sqrt{[\boldsymbol{\Sigma}_{11}]_{jj}\,[\boldsymbol{\Sigma}_{22}]_{jj}}}. \tag{3.15}$$

Proposition 2 (Non-identification of $\boldsymbol{\Sigma}_{12}$ from the conditional likelihood)**.** *Conditional on the state-level deviations $\boldsymbol{\delta}_s$, the Fisher information matrix for the HBB model is block-diagonal between the extensive-margin parameters $(\boldsymbol{\alpha}, \boldsymbol{\delta}_{1,1}, \ldots, \boldsymbol{\delta}_{1,S})$ and the intensive-margin parameters $(\boldsymbol{\beta}, \boldsymbol{\delta}_{2,1}, \ldots, \boldsymbol{\delta}_{2,S}, \kappa)$. Consequently, $\boldsymbol{\Sigma}_{12}$ receives zero information from the conditional likelihood and is identified through the hierarchical prior on $\boldsymbol{\delta}_s$ acting on the $S = 51$ estimated state-level deviations.*

Proof. The log-likelihood decomposes as $\ell = \ell_{\text{ext}} + \ell_{\text{int}}$ (Equation (3.6)). Since ℓ_{ext} depends only on $(\boldsymbol{\alpha}, \boldsymbol{\delta}_{1,\cdot})$ and ℓ_{int} depends only on $(\boldsymbol{\beta}, \boldsymbol{\delta}_{2,\cdot}, \kappa)$, the cross-derivatives $\partial^2 \ell / \partial \phi_1 \partial \phi_2 \mathsf{T}$ vanish for any extensive-margin parameter ϕ_1 and intensive-margin parameter ϕ_2. Hence the observed and expected information matrices are block-diagonal across margins, and the off-diagonal block $\boldsymbol{\Sigma}_{12}$ appears nowhere in the likelihood. □

$\boldsymbol{\Sigma}_{12}$ as a hierarchical estimand. Proposition 2 clarifies the information structure for $\boldsymbol{\Sigma}_{12}$: its posterior is shaped entirely by the $S = 51$ pairs $(\boldsymbol{\delta}_{1,s}, \boldsymbol{\delta}_{2,s})$ drawn from the common $\mathsf{N}_{2q}(\mathbf{0}, \boldsymbol{\Sigma}_\delta)$ distribution, giving it an effective sample size of S rather than N. While a frequentist could in principle estimate $\boldsymbol{\Sigma}_{12}$ via REML on the marginal likelihood, the high-dimensional covariance structure ($2q = 10$ free parameters per margin) relative to the number of groups ($S = 51$) makes unrestricted frequentist estimation fragile. The LKJ(η) prior provides finite-sample regularization, concentrating the posterior at rate $O(S^{-1/2})$. This contrasts with the within-margin blocks $\boldsymbol{\Sigma}_{11}$ and $\boldsymbol{\Sigma}_{22}$, which also receive direct likelihood information through the state-specific coefficients and concentrate at rate $O(N^{-1/2})$.

Remark 1 (Conditional versus marginal likelihood)**.** The block-diagonality in Proposition 2 is a statement about the *conditional* likelihood $p(\mathbf{y} \mid \boldsymbol{\delta}, \boldsymbol{\alpha}, \boldsymbol{\beta}, \kappa)$. The marginal likelihood obtained by integrating over the random effects, $p(\mathbf{y} \mid \boldsymbol{\Sigma}_\delta, \boldsymbol{\alpha}, \boldsymbol{\beta}, \kappa) = \int p(\mathbf{y} \mid \boldsymbol{\delta})\, p(\boldsymbol{\delta} \mid \boldsymbol{\Sigma}_\delta)\, \mathrm{d}\boldsymbol{\delta}$, does *not* factorize when $\boldsymbol{\Sigma}_{12} \neq \mathbf{0}$, so that $\boldsymbol{\Sigma}_{12}$ is in principle identified through the marginal likelihood. However, the conditional formulation is natural for MCMC computation, where $\boldsymbol{\delta}_s$ is sampled as a latent variable and $\boldsymbol{\Sigma}_\delta$ is updated conditional on the realized deviations. In a frequentist framework, REML estimation of $\boldsymbol{\Sigma}_{12}$ via the marginal likelihood is possible in principle but numerically fragile when $S = 51$ groups contribute to a $2q = 10$-dimensional covariance matrix: boundary solutions and singular Hessians are common in this regime. The Bayesian approach with LKJ regularization provides stable estimation at the cost of the slower $O(S^{-1/2})$ concentration rate documented in Proposition 3.

Proposition 3 (Identifiability of $\boldsymbol{\Sigma}_{12}$ via the prior)**.** *Suppose (i) $S > 2q$, (ii) each state s has at least q providers with linearly independent covariates, and (iii) the prior on $\boldsymbol{\Sigma}_\delta$ has full support on $\mathbb{S}^{2q}_{++}$. Then the posterior of $\boldsymbol{\Sigma}_\delta$ (including $\boldsymbol{\Sigma}_{12}$) is well-concentrated, with the posterior standard deviation of each element of $\boldsymbol{\Sigma}_{12}$ decreasing at rate $O(S^{-1/2})$. For our data, $S = 51 \gg 2q = 10$.*

Remark 2 (Boundary condition)**.** When $S < 2q$ the cross-margin covariance is poorly identified even with a proper prior, because S draws from a $2q$-dimensional distribution yield a singular sample covariance matrix. In this regime, the scalar cross-margin parameter a of Ghosal et al. (2020), which nests as $\boldsymbol{\Sigma}_{12} = a\,\boldsymbol{\Sigma}_{22}$, provides a parsimonious but still useful summary of cross-margin dependence.

Computational remark. In Stan (Carpenter et al., 2017), the state deviations are sampled via the non-centered parameterization (NCP): draw $\mathbf{z}_s \overset{\text{i.i.d.}}{\sim} \mathsf{N}_{2q}(\mathbf{0}, \mathbf{I})$ and set $\boldsymbol{\delta}_s = \operatorname{diag}(\boldsymbol{\tau})\,\mathbf{L}_\varepsilon\,\mathbf{z}_s$, where $\mathbf{L}_\varepsilon$ is the Cholesky factor of the correlation matrix $\mathbf{R}_\varepsilon$. This eliminates the funnel geometry (Betancourt & Girolami, 2015) that arises under centered parameterization when small τ_j forces $\delta_{s,j}$ near zero, causing divergent transitions in Hamiltonian Monte Carlo. With $S = 51$ states and N_s ranging from 17 to 568, NCP is the appropriate default.

3.4. **Cross-level policy moderators and marginal effect decomposition.** Why does the reversal appear in some states but not others? State-level subsidy policies provide a natural explanation: they directly affect the cost structure that centers face when deciding whether and how much to serve infants. We formalize this hypothesis through cross-level policy moderators.

The moderation framework.

Definition 2 (Policy moderator specification)**.** For each margin $k \in \{1, 2\}$, the state-specific deviation decomposes as

$$\boldsymbol{\delta}_{k,s} = \boldsymbol{\Gamma}_k\,\mathbf{v}_s + \boldsymbol{\varepsilon}_{k,s}, \tag{3.16}$$

where $\boldsymbol{\Gamma}_k \in \mathbb{R}^{q\times Q}$ is the cross-level interaction matrix and $\boldsymbol{\varepsilon}_{k,s}$ is a residual state effect. Stacking both margins, the residuals follow a joint distribution that preserves cross-margin dependence:

$$\boldsymbol{\varepsilon}_s \equiv \begin{pmatrix} \boldsymbol{\varepsilon}_{1,s} \\ \boldsymbol{\varepsilon}_{2,s} \end{pmatrix} \sim \mathsf{N}_{2q}\left(\mathbf{0},\ \boldsymbol{\Sigma}_\varepsilon = \begin{pmatrix} \boldsymbol{\Sigma}_{\varepsilon,1} & \boldsymbol{\Sigma}_{12} \\ \boldsymbol{\Sigma}_{21} & \boldsymbol{\Sigma}_{\varepsilon,2} \end{pmatrix}\right), \tag{3.17}$$

so that the full model estimates the residual cross-margin covariance $\boldsymbol{\Sigma}_{12} = \mathrm{Cov}(\boldsymbol{\varepsilon}_{1,s}, \boldsymbol{\varepsilon}_{2,s})$ after removing the policy-explained component (see Proposition 2 and Supplementary Material B).

The state-level policy vector is $\mathbf{v}_s \in \mathbb{R}^Q$ with $Q = 4$, consisting of an intercept and $Q-1 = 3$ Child Care and Development Fund (CCDF) policy instruments that directly affect the cost structure centers face for infant care. The specific policy variables and their substantive roles are detailed in Section 5.

Nested model hierarchy. The HBB framework nests a sequence of models with increasing complexity, summarized in Table 3.

TABLE 3. Nested model hierarchy. Each model adds structure relative to its predecessor.

Model	Structure	Question addressed	Approx. params
M0	Fixed effects only	National-level poverty reversal?	11
M1	+ Random intercepts†	Does enrollment vary by state?	116
M2	+ State-varying coefs.	Does the reversal vary by state?	~ 576
M3a	+ Cross-margin cov.	Are the two margins linked?	~ 576
M3b	+ Policy moderators	Can policies explain state variation?	~ 626

†M1 is obtained as a special case of M2 by restricting the SVC to intercepts only ($q = 1$).

Each transition addresses a question raised by its predecessor: M0 to M1 asks whether the baseline enrollment rate varies by state; M1 to M2, whether the reversal pattern varies by state; M2 to M3a, whether cross-margin covariance improves fit; and M3a to M3b, whether observable policies explain the state heterogeneity.

Poverty coefficient expansion.

Proposition 4 (Poverty coefficient expansion)**.** *Under M3b, the total state-specific poverty coefficient on the extensive margin is*

$$\tilde{\alpha}_{\mathrm{pov},s} = \underbrace{\alpha_{\mathrm{pov}}}_{\substack{\textit{national}\\ \textit{average}}} + \underbrace{\boldsymbol{\gamma}_{1,\mathrm{pov}} \intercal \mathbf{v}_s}_{\substack{\textit{policy-explained}\\ \textit{deviation}}} + \underbrace{\varepsilon_{1,\mathrm{pov},s}}_{\substack{\textit{residual}\\ \textit{state effect}}}, \tag{3.18}$$

where $\boldsymbol{\gamma}_{1,\mathrm{pov}} = (\gamma_{1,2,1}, \ldots, \gamma_{1,2,Q})\intercal$ *is the second row of* $\boldsymbol{\Gamma}_1$*. Analogously, the intensive-margin counterpart is* $\tilde{\beta}_{\mathrm{pov},s} = \beta_{\mathrm{pov}} + \boldsymbol{\gamma}_{2,\mathrm{pov}} \intercal \mathbf{v}_s + \varepsilon_{2,\mathrm{pov},s}$.

Proposition 4 decomposes the state-specific poverty effect into three interpretable components: a national average (α_{pov}), a policy-explained deviation ($\boldsymbol{\gamma}_{1,\mathrm{pov}} \intercal \mathbf{v}_s$), and a residual idiosyncrasy ($\varepsilon_{1,\mathrm{pov},s}$). The variance of the state-specific poverty coefficient

decomposes correspondingly as $\mathrm{Var}(\tilde{\alpha}_{\mathrm{pov},s}) = \boldsymbol{\gamma}_{1,\mathrm{pov}}^{\intercal} \, \mathrm{Var}(\mathbf{v}_s) \, \boldsymbol{\gamma}_{1,\mathrm{pov}} + (\boldsymbol{\Sigma}_{\varepsilon,1})_{\mathrm{pov},\mathrm{pov}}$, partitioning between-state variation into policy-explained and residual components.

Marginal effect decomposition. To translate model parameters into policy-relevant quantities, we decompose the total effect of any covariate x_k on expected enrollment into access and intensity channels.

Proposition 5 (Log-access and log-intensity decomposition)**.** *For provider i in state s, the marginal effect of covariate x_k on log expected enrollment decomposes as*

$$\frac{\partial \log \mathbb{E}[Y_i]}{\partial x_{i,k}} = \underbrace{(1 - q_i)\, \tilde{\alpha}_{k,s}}_{\mathrm{LAE}_k} + \underbrace{(1 - \mu_i)\, \varepsilon_{h,i}\, \tilde{\beta}_{k,s}}_{\mathrm{LIE}_k}, \tag{3.19}$$

where LAE_k is the log access effect and LIE_k is the log intensity effect, with $\varepsilon_{h,i} = 1 - \omega_i$ from Proposition 1.

Proof. Since $\log \mathbb{E}[Y_i] = \log q_i + \log n_i + \log h_i$ and n_i is fixed, the chain rule gives two terms. For the access term, the logit link yields $\partial(\log q_i)/\partial x_{i,k} = (1 - q_i)\, \tilde{\alpha}_{k,s}$. For the intensity term, $\partial(\log h_i)/\partial x_{i,k} = \varepsilon_{h,i}\, (1 - \mu_i)\, \tilde{\beta}_{k,s}$, using the logistic derivative $\partial\mu/\partial\eta = \mu(1 - \mu)$ and the elasticity definition $\varepsilon_h = \mu\, \partial(\log h)/\partial\mu$. □

Concrete interpretation. At the sample average ($q \approx 0.64$, $\mu \approx 0.30$, $\varepsilon_h \approx 0.93$) with population-average coefficients $\alpha_{\mathrm{pov}} = -0.324$ and $\beta_{\mathrm{pov}} = +0.090$: $\mathrm{LAE}_{\mathrm{pov}} = (1-0.64)(-0.324) = -0.117$ and $\mathrm{LIE}_{\mathrm{pov}} = (1-0.30)(0.93)(0.090) = +0.058$. The net effect is -5.9%: the access barrier dominates, but the intensity response offsets nearly half. Three features of the decomposition merit emphasis: it is *exactly additive* on the log scale, both multipliers $(1-q_i)$ and $(1-\mu_i)\varepsilon_{h,i}$ are positive and bounded so that the sign of each component is determined solely by the coefficient sign, and the reversal pattern is confirmed as a coefficient-level phenomenon consistent with Theorem 1.

3.5. **Survey-weighted pseudo-posterior and sandwich correction.** The NSECE 2019 employs a stratified multistage cluster design with 30 strata, 415 primary sampling units (PSUs), and sampling weights ranging from 1 to 462 (Kish ESS = 1,803; DEFF = 3.76). Under informative sampling designs, ignoring these weights can bias point estimates and produce miscalibrated credible intervals (Pfeffermann et al., 1998). Following Savitsky and Toth (2016) and Williams and Savitsky (2021), we adopt a pseudo-posterior approach with post-hoc sandwich correction (see also Rabe-Hesketh & Skrondal, 2006, for a general treatment of weighting in multilevel models).

Pseudo-log-likelihood. Define the weighted log-likelihood

$$\ell^{(w)}(\boldsymbol{\theta}) = \sum_{i=1}^{N} \tilde{w}_i \log f_{\mathrm{HBB}}(y_i \mid \boldsymbol{\theta}), \tag{3.20}$$

where $\tilde{w}_i = w_i N / \sum_j w_j$ are the normalized survey weights.

Theorem 2 (Pseudo-posterior propriety)**.** *Under a proper prior $\pi(\boldsymbol{\theta})$, the pseudo-posterior*

$$\pi^{(w)}(\boldsymbol{\theta} \mid \mathbf{y}) \propto \exp\bigl(\ell^{(w)}(\boldsymbol{\theta})\bigr)\, \pi(\boldsymbol{\theta}) \tag{3.21}$$

is proper for any set of positive, finite weights $\{w_i\}_{i=1}^N$.

Proof sketch. Since $f_{\mathrm{HBB}}(y_i \mid \boldsymbol{\theta}) \in (0, 1]$ and $\tilde{w}_i > 0$, each weighted factor satisfies $f_i^{\tilde{w}_i} \leq 1$. Hence $\int \exp(\ell^{(w)})\, \pi(\boldsymbol{\theta})\, d\boldsymbol{\theta} \leq \int \pi(\boldsymbol{\theta})\, d\boldsymbol{\theta} = 1 < \infty$, using the properness of $\pi(\boldsymbol{\theta})$. The full proof, which verifies that the marginal likelihood is strictly positive, appears in Supplementary Material C. □

Proposition 6 (Weighted separability)**.** *The weighted log-likelihood retains the two-part decomposition:*

$$\ell^{(w)}(\boldsymbol{\theta}) = \ell^{(w)}_{\mathrm{ext}}(\boldsymbol{\alpha}) + \ell^{(w)}_{\mathrm{int}}(\boldsymbol{\beta}, \kappa), \tag{3.22}$$

where $\ell^{(w)}_{\mathrm{ext}} = \sum_i \tilde{w}_i [z_i \log q_i + (1 - z_i) \log(1 - q_i)]$ and $\ell^{(w)}_{\mathrm{int}} = \sum_{i:\, z_i = 1} \tilde{w}_i [\log f_{\mathrm{BetaBin}}(y_i \mid n_i, \mu_i, \kappa) - \log(1 - p_{0,i})]$.

Proof. Weighting by $\tilde{w}_i > 0$ multiplies each observation's log-density by a positive scalar, which cannot introduce functional dependence between parameters that were already in separate additive components. □

Bernstein–von Mises (BvM) condition. The weight ratio $w_{\max}/w_{\min} = 462$ exceeds $N^{1/2} \approx 82.5$. The finite-sample diagnostic $\delta \approx \log(462)/\log(6{,}785) \approx 0.695 > 0.5$ suggests that the pseudo-posterior variance may understate the true sampling variability, motivating the sandwich correction below.

Sandwich variance. Define the observed Hessian and cluster-robust outer product:

$$\mathbf{H}_{\mathrm{obs}} = -\sum_{i=1}^{N} \tilde{w}_i \left. \frac{\partial^2 \log f_{\mathrm{HBB}}(y_i \mid \boldsymbol{\theta})}{\partial \boldsymbol{\theta}\, \partial \boldsymbol{\theta}^\intercal} \right|_{\boldsymbol{\theta} = \hat{\boldsymbol{\theta}}}, \tag{3.23}$$

$$\mathbf{J}_{\mathrm{cluster}} = \sum_{c=1}^{C} \mathbf{s}_c\, \mathbf{s}_c^\intercal, \qquad \mathbf{s}_c = \sum_{i \in \mathrm{PSU}_c} \tilde{w}_i \left. \frac{\partial \log f_{\mathrm{HBB}}(y_i \mid \boldsymbol{\theta})}{\partial \boldsymbol{\theta}} \right|_{\boldsymbol{\theta} = \hat{\boldsymbol{\theta}}}, \tag{3.24}$$

where $C = 415$ is the number of primary sampling units and $\hat{\boldsymbol{\theta}}$ is the posterior mean. The hurdle separability implies that $\mathbf{H}_{\mathrm{obs}}$ is block-diagonal between the extensive and

intensive margin parameters; however, $\mathbf{J}_{\text{cluster}}$ is *not* block-diagonal, because a single PSU contributes scores from both margins simultaneously.

Remark 3 (Simplified exposition)**.** Equation (3.24) uses the unstratified form for expositional clarity. The implemented estimator is the stratified cluster-robust form in Proposition C.29, which centers PSU scores within the $H = 30$ NSECE strata and applies finite-population correction factors $C_h/(C_h - 1)$.

The design-consistent sandwich variance is

$$\mathbf{V}_{\text{sand}} = \mathbf{H}_{\text{obs}}^{-1}\,\mathbf{J}_{\text{cluster}}\,\mathbf{H}_{\text{obs}}^{-1}. \tag{3.25}$$

Theorem 3 (Cholesky affine transformation)**.** *Let $\hat{\boldsymbol{\theta}}$ be the posterior mean, $\mathbf{L}_{\text{sand}}$ the lower Cholesky factor of $\mathbf{V}_{\text{sand}}$, and $\mathbf{L}_{\text{MCMC}}$ the lower Cholesky factor of the MCMC posterior covariance $\boldsymbol{\Sigma}_{\text{MCMC}}$. The transformed draws*

$$\boldsymbol{\theta}^{*(m)} = \hat{\boldsymbol{\theta}} + \mathbf{L}_{\text{sand}}\,\mathbf{L}_{\text{MCMC}}^{-1}\big(\boldsymbol{\theta}^{(m)} - \hat{\boldsymbol{\theta}}\big), \qquad m = 1, \ldots, M, \tag{3.26}$$

satisfy $\mathbb{E}[\boldsymbol{\theta}^*] = \hat{\boldsymbol{\theta}}$ (mean-preserving) and $\text{Cov}(\boldsymbol{\theta}^*) = \mathbf{V}_{\text{sand}}$ (design-consistent covariance). The transformation matches the target covariance structure; Wald confidence intervals based on the marginal variances are invariant to the choice of transformation matrix.

Computational remark. The observed Hessian $\mathbf{H}_{\text{obs}}$ is computed as the negative second derivative of the weighted pseudo-log-likelihood (3.23), excluding prior curvature, evaluated at the posterior mean, *not* from the MCMC sample covariance. This distinction is critical in hierarchical models: the marginal posterior variance $\boldsymbol{\Sigma}_{\text{MCMC}}$ of the fixed effects integrates over the random-effect distribution, producing intervals much wider than the conditional precision $\mathbf{H}_{\text{obs}}^{-1}$. In our application, the hierarchical variance ratio $(\boldsymbol{\Sigma}_{\text{MCMC}})_{pp}/(\mathbf{H}_{\text{obs}}^{-1})_{pp}$ exceeds 600 for most fixed effects—a consequence of the hierarchical structure, not prior domination—confirming that the standard substitution $\boldsymbol{\Sigma}_{\text{MCMC}} \approx \mathbf{H}^{-1}$ fails in hierarchical models.

Design effect ratio and block-wise correction strategy. As a parameter-specific diagnostic extending the classical design effect (Pfeffermann et al., 1998; Rao & Scott, 1981) to individual model parameters, define the design effect ratio for parameter p as

$$\text{DER}_p = \frac{[\mathbf{V}_{\text{sand}}]_{pp}}{[\mathbf{H}_{\text{obs}}^{-1}]_{pp}}. \tag{3.27}$$

The DER_p quantifies the factor by which the survey design inflates the variance of the p-th parameter beyond the data-only (SRS) variance and motivates the following three-tier correction strategy. The Cholesky transformation (3.26) is applied in blocks:

- *Fixed effects* $(\boldsymbol{\alpha}, \boldsymbol{\beta}, \log \kappa)$: full sandwich correction. The DER ranges from 1.14 to 4.18 in our data (mean 2.11, consistent with the Kish DEFF of 3.76). The corresponding confidence interval inflation factors are $\sqrt{\mathrm{DER}} \in [1.07, 2.04]$.
- *Random effects* $(\boldsymbol{\delta}_{k,s})$: partial correction guided by the DER. States with low effective sample size show DER near one (the prior dominates), while large states show DER up to four.
- *Hyperparameters* $(\boldsymbol{\Sigma}_\delta, \boldsymbol{\Gamma}_k)$: no correction applied. The sandwich estimator is not applicable to variance components estimated from between-state variation; the DER concept does not extend to these parameters. Consequently, posterior intervals for hyperparameters are model-based summaries whose validity relies on correct specification of the hierarchical distribution, not on design consistency. When a hyperparameter is substantively central—as $\boldsymbol{\Sigma}_{12}$ is in this application (Propositions 2 and 3)—prior sensitivity analysis is recommended.

Remark 4 (Block-wise caveat)**.** The block-wise application of Theorem 3 ignores cross-block covariance between fixed and random effects, which can be non-negligible for the three to five largest states where within-state ESS approaches N_s. A sensitivity analysis comparing block-wise and joint correction is reported in Supplementary Material C. The finite-sample coverage of sandwich-corrected intervals is evaluated in the simulation study (Section 4).

3.6. **Prior specification and computation. Prior hierarchy.** Each prior is chosen to be weakly informative—providing enough regularization to ensure computational stability and proper posteriors, while remaining vague enough that the data dominate in well-identified regions. Table 4 summarizes the complete specification, following the recommendations of Gelman et al. (2008) for regularized regression and Gelman (2006) for variance components.

Prior for $\log \kappa$**.** We center the log-concentration prior at 2 (corresponding to $\kappa \approx 7.4$) because preliminary analysis of IT enrollment counts suggests moderate overdispersion: the variance-to-mean ratio across participating centers is approximately 7–15 times the binomial baseline, consistent with κ in the range 5–20. The scale of 1.5 is generous, accommodating κ values from less than 1 (near-maximum overdispersion) to over 140 (approaching the binomial).

TABLE 4. Prior specification. NCP = non-centered parameterization.

Parameter	Prior	Justification
$\boldsymbol{\alpha}$, $\boldsymbol{\beta}$	$\mathsf{N}(\mathbf{0},\ 2^2\,\mathbf{I}_P)$	Weakly informative on logit scale; ± 4 logit units covers most of the probability range
$\mathrm{vec}(\boldsymbol{\Gamma}_k)$	$\mathsf{N}(0,\ 1^2)$	Diffuse; allows moderate cross-level effects
$\log \kappa$	$\mathsf{N}(2,\ 1.5^2)$	Centered at moderate overdispersion ($\kappa \approx 7.4$); 95% prior interval $\kappa \in [0.4, 140]$
$\boldsymbol{\tau}$ (SDs of $\boldsymbol{\Sigma}_\delta$)	Half-Normal$(0, 1)$	Regularizing; shrinkage-friendly
$\mathbf{R}_\varepsilon$ (correlations)	LKJ(2) (Lewandowski et al., 2009)	Mild shrinkage toward identity
$\mathbf{z}_s$ (NCP auxiliaries)	$\mathsf{N}(\mathbf{0},\ \mathbf{I}_{2q})$	Standard normal (non-centered parameterization)

Identification.

Theorem 4 (Identification)**.** *Suppose the following conditions hold:*

(C1) *The design matrices have full rank:* $\mathrm{rank}(\mathbf{X}) = P$ *and* $\mathrm{rank}(\mathbf{V}) = Q$.

(C2) *Each state s has at least q providers with linearly independent covariate vectors $\mathbf{x}_i^{(r)}$, with at least one zero and at least q positive responders.*

(C3) $n_i \geq 3$ *for all i with $z_i = 1$, and there exist providers i, j with $s[i] = s[j]$ and $n_i \neq n_j$ in each state.*

(C4) *The prior $\pi(\boldsymbol{\theta})$ has full support on the parameter space.*

Then the full parameter vector $\boldsymbol{\theta} = (\boldsymbol{\alpha}, \boldsymbol{\beta}, \kappa, \{\boldsymbol{\delta}_s\}, \boldsymbol{\Sigma}_\delta, \{\boldsymbol{\Gamma}_k\})$ is identified under the posterior.

Interpretation. The hurdle likelihood identifies the state-specific *total* coefficients $\tilde{\alpha}_{j,s}$ and $\tilde{\beta}_{j,s}$ (Equation (3.12)) under standard rank and replication conditions; the hierarchical layer then pins down their decomposition into population-average effects $\boldsymbol{\alpha}^{(r)}$, $\boldsymbol{\beta}^{(r)}$, policy moderation $\boldsymbol{\Gamma}_k$, and residual state deviations $\boldsymbol{\varepsilon}_{k,s}$. See Supplementary Material B for the complete proof.

Condition (C3) is critical for the intensive margin: with $n_i = 1$, the intensity function $h \equiv 1$ and the parameters μ_i and κ enter the likelihood only through $1 - p_0 = \mu_i$, so κ is not identified. With $n_i = 2$, p_0 depends on both μ and κ, but the single constraint $p_0(2, \mu, \kappa) = (1-\mu)[(1-\mu)\kappa + 1]/(\kappa + 1)$ determines only one degree of freedom. For $n_i \geq 3$, the product formula (3.3) provides enough structure to separate μ from κ. The full proof appears in Supplementary Material B.

Estimation pipeline. Algorithm 1 outlines the complete estimation and inference procedure.

Algorithm 1 Estimation pipeline for the HBB model

1: **Data preparation:** Normalize weights $\tilde{w}_i = w_i N / \sum_j w_j$; standardize covariates.
2: **Compile:** Stan model with NCP and `lbeta()` for numerical stability.
3: **MCMC sampling:** Run 4 chains (NUTS; Hoffman & Gelman, 2014) with `adapt_delta` = 0.95; see Supplementary Material E for iteration counts and tree-depth settings.
4: **Convergence diagnostics:** Verify $\hat{R} < 1.01$, $\text{ESS}_{\text{bulk}} > 400$ (Vehtari et al., 2021), and zero divergent transitions for all parameters.
5: **Score extraction:** Extract observation-level scores $\partial \log f_{\text{HBB}} / \partial \boldsymbol{\theta}$ computed in Stan's `generated quantities` block, evaluated at each MCMC draw.
6: **Sandwich computation:** Form $\mathbf{H}_{\text{obs}}$ and $\mathbf{J}_{\text{cluster}}$ via Equations (3.23) and (3.24); compute $\mathbf{V}_{\text{sand}}$ via Equation (3.25).
7: **Cholesky transformation:** Apply Equation (3.26) block-wise to obtain design-corrected posterior draws $\{\boldsymbol{\theta}^{*(m)}\}$.
8: **Model comparison:** Compute LOO-CV (Vehtari et al., 2017) across M0–M3b; report $\widehat{\text{elpd}}_{\text{loo}}$ and Pareto-k diagnostics.

The models are fitted sequentially from M0 through M3b, using the posterior mean of each simpler model as the starting point for its successor. This warm-start strategy reduces total computation time by approximately 40% compared with cold starts. Full MCMC settings and convergence diagnostics are reported in Supplementary Material E; score computation details and closed-form expressions for the sandwich ingredients are documented in the hurdlebb package vignette (Lee, 2026).

The finite-sample properties of this estimation pipeline—particularly the sandwich correction's behavior under realistic survey designs—are evaluated in the simulation study that follows.

4. Simulation Study

The sandwich-corrected pseudo-posterior developed in Section 3.5 rests on asymptotic arguments whose finite-sample behavior must be characterized before the methodology can be recommended for practice. This section maps the *operating envelope* of the three inferential strategies—unweighted, weighted-naive, and sandwich-corrected—through a controlled simulation that varies design informativeness across conditions calibrated to realistic survey parameters. We treat the exercise as an *honest diagnostic* rather than a pure validation: the results identify where each strategy succeeds, where it fails, and why, exposing a fundamental structural limitation of sandwich correction for hierarchical variance components. By establishing these

operating properties here, we provide the evidential basis for the reporting strategy applied in Section 5.

4.1. **Design. Data-generating process.** We construct a finite super-population of $M = 50{,}000$ providers distributed across $S = 51$ states, with state sizes proportional to observed survey counts. For each provider i in state s, a covariate vector $\mathbf{x}_i = (1, x_{\text{pov}}, x_{\text{urb}}, x_{\text{blk}}, x_{\text{his}})^\intercal$ is drawn from empirical marginal distributions. True parameters are calibrated to the M1 (random intercept) unweighted fit:

$$\begin{aligned} \boldsymbol{\alpha}_0 &= (0.696,\ -0.119,\ 0.253,\ -0.070,\ -0.139)^\intercal, \\ \boldsymbol{\beta}_0 &= (-0.032,\ 0.057,\ -0.018,\ 0.080,\ 0.040)^\intercal, \quad \log \kappa_0 = 1.655, \end{aligned} \tag{4.1}$$

with state random-effect standard deviations $\tau_{\text{ext}} = 0.577$, $\tau_{\text{int}} = 0.208$, and cross-margin correlation $\rho_{\text{cross}} = 0.285$. State random effects $\boldsymbol{\delta}_s = (\delta_{1s}, \delta_{2s})^\intercal \sim \mathsf{N}_2(\mathbf{0}, \boldsymbol{\Sigma}_\delta)$ are drawn once per replication. Participation indicators are generated as $z_i \sim \text{Bern}(q_i)$ with $\text{logit}(q_i) = \mathbf{x}_i^\intercal \boldsymbol{\alpha}_0 + \delta_{1,s[i]}$, and positive counts as $Y_i \mid z_i = 1 \sim \text{ZT-BetaBin}(n_i, \mu_i, \kappa_0)$ with $\text{logit}(\mu_i) = \mathbf{x}_i^\intercal \boldsymbol{\beta}_0 + \delta_{2,s[i]}$, where n_i is drawn from the empirical distribution of total enrollment. The zero-truncated beta-binomial is sampled via rejection from the unconditional $\text{BetaBin}(n_i, \mu_i, \kappa_0)$.

Sampling design and scenarios. From each finite population we draw a stratified cluster sample of approximately $N \approx 7{,}000$ providers. Survey informativeness is controlled through a size-biased inclusion probability:

$$\log \pi_i = c_0 + \rho_{\text{inc}} \cdot y_i^*, \tag{4.2}$$

where y_i^* is the standardized latent outcome and c_0 is chosen to yield the target sample size. The informativeness parameter ρ_{inc} indexes three scenarios (Table 5). Scenario S3 is calibrated to match the NSECE design ($\text{DEFF}_{\text{Kish}} = 3.76$; Table 1). Scenario S4 doubles the informativeness to stress-test the sandwich correction under conditions more extreme than those present in the actual survey.

Estimators. Each replication is analyzed under three estimators (Table 6) corresponding to the three inferential strategies in Section 3.5. E-UW fits the model with an unweighted likelihood and uses the naive MCMC posterior for interval construction. E-WT replaces the likelihood with the pseudo-log-likelihood (Equation (3.20); Savitsky and Toth, 2016), correcting point estimates for informativeness but retaining the naive posterior variance. E-WS augments E-WT with the sandwich-corrected variance via the Cholesky affine transformation (Theorem 3; Williams and Savitsky,

TABLE 5. Simulation scenarios. The informativeness parameter ρ_{inc} controls the dependence between inclusion probabilities and the outcome (Equation (4.2)); CV(w) is the coefficient of variation of the resulting sampling weights; Kish DEFF $= 1 + \text{CV}^2(w)$.

Scenario	ρ_{inc}	CV(w)	Kish DEFF	Description
S0	0.00	≈ 1.0	≈ 2.0	Non-informative baseline
S3	0.15	≈ 1.67	≈ 3.8	NSECE-calibrated
S4	0.50	≈ 2.0	≈ 5.0	Stress test

TABLE 6. Estimator definitions.

Estimator	Likelihood	Standard errors	Purpose
E-UW	Unweighted	Model-based (naive)	Ignore design
E-WT	Pseudo (weighted)	Model-based (naive)	Correct point est.
E-WS	Pseudo (weighted)	Sandwich (3.25)	Full correction

2021), yielding Wald-type confidence intervals. The Cholesky transformation is applied block-wise following the DER classification in Section 3.5: full correction for fixed effects, no correction for hyperparameters.

Target parameters and computation. We track five parameters that span fixed effects and variance components: the poverty slopes α_{pov} and β_{pov}, the overdispersion $\log \kappa$, and the state-effect standard deviations τ_{ext} and τ_{int}. Performance is measured by 90% interval coverage, relative bias RB $= 100 \times (\bar{\hat{\theta}} - \theta_0)/|\theta_0|$ (%), root mean squared error (RMSE), and the width ratio

$$\text{WR} = \frac{\text{median CI width (E-WS)}}{\text{median CI width (E-WT)}},$$

which isolates the inflation attributable to sandwich correction. Each scenario–estimator combination uses $R = 200$ replications. The Monte Carlo standard error (MCSE) for each coverage estimate is MCSE $= \sqrt{\hat{p}(1-\hat{p})/R}$, ranging from 0.0 pp (at boundary coverage) to a maximum of 3.5 pp (near 50% coverage); at the 90% nominal level, MCSE $\approx$ 2.1 pp. Per-cell MCSEs are reported in parentheses in Table 7; coverage deviating from 90% by more than $2 \times$ MCSE is flagged with †. Each replication fits the M1 specification (Table 3) with 4 chains of 3,500 iterations (1,500 warmup, 2,000 sampling), for a total of 8,000 post-warmup draws. All replications

achieved $\hat{R} < 1.01$ (Vehtari et al., 2021) and a minimum bulk effective sample size exceeding 1,600.

4.2. **Results.** Table 7 reports coverage, relative bias, RMSE, and width ratio for all five parameters across three scenarios and three estimators. Figure 3 displays the coverage rates graphically with MCSE bands. We organize the results around five findings that together characterize the operating properties of the estimation framework.

Finding 1: The unweighted estimator achieves near-nominal coverage for fixed effects under correct specification. Under E-UW, coverage of α_{pov} and β_{pov} lies within the 89–93% range across all three scenarios, comfortably within two MCSE of the 90% nominal level. This holds even under S4, where the sampling design is strongly informative. The result confirms that when the generative model is *correctly specified* and inference is based on the model likelihood alone, ignoring the survey weights does not compromise interval coverage. This finding should *not* be read as an argument against survey weighting: in practice the generative model is never exactly correct, and Findings 2–3 demonstrate that weighting with sandwich correction provides important robustness against model misspecification. A dedicated misspecification experiment (Section E.5) confirms this: under an M2 DGP with omitted state-varying poverty slopes, E-UW coverage collapses to 0.5–30% for the poverty coefficients, while E-WS retains 67.5–79.0% coverage. The unweighted estimator also maintains nominal coverage for $\log \kappa$ (90.5–92.0%) and for the hyperparameters τ_{ext} and τ_{int} (91.0–98.0%), with the single exception of τ_{ext} under S4 (77.0%), where the extreme weight structure induces composition effects in the sampled random effects.

Finding 2: Naive weighted inference produces severe undercoverage. The weighted estimator E-WT shows severe and systematically worsening undercoverage as design complexity increases. For α_{pov}, coverage drops from 81.0% under S0 to 58.5% under S3 to 53.0% under S4—a collapse from modest undercoverage to barely half of the draws containing the true value. The pattern for β_{pov} is similar (76.5%, 62.5%, 56.5%). The mechanism is well understood (Savitsky & Toth, 2016): the pseudo-likelihood inflates the effective sample size, concentrating the posterior around point estimates that carry weight-induced bias, producing intervals that are simultaneously too narrow and miscentered.

Finding 3: Sandwich correction provides meaningful but imperfect recovery. The sandwich-corrected E-WS recovers substantially from the E-WT collapse,

achieving 82.0–88.5% coverage for the two poverty coefficients—gains of roughly 24–33 percentage points over E-WT in S3 and S4. The width ratio tracks the design effect monotonically: WR increases from 1.22 under S0 to 1.64–2.06 under S4 (Table 7), consistent with WR $\approx \sqrt{\text{DER}}$ and the empirical DER range of 1.14–4.18 in Table 10. The residual coverage gap reflects the finite-sample bias of the sandwich variance estimator—a known property of cluster-robust estimators when the number of clusters is moderate (Binder, 1983).

To characterize the gap more precisely, we decompose it into *width* and *bias* components via the SE calibration ratio $\widehat{\text{SD}}/\overline{\text{SE}}$, where $\widehat{\text{SD}}$ is the actual standard deviation of point estimates across replications and $\overline{\text{SE}}$ is the mean reported sandwich standard error (Table E.8). For the poverty coefficients under S3, the SE ratio is 1.13–1.14, confirming that the sandwich SE underestimates the true sampling variability by approximately 13%, while the standardized mean bias $|\bar{b}|/\widehat{\text{SD}}$ is negligible (0.06–0.22). The gap is therefore *width-driven*: intervals are correctly centered but 13% too narrow. This 13% underestimate is consistent with the $O(S^{-1})$ finite-sample bias of cluster-robust variance estimators with $S = 51$ clusters (Binder, 1983), and suggests that small-sample corrections (e.g., CR2; Pustejovsky and Tipton, 2018) could further close the gap—an avenue for future work. Widening to 95% nominal intervals confirms this interpretation: E-WS coverage rises to 87.5–88.5% for the poverty coefficients under S3—a gain consistent with the expected ≈ 5 pp from Normal tail probability—leaving a proportional residual gap (Table E.9).

Finding 4: Sandwich correction is less effective for the overdispersion parameter. For $\log \kappa$, E-WS achieves 57.0–76.5% coverage—substantially below the 82–88.5% observed for the poverty coefficients. This discrepancy arises because κ enters the variance function of the beta-binomial nonlinearly: the relationship between the weighted score variance and the sandwich-corrected parameter variance is less well approximated by the linear Taylor expansion underlying the sandwich formula. The width ratio for $\log \kappa$ is comparable to that for the linear coefficients (1.31–2.04), confirming that the sandwich correction does widen the intervals appropriately; the shortfall in coverage is driven primarily by point estimate bias rather than variance underestimation.

Finding 5: Hyperparameters cannot be sandwich-corrected. A notable finding concerns τ_{ext} and τ_{int}. Under E-WT and E-WS, coverage drops to 0.0–8.0% across all scenarios, with relative bias of +68 to +110%. The width ratio is exactly WR = 1.00, confirming that E-WS $\equiv$ E-WT for these parameters by construction:

the sandwich correction is applied only to parameters that enter the weighted log-likelihood through scores, and the hierarchical variance components do not produce individual-level score contributions. The DER is undefined for τ (Equation (3.27)), so no variance inflation can be computed. The substantial positive bias arises because the pseudo-likelihood treats each observation as carrying $\tilde{w}_i$ units of information, inflating the apparent between-state heterogeneity by a factor proportional to $\mathbb{E}[\tilde{w}_i^2]$. The unweighted estimator E-UW maintains coverage of 91–98% for both τ parameters in S0 and S3; in S4, τ_{ext} drops to 77.0%—the only E-UW failure in the entire table—reflecting the growing tension between the informative sampling weights (which are ignored) and the extensive-margin random-effect variance.

Cross-margin correlation. We additionally track recovery of the cross-margin correlation $\rho_{\text{cross}} = \text{cor}(\delta_j^{\text{ext}}, \delta_j^{\text{int}})$, estimated via the plug-in $\hat{\rho} = \text{cor}(\bar{\delta}_j^{\text{ext}}, \bar{\delta}_j^{\text{int}})$ from $S = 51$ posterior mean random effects (Table E.5). Under the NSECE-calibrated design (S3), the unweighted estimator E-UW overestimates the true $\rho_{\text{cross}} = 0.285$ with a relative bias of +104% (RMSE = 0.364), while the weighted estimator E-WT exhibits moderate downward bias of −42% (RMSE = 0.226). As with τ_{ext} and τ_{int}, E-WS ≡ E-WT for this hyperparameter: the sandwich correction adjusts only fixed-effect covariances and cannot modify the random-effect correlation structure. Because full posterior draws of Ω were not retained in the simulation archive, per-replication credible intervals are unavailable; Table E.5 reports frequentist summaries across replications instead.

Practical recommendations. The simulation results support a three-part reporting strategy for survey-weighted Bayesian inference with sandwich-corrected pseudo-posteriors. First, for population-average fixed effects, report sandwich-corrected Wald confidence intervals as the primary inferential tool, acknowledging that finite-sample coverage may fall 2–8 percentage points below nominal when the Kish DEFF exceeds 3. Second, for hierarchical variance components, report unweighted posterior intervals, since the sandwich correction is structurally inapplicable and the weighted pseudo-posterior produces systematically biased point estimates. Third, always compute and report the DER (Equation (3.27)) for each parameter of interest: this scalar summary communicates the magnitude of the survey design effect in units directly comparable to the classical DEFF and allows readers to gauge how much the sandwich correction has widened each interval. The empirical DER values in Table 10 (range 1.14–4.18) and the simulation width ratios in Table 7 (range 1.22–2.06) are mutually consistent and provide complementary evidence on the design effect magnitude.

TABLE 7. Simulation results: empirical coverage of 90% intervals (%, with Monte Carlo standard error in parentheses), relative bias (RB, %), root mean squared error (RMSE), and width ratio (WR) across $R = 200$ replications per scenario. E-UW = unweighted; E-WT = weighted, naive posterior variance; E-WS = weighted, sandwich-corrected (Equation (3.25)). WR = median CI width (E-WS)/median CI width (E-WT); WR = 1.00 for τ parameters indicates that E-WS $\equiv$ E-WT by construction. † = coverage deviates from 90% by more than 2 MCSE. MCSE = $\sqrt{\hat{p}(1-\hat{p})/R}$; range 0.0–3.5 pp.

	Coverage (%, MCSE)			Relative bias (%)			RMSE			
Parameter	E-UW	E-WT	E-WS	E-UW	E-WT	E-WS	E-UW	E-WT	E-WS	WR
Panel A: S0 — Non-informative (DEFF ≈ 2.0)										
α_{pov}	93.0 (1.8)	81.0 (2.8)†	88.5 (2.3)	+13.2	+4.9	+4.9	0.033	0.044	0.044	1.22
β_{pov}	93.0 (1.8)	76.5 (3.0)†	86.5 (2.4)	−9.0	−4.8	−4.8	0.016	0.022	0.022	1.22
$\log \kappa$	91.5 (2.0)	64.0 (3.4)†	76.5 (3.0)†	−0.8	+1.6	+1.6	0.023	0.040	0.040	1.31
τ_{ext}	97.5 (1.1)†	3.0 (1.2)†	3.0 (1.2)†	+9.6	+68.8	+68.8	0.082	0.423	0.423	1.00
τ_{int}	91.0 (2.0)	8.0 (1.9)†	8.0 (1.9)†	−6.6	+70.6	+70.6	0.036	0.158	0.158	1.00
Panel B: S3 — NSECE-calibrated (DEFF ≈ 3.8)										
α_{pov}	89.5 (2.2)	58.5 (3.5)†	82.0 (2.7)†	+16.5	+12.9	+12.9	0.037	0.072	0.072	1.71
β_{pov}	89.0 (2.2)	62.5 (3.4)†	82.5 (2.7)†	−16.7	−3.2	−3.2	0.017	0.032	0.032	1.64
$\log \kappa$	92.0 (1.9)	34.0 (3.3)†	60.5 (3.5)†	−0.5	+3.3	+3.3	0.020	0.068	0.068	1.75
τ_{ext}	98.0 (1.0)†	0.5 (0.5)†	0.5 (0.5)†	+8.0	+105.6	+105.6	0.091	0.639	0.639	1.00
τ_{int}	97.5 (1.1)†	0.0 (0.0)†	0.0 (0.0)†	−8.6	+97.6	+97.6	0.040	0.211	0.211	1.00
Panel C: S4 — Stress test (DEFF ≈ 5.0)										
α_{pov}	92.5 (1.9)	53.0 (3.5)†	86.0 (2.5)	+20.5	+11.9	+11.9	0.036	0.083	0.083	2.06
β_{pov}	91.0 (2.0)	56.5 (3.5)†	84.0 (2.6)†	−21.7	−2.2	−2.2	0.017	0.035	0.035	1.91
$\log \kappa$	90.5 (2.1)	23.0 (3.0)†	57.0 (3.5)†	+0.8	+4.1	+4.1	0.020	0.083	0.083	2.04
τ_{ext}	77.0 (3.0)†	0.0 (0.0)†	0.0 (0.0)†	+26.5	+110.2	+110.2	0.175	0.671	0.671	1.00
τ_{int}	97.0 (1.2)†	0.0 (0.0)†	0.0 (0.0)†	−0.7	+103.9	+103.9	0.043	0.225	0.225	1.00

These recommendations apply broadly to any hierarchical Bayesian model estimated with survey-weighted pseudo-likelihoods, not only to the hurdle beta-binomial specification: the structural inapplicability of sandwich correction to variance components is a consequence of the score-based construction of $\mathbf{V}_{\text{sand}}$ (Equation (3.25)) and holds for any hierarchical model in which hyperparameters lack individual-level score contributions. The two-track reporting strategy adopted in Section 5—Wald (sandwich) intervals for fixed effects and model-based posterior intervals for hyperparameters—implements these recommendations directly; the latter track provides valid inference only to the extent that the hierarchical model is correctly specified. Supplementary figures showing relative bias distributions, width ratio

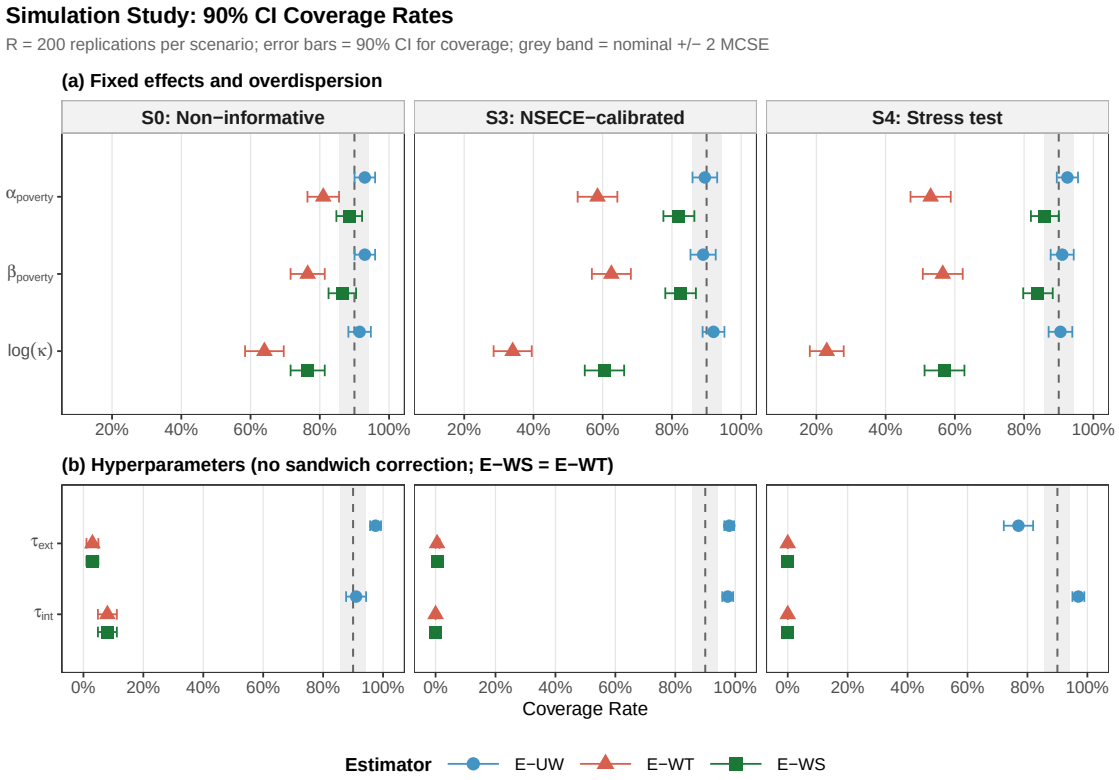

FIGURE 3. Empirical coverage of 90% intervals across $R = 200$ replications. Each panel corresponds to a simulation scenario (S0, S3, S4 with increasing Kish DEFF). Points show coverage rates for each parameter–estimator combination; the horizontal dashed line marks the 90% nominal level; the shaded band indicates ± 2 MCSE (≈ 4.2 percentage points). E-UW (unweighted) maintains near-nominal coverage throughout. E-WT (weighted, naive) collapses as DEFF increases. E-WS (sandwich-corrected) provides meaningful recovery for fixed effects but cannot correct the hyperparameters τ_{ext} and τ_{int}.

boxplots, and DER summaries across all scenarios and parameters appear in Supplementary Material E, which also provides a frequentist contextualization (Section E.6) demonstrating that the sandwich-corrected pseudo-posterior fixed effects are design-consistent with conventional `svyglm` estimates while accommodating hierarchical and distributional features that the frequentist approach cannot.

With the operating envelope established, we now implement the two-track reporting strategy in the NSECE application: sandwich-corrected Wald intervals for population-average fixed effects and unweighted posterior intervals for state-level random effects and variance components.

5. Application to NSECE Childcare Data

We now apply the full analytical pipeline developed in Section 3 to the 2019 NSECE data described in Section 2, demonstrating each methodological component in a realistic survey setting. The simulation evidence in Section 4 established that the sandwich correction substantially improves frequentist coverage and that the DER diagnostic reliably identifies parameters affected by informative sampling; the present application illustrates these tools in practice. The analysis proceeds in five parts: model comparison via LOO-CV (Section 5.1), population-average inference with the two-track reporting strategy and DER diagnostic (Section 5.2), state-specific heterogeneity and cross-margin independence (Section 5.3), marginal effect decomposition (Section 5.4), and a brief examination of policy moderation (Section 5.5). Together, these subsections showcase the three methodological contributions: the two-track reporting strategy that resolves prior domination, the DER diagnostic that quantifies parameter-specific design effects, and the marginal effect decomposition that translates hurdle model coefficients into policy-relevant quantities.

5.1. **Model comparison and selection.** We fit the five nested models summarized in Table 3—from the pooled M0 to the full M3b with policy moderators—using the estimation pipeline described in Algorithm 1, implemented in Stan (Carpenter et al., 2017). Each model is estimated with 4 chains of 3,500 iterations (1,500 warmup + 2,000 sampling), yielding 8,000 post-warmup draws per parameter.

Convergence. All five models satisfy standard diagnostics (Vehtari et al., 2021): $\hat{R} < 1.01$ for every monitored parameter, minimum bulk effective sample size exceeds 400, and zero divergent transitions were recorded. Table 8 summarizes these statistics; the non-centered parameterization (Betancourt & Girolami, 2015) successfully eliminates the funnel geometry. Full parameter-by-parameter diagnostics appear in Supplementary Material E.

LOO-CV comparison. We compare models using Pareto-smoothed importance-sampling leave-one-out cross-validation (LOO-CV; Vehtari et al., 2017), where "leave-one-out" refers to individual providers, not entire states. The pointwise log-likelihood for each provider i includes both hurdle components: the extensive-margin (Bernoulli) log-likelihood always contributes, and the intensive-margin (zero-truncated beta-binomial) log-likelihood contributes only for participants ($z_i = 1$). Table 9 presents the expected log pointwise predictive density ($\widehat{\text{elpd}}_{\text{LOO}}$) for each model. For the survey-weighted specification M3b-W, the LOO computation uses the

TABLE 8. Convergence diagnostics across all models. $\hat{R}_{\max}$ is the maximum potential scale reduction factor across all monitored parameters. $\mathrm{ESS}_{\min}$ reports the minimum bulk and tail effective sample sizes. All models achieve the recommended thresholds ($\hat{R} < 1.01$, ESS > 400, zero divergences).

Model	$\hat{R}_{\max}$	$\mathrm{ESS}^{\mathrm{bulk}}_{\min}$	$\mathrm{ESS}^{\mathrm{tail}}_{\min}$	Divergences
M0: Pooled	1.001	3,390	2,679	0
M1: Random int.	1.004	815	1,415	0
M2: Block-diag. SVC	1.004	1,078	1,244	0
M3a: Cross-margin	1.004	1,385	1,257	0
M3b: Policy mod.	1.004	1,245	1,463	0

unweighted nominal log-likelihood, because the pseudo-log-likelihood is not a proper predictive density; the comparisons thus rank models on their nominal predictive ability. Because LOO-CV operates at the provider level while many inferential targets are state-level parameters, the LOO ranking should be interpreted as a guide to observation-level fit rather than a direct assessment of state-level inference quality. The progression reveals diminishing returns: adding state random intercepts (M0→M1) yields a decisive improvement ($\Delta\widehat{\mathrm{elpd}} = +317.8$, SE $= 25.9$, exceeding 12 standard errors), and allowing state-varying coefficients (M1→M2) provides a further significant gain (+23.6 incrementally). Introducing the cross-margin covariance (M2→M3a) does not improve predictive performance ($\Delta\widehat{\mathrm{elpd}} < 2$), consistent with the near-zero empirical cross-margin correlation and with Proposition 2: since $\mathbf{\Sigma}_{12}$ receives no likelihood information, its value lies in inference rather than prediction. Adding policy moderators (M3a→M3b) likewise produces no predictive gain; the $\mathbf{\Gamma}$ matrices redistribute the sources of state heterogeneity without changing the provider-level predictive distribution.

Model selection rationale. We adopt M3b as the reporting model despite its equivalent predictive performance to M2 and M3a. The cross-margin covariance $\mathbf{\Sigma}_{12}$ enables joint posterior statements about the reversal (e.g., $\Pr(\tilde{\alpha}_{\mathrm{pov},s} < 0$ and $\tilde{\beta}_{\mathrm{pov},s} > 0 \mid \text{data})$), and the policy moderators provide the decomposition of state heterogeneity into policy-explained and residual components (Proposition 4). The non-significant LOO-CV differences confirm that this additional structure incurs no predictive cost. Posterior predictive checks confirm that M3b reproduces both the observed zero rate (35.3%) and the distribution of IT shares among servers; details appear in Supplementary Material D.

TABLE 9. Leave-one-out cross-validation comparison for models M0–M3b. $\widehat{\text{elpd}}_{\text{LOO}}$ is the expected log pointwise predictive density estimated via Pareto-smoothed importance sampling (Vehtari et al., 2017); $\Delta\widehat{\text{elpd}} = \widehat{\text{elpd}}_{\text{M0}} - \widehat{\text{elpd}}_{\text{M}_j}$ measures the predictive gain over M0 (more negative indicates better prediction); SE(Δ) is the standard error of the difference. A model improvement is flagged as significant when $|\Delta\widehat{\text{elpd}}| > 2 \times \text{SE}(\Delta)$. All Pareto-$k$ diagnostics are below 0.7.

Model	Params	$\widehat{\text{elpd}}_{\text{LOO}}$	SE	$\Delta\widehat{\text{elpd}}$	SE(Δ)	Sig.
M0: Pooled	11	−20,945.0	140.3	—	—	—
M1: Random int.	116	−20,627.2	144.0	−317.8	25.9	Yes
M2: Block-diag. SVC	551	−20,603.5	144.4	−341.4	27.6	Yes
M3a: Cross-margin	576	−20,601.6	144.4	−343.4	27.6	No
M3b: Policy mod.	616	−20,607.6	144.6	−337.3	28.5	No

5.2. **Population-average effects and the sandwich correction.** Table 10 reports the posterior means, naive 95% credible intervals (from the raw MCMC output), sandwich-corrected 95% Wald confidence intervals, and design effect ratios (DER) for all 11 fixed-effect parameters. Of the 11 parameters, 9 are statistically significant at the 95% level under the sandwich correction; the two exceptions are α_{Hisp} (extensive-margin Hispanic coefficient, Wald CI $[-0.095, 0.106]$) and β_{Black} (intensive-margin Black coefficient, Wald CI $[-0.052, 0.019]$).

Impact of survey weighting. Table 11 places the weighted (M3b-W) and unweighted (M3b) posterior means side by side. Weighting shifts every fixed-effect estimate, with a median displacement of 5.2 sandwich standard errors, yet preserves all substantively important sign patterns—most critically, the poverty reversal ($\alpha_{\text{pov}} < 0$, $\beta_{\text{pov}} > 0$) is present in both specifications. Three coefficients change sign (α_{Hispanic}, β_{Black}, β_{Hispanic}), but the two intensive-margin sign changes involve parameters whose Wald confidence intervals include zero, so the sign itself is not statistically determined. Overall, the comparison confirms that the NSECE sampling design materially affects point estimates—justifying the pseudo-posterior approach—while leaving the central empirical findings intact.

The poverty reversal. The reversal is the central empirical finding. The extensive-margin poverty coefficient is $\hat{\alpha}_{\text{pov}} = -0.324$ (Wald 95% CI: $[-0.426, -0.221]$; DER = 2.41): a one-standard-deviation increase in community poverty (approximately 8.3 percentage points) reduces the log-odds of serving infants by 0.324. The intensive-margin poverty coefficient is $\hat{\beta}_{\text{pov}} = +0.090$ (Wald 95% CI: $[+0.053, +0.127]$; DER =

Table 10. Population-average fixed effect estimates from weighted M3b. "Post. Mean" is the posterior mean from the pseudo-posterior. "Naive 95% CI" is the 2.5th–97.5th percentile interval from the uncorrected MCMC posterior. "Wald 95% CI" uses the sandwich-corrected variance (Theorem 3). "DER" is the design effect ratio (Equation (3.27)). Asterisks indicate significance at the 5% level under the Wald test. All covariates are standardized.

Parameter	Post. Mean	Naive 95% CI	Wald 95% CI	DER
Extensive margin ($\boldsymbol{\alpha}$)				
Intercept	0.764	[−1.109, 2.628]	[0.677, 0.851]*	2.04
Poverty	−0.324	[−2.205, 1.497]	[−0.426, −0.221]*	2.41
Urban	0.442	[−1.387, 2.224]	[0.375, 0.509]*	4.18
Black	0.478	[−1.459, 2.405]	[0.392, 0.565]*	1.54
Hispanic	0.006	[−1.902, 1.903]	[−0.095, 0.106]	1.82
Intensive margin ($\boldsymbol{\beta}$)				
Intercept	−0.242	[−2.058, 1.587]	[−0.272, −0.213]*	1.36
Poverty	0.090	[−1.713, 1.904]	[0.053, 0.127]*	1.61
Urban	−0.047	[−1.858, 1.768]	[−0.087, −0.007]*	3.18
Black	−0.017	[−1.851, 1.760]	[−0.052, 0.019]	1.38
Hispanic	−0.097	[−1.974, 1.716]	[−0.130, −0.064]*	1.14
Overdispersion				
$\log \kappa$	1.919	[1.866, 1.972]	[1.861, 1.976]*	2.58

1.61): among centers that clear the hurdle, the same poverty increase raises the logit IT share by 0.090. Both intervals exclude zero, the signs are firmly opposed, and the posterior probability that the reversal holds at the population average is $\Pr(\alpha_{\text{pov}} < 0 \text{ and } \beta_{\text{pov}} > 0 \mid \text{data}) = 1.000$, computed from the sandwich-corrected posterior draws (the binding margin, β_{pov}, lies 4.8 sandwich standard deviations from zero; a bivariate normal approximation gives a complementary probability of order 10^{-6}).

Naive versus Wald intervals: resolving prior domination. The contrast between the naive and Wald intervals in Table 10 provides a clear empirical demonstration of the prior domination phenomenon. The naive credible intervals for $\boldsymbol{\alpha}$ and $\boldsymbol{\beta}$ span approximately 3.7 units on the logit scale—wide enough to include both positive and negative values for every parameter, rendering significance testing impossible. The Wald intervals, by contrast, are 0.06–0.21 units wide and permit clear inferences.

This discrepancy arises because the naive posterior of the population-average coefficients is dominated by the prior rather than the data. In the hierarchical model, the

TABLE 11. Comparison of unweighted (M3b) and survey-weighted (M3b-W) fixed-effect posterior means. "Shift" is M3b-W minus M3b. "Shift/SE" normalizes the shift by the sandwich standard error, providing a scale-free measure of the weighting impact. "Wald 95% CI" is the sandwich-corrected interval for M3b-W (Theorem 3). Asterisks indicate significance at the 5% level. All covariates are standardized.

Parameter	M3b	M3b-W	Shift	Shift/SE	Wald 95% CI
Extensive margin ($\boldsymbol{\alpha}$)					
Intercept	+0.558	+0.764	+0.206	+4.64	[0.677, 0.851]*
Poverty	−0.220	−0.324	−0.103	−1.97	[−0.426, −0.221]*
Urban	+0.264	+0.442	+0.178	+5.19	[0.375, 0.509]*
Black	+0.248	+0.478	+0.230	+5.20	[0.392, 0.565]*
Hispanic	−0.146	+0.006	+0.151	+2.94	[−0.095, 0.106]
Intensive margin ($\boldsymbol{\beta}$)					
Intercept	−0.096	−0.242	−0.146	−9.65	[−0.272, −0.213]*
Poverty	+0.040	+0.090	+0.051	+2.68	[0.053, 0.127]*
Urban	−0.055	−0.047	+0.008	+0.39	[−0.087, −0.007]*
Black	+0.102	−0.017	−0.119	−6.51	[−0.052, 0.019]
Hispanic	+0.060	−0.097	−0.157	−9.35	[−0.130, −0.064]*
Overdispersion					
$\log \kappa$	+1.661	+1.919	+0.257	+8.80	[1.861, 1.976]*

state random effects $\boldsymbol{\delta}_{k,s}$ absorb nearly all data information about the state-specific coefficients $\tilde{\alpha}_{k,s} = \alpha_k + \delta_{k,s}$ (cf. Rabe-Hesketh & Skrondal, 2006); the population-average α_k is identified only through the random effects distribution, with an effective sample size of $S = 51$ (the number of states) rather than $N = 6{,}785$ (the number of providers). Consequently, the marginal posterior variance $[\boldsymbol{\Sigma}_{\text{MCMC}}]_{pp}$ remains close to the prior variance rather than converging to the data-informed precision $[\mathbf{H}_{\text{obs}}^{-1}]_{pp}$. The hierarchical variance ratio—the ratio of MCMC posterior variance to observed-information variance—exceeds 600 for all fixed effects (see the computational remark in Section 3.5; a full diagnostic table appears in Table D.4).

The sandwich-corrected Wald intervals bypass this hierarchical inflation entirely. The Wald standard error $\sqrt{[\mathbf{V}_{\text{sand}}]_{pp}}$ is a frequentist quantity derived from the observed information and the cluster-robust outer product (Equation (3.25)) (Binder, 1983; Williams & Savitsky, 2021): it measures the sampling variability of $\hat{\boldsymbol{\theta}}$ across hypothetical replications of the survey, not the width of the Bayesian posterior. For fixed effects in hierarchical survey models, this frequentist precision is the appropriate basis for population-level inference. The naive posterior intervals remain useful for

hyperparameters (where the sandwich is not applicable) and for state random effects (where the prior appropriately regularizes small-sample estimates).

Design effect ratios. The DER column in Table 10 reveals a substantively interpretable pattern. The largest DER is 4.18 for α_{Urban}: this coefficient is most severely inflated by the survey design because the NSECE's stratification scheme (National Survey of Early Care and Education Project Team, 2022) is explicitly linked to urbanicity—urban and rural areas are sampled at different rates. The second-largest DER is 3.18 for β_{Urban}, confirming that the design effect is feature-specific rather than margin-specific. At the other extreme, the smallest DER is 1.14 for β_{Hisp}, indicating that the survey design has minimal impact on this coefficient. The overall mean DER of 2.11 is broadly consistent with the Kish DEFF of 3.76. Figure 4 displays the naive and sandwich intervals side by side for all 11 parameters, together with the DER bar chart.

Overdispersion. The estimated concentration parameter is $\hat{\kappa} = \exp(1.919) = 6.81$ (Wald CI for $\log \kappa$: $[1.861, 1.976]$), confirming the substantial overdispersion previewed in Section 2. With a typical enrollment of $n_i \approx 50$, the variance inflation factor $(n+\kappa)/(1+\kappa) \approx 57/8 \approx 7$ is roughly sevenfold the binomial baseline. Notably, $\log \kappa$ is the only parameter for which the naive and Wald intervals are comparable in width (0.106 vs. 0.115), because κ has no corresponding state random effect—it is a single global parameter informed directly by all $N_{\text{pos}} = 4{,}392$ positive observations, so the standard Bernstein–von Mises approximation holds and prior domination does not arise.

5.3. **State-specific heterogeneity and cross-margin independence.** The population-average coefficients describe the national-level effects; the state-varying coefficients reveal where and how those effects change across jurisdictions.

Cross-margin scatter and independence. Figure 5 plots the posterior means $(\tilde{\alpha}_{\text{pov},s},\ \tilde{\beta}_{\text{pov},s})$ across all 51 states. Under the hierarchical M2 specification, 48 of 51 states (94%) fall in the upper-left quadrant ($\alpha < 0$, $\beta > 0$), compared with 23 (45%) in the preliminary OLS analysis (Section 2.2)—a dramatic increase driven by hierarchical shrinkage pulling small-state estimates toward the negative population-average extensive-margin coefficient. The posterior cross-margin correlation for the poverty coefficient is $\hat{\varrho}^{\text{cross}}_{\text{pov}} \approx 0.021$, indistinguishable from zero, implying that the extensive and intensive margins of the poverty effect operate through largely independent

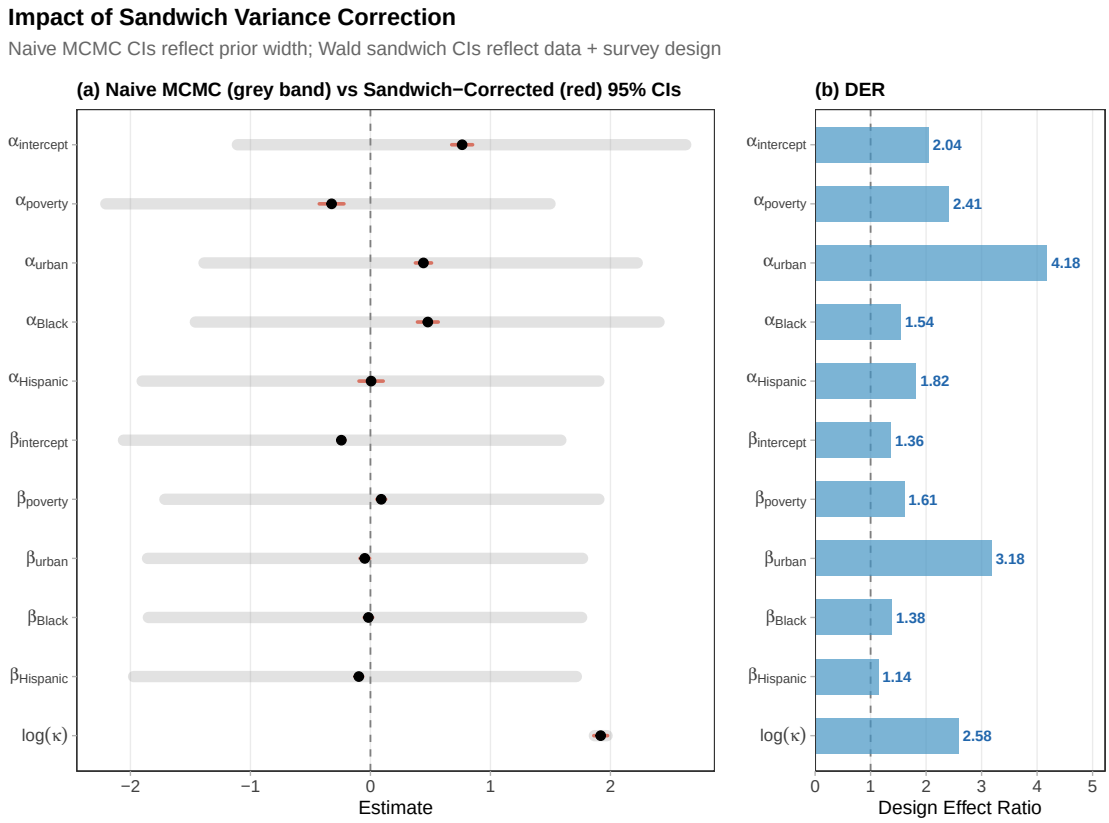

FIGURE 4. Impact of the sandwich variance correction on fixed-effect inference. *Left panel:* naive MCMC 95% credible intervals (gray) versus sandwich-corrected Wald 95% confidence intervals (colored) for all 11 fixed effects. The naive intervals reflect prior width; the Wald intervals reflect data information adjusted for the survey design. *Right panel:* design effect ratio (DER) for each parameter. A DER of 1 indicates no survey design effect; the observed range is 1.14–4.18 (mean 2.11).

mechanisms—a structural finding consistent with the block-diagonal identification result of Proposition 2. Full state-specific coefficient estimates appear in Supplementary Material D (Section D.3); the reversal probability map is provided in Section D.6.

5.4. **Marginal effect decomposition.** This subsection demonstrates the marginal effect decomposition—an analytical framework enabled by the two-part structure—by translating logit-scale coefficients into probability-scale quantities, building on the log-scale decomposition in Proposition 5. The decomposition separates each covariate's total effect into extensive (access) and intensive (share) components, a partition available only under the two-part specification. Table 12 reports the results.

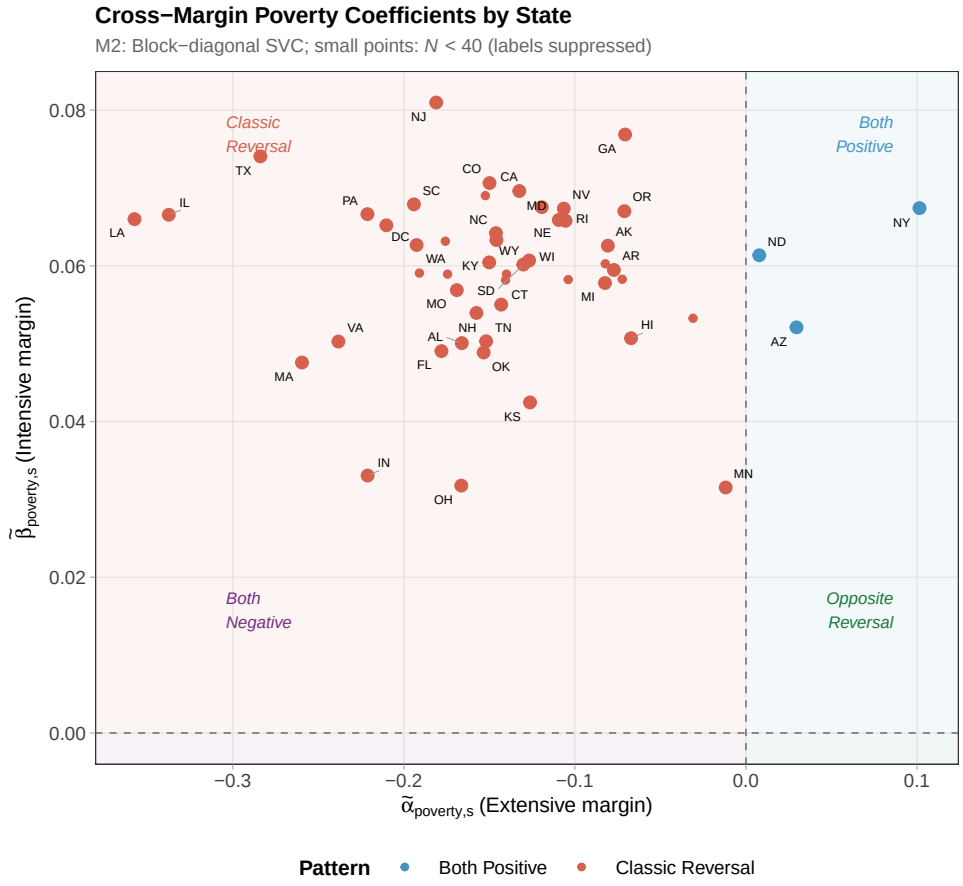

FIGURE 5. Cross-margin scatter plot of state-specific poverty coefficients from the unweighted M2 model (block-diagonal SVC). Each point represents the posterior mean $(\tilde{\alpha}_{\text{pov},s},\ \tilde{\beta}_{\text{pov},s})$ for one state. Small points without labels denote states with $N < 40$. The four quadrants correspond to different poverty–enrollment patterns: the upper-left quadrant (shaded) contains 48 of 51 states (94%), indicating a near-universal reversal pattern under hierarchical shrinkage. The dashed lines indicate the population-average coefficients $(\alpha_{\text{pov}},\ \beta_{\text{pov}})$. The near-zero cross-margin correlation ($\hat{\varrho} \approx 0.021$) implies that the two margins respond to poverty through largely independent mechanisms.

From log to probability scale. Proposition 5 decomposes the marginal effect on the log-expected-enrollment scale, where the two channels are exactly additive. To translate to the probability scale, apply the product rule to $E[Y_i/n_i] = q_i\, h_i$:

$$\frac{\partial\,(q_i\, h_i)}{\partial x_{i,k}} = \underbrace{h_i\, q_i(1-q_i)\,\tilde{\alpha}_{k,s}}_{\text{extensive}} + \underbrace{q_i\, h_i\, \varepsilon_{h,i}\,(1-\mu_i)\,\tilde{\beta}_{k,s}}_{\text{intensive}} = q_i\, h_i\big(\text{LAE}_k + \text{LIE}_k\big). \tag{5.1}$$

The AME entries in Table 12 average each component over providers and Cholesky-calibrated posterior draws $\boldsymbol{\theta}^{*(m)}$ (Theorem 3), mapping the log-scale decomposition of Proposition 5 onto the percentage-point scale.

Poverty. On the extensive margin, the AME of poverty is -0.029 (95% CI: $[-0.038, -0.020]$): a one-SD increase in community poverty (≈ 8.3 percentage points) reduces the probability of serving any infants by 2.9 percentage points. On the intensive margin, AME $= +0.015$ (95% CI: $[+0.009, +0.021]$): among centers that do serve infants, the same poverty increase raises the IT share by 1.5 percentage points. The total AME is -0.014 (95% CI: $[-0.027, -0.001]$), with the extensive margin accounting for 66.0% of the total absolute effect: the access barrier dominates, but the intensity response offsets roughly half of it. This decomposition—unavailable under any single-equation specification—illustrates the analytical value of the two-part structure (Proposition 5): the hurdle identifies the access barrier as the dominant channel, a finding that a single-equation model would collapse into a muted net effect of -1.4 percentage points.

Other covariates. A notable pattern in Table 12 is that all four covariates exhibit opposing extensive and intensive effects—the reversal pattern is not unique to poverty. Urbanicity increases access (+3.9 pp) but decreases intensity (-0.8 pp), with the extensive margin accounting for 83.6% of the total. The Black population share increases access (+4.3 pp) with a negligible intensity effect (Ext. Share = 93.9%). Hispanic composition stands out: the extensive AME is effectively zero (+0.001), so the entire effect operates through the intensive margin, where higher Hispanic composition is associated with a lower IT share (-1.6 pp). This intensive-margin-dominated pattern contrasts sharply with the extensive-margin-dominated patterns for the other covariates. In every case, a single-equation model would report only the net total effect, concealing the offsetting channels—precisely the decomposition that the hurdle framework and the AME formulas in Proposition 5 are designed to reveal.

5.5. **Policy moderation.** The reversal rate and the geographic clustering documented in Section 5.3 raise a natural question: can observable CCDF policies explain why the reversal is strong in some states and absent in others? The $\boldsymbol{\Gamma}_k$ matrices in M3b decompose the state-level variation in covariate effects into policy-explained and residual components (Proposition 4). The full 40-element $\boldsymbol{\Gamma}$ matrices appear in Supplementary Material D (Section D.4); here we summarize three key patterns.

TABLE 12. Average marginal effect (AME) decomposition from model M3b. Effects are on the probability scale (change per one-SD increase in the covariate). "Ext. Share" is the percentage of the total absolute effect attributable to the extensive (access) margin, computed as |Extensive|/(|Extensive| + |Intensive|). All four covariates exhibit opposing extensive and intensive effects. The 95% intervals are 2.5th and 97.5th posterior percentiles of each AME component evaluated at the Cholesky-calibrated draws $\boldsymbol{\theta}^{*(m)}$ (Theorem 3).

Covariate	Extensive	Intensive	Total	Ext. Share
Poverty	−0.029	+0.015	−0.014	66.0%
	[−0.038, −0.020]	[+0.009, +0.021]	[−0.027, −0.001]	
Urban	+0.039	−0.008	+0.032	83.6%
	[+0.034, +0.045]	[−0.014, −0.001]	[+0.022, +0.041]	
Black	+0.043	−0.003	+0.040	93.9%
	[+0.035, +0.051]	[−0.009, +0.003]	[+0.029, +0.052]	
Hispanic	+0.001	−0.016	−0.016	2.8%
	[−0.009, +0.010]	[−0.021, −0.010]	[−0.027, −0.004]	

First, tiered reimbursement attenuates the extensive-margin poverty barrier ($\hat{\gamma}_{1,\text{pov},\text{TR}} = +0.495$, $\Pr(> 0) = 0.962$) and raises baseline intensive-margin enrollment ($\hat{\gamma}_{2,\text{int},\text{TR}} = +0.271$, $\Pr(> 0) = 0.985$). Second, IT rate add-ons raise baseline participation ($\hat{\gamma}_{1,\text{int},\text{IT}} = +0.728$, $\Pr(> 0) = 0.996$) but paradoxically *strengthen* the poverty gradient on the extensive margin ($\hat{\gamma}_{1,\text{pov},\text{IT}} = -0.488$, $\Pr(< 0) = 0.982$), suggesting that the add-on primarily benefits moderate-poverty areas. Third, IT add-ons are most effective at closing demographic gaps in infant enrollment, particularly for centers in Black communities on the intensive margin ($\hat{\gamma}_{2,\text{Black},\text{IT}} = +0.256$, $\Pr(> 0) = 1.000$).

The three observable policy variables explain between 4% and 16% of the between-state variation in poverty coefficients. These associations are cross-sectional and do not support causal claims; nonetheless, the $\boldsymbol{\Gamma}$ decomposition illustrates how the hierarchical structure in Section 3 partitions state heterogeneity into policy-attributable and residual components.

6. Discussion

Summary. This paper develops a hierarchical hurdle beta-binomial (HBB) framework for survey-weighted bounded counts with structural zeros, extending the pseudo-posterior methodology of Williams and Savitsky (2021) to two-part hierarchical models. The two primary methodological contributions—cross-margin covariance identification (Proposition 2) and DER-guided sandwich correction (Theorem 3)—are discussed below alongside generalizability, practical recommendations, and limitations.

Generalizability. The HBB framework developed in Section 3 applies whenever the outcome is a bounded discrete proportion—a count $Y_i \in \{0, 1, \ldots, n_i\}$ with a known denominator—embedded in a hierarchical structure with structural zeros. Five domains illustrate the breadth of applicability beyond childcare enrollment. (i) In *dental epidemiology*, Y_i is the number of carious surfaces out of n_i surfaces examined, with structural zeros for caries-free individuals whose zero counts reflect disease absence rather than low risk (Bandyopadhyay et al., 2011). (ii) In *species occupancy modeling*, Y_i is the number of detection events out of n_i replicate surveys at a site, with structural zeros at sites where the species is truly absent (MacKenzie et al., 2002). (iii) In *hospital bed management*, Y_i is the number of beds occupied out of n_i staffed beds on a ward, with structural zeros for wards temporarily closed to admissions due to staffing shortages or infection control measures. (iv) In *insurance enrollment*, Y_i is the number of enrollees out of n_i eligible individuals within a plan–region cell, with structural zeros for plan–region combinations in which the plan is not offered. (v) In *survey item endorsement*, Y_i is the number of items endorsed out of n_i items administered on a subscale, with structural zeros for respondents to whom the construct does not apply (e.g., substance-use items for lifetime abstainers). Each domain exhibits the three features that jointly motivate the HBB specification: bounded support with a known denominator, extra-binomial variation within the active population, and a mass of structural zeros generated by a qualitatively distinct mechanism from the count process. In every case, the hurdle separates the structural participation decision from the intensity conditional on participation, and the beta-binomial kernel accommodates overdispersion relative to the binomial baseline.

Inference contributions. Beyond the modeling framework, this paper makes two specific contributions to inference. First, Proposition 2 establishes that the observation-level conditional likelihood is block-diagonal across margins, so the cross-margin covariance $\mathbf{\Sigma}_{12}$ (Section 3.3) is a hierarchical estimand whose effective sample size is S, not N. While a frequentist could in principle target $\mathbf{\Sigma}_{12}$ via REML on the

marginal likelihood (Remark 1), unrestricted estimation of a $q \times q$ cross-covariance from $S = 51$ groups is fragile when $2q = 10$; Proposition 3 shows that the LKJ prior provides finite-sample regularization, yielding posterior concentration at rate $O(S^{-1/2})$. Second, the sandwich correction pipeline with DER-guided block-wise application (Theorem 3) extends the pseudo-posterior framework of Williams and Savitsky (2021) to two-part hierarchical models, nesting the scalar design-effect multiplier of Ghosal et al. (2020) as a special case when the DER is constant across parameters (Equation (3.27)).

Practical recommendations. The combined evidence from the application (Section 5) and simulation (Section 4) suggests three guidelines for applied researchers fitting Bayesian hierarchical models to complex survey data. First, *test for informativeness before deciding on weighting.* Following Pfeffermann (1993), regress the sampling weight on the outcome conditional on the full model covariate vector and test whether the weight coefficient is significant; conditioning on the full covariate set is essential, because a univariate test conflates informativeness with confounding (Pfeffermann et al., 1998). For two-part models, test each margin separately, as the design can be non-informative for one margin and informative for the other (Section 2).

Second, *report both naive and sandwich-corrected confidence intervals alongside the* DER. The empirical DER range in this study is 1.14–4.18 (Table 10), and the simulation width ratios of 1.22–2.06 (Table 7) provide independent validation. The linear interpolation heuristic overestimates the DER at intermediate shrinkage levels by up to 50%; the exact conjugate formula in Supplementary Material C (Section C) provides a more accurate benchmark.

Third, *for hierarchical variance components, report model-based posterior intervals and conduct prior sensitivity analysis.* The sandwich correction is structurally inapplicable to parameters—such as the state-effect standard deviations τ and the cross-margin correlation ρ—that do not produce individual-level score contributions (Section 4.2), and the simulation confirms that the width ratio is identically 1.00 for these parameters. We conduct a sensitivity analysis in Section D.7, refitting under LKJ(η) with $\eta \in \{1, \ldots, 8\}$: the cross-margin correlation remains near zero ($\hat{\rho} \in [0.10, 0.19]$) and all fixed-effect estimates are stable, confirming that the posterior is data-dominated. For the 3–5 largest clusters, a sensitivity analysis comparing block-wise and joint Cholesky correction is also recommended (Remark 4; Supplementary Material C).

Limitations and future directions. Several limitations merit discussion. The weight ratio $w_{\max}/w_{\min} = 462$ exceeds $N^{1/2} \approx 82.4$ (Section 3.5), so the sufficient condition C3 for the Bernstein–von Mises theorem under informative sampling is not comfortably satisfied; this motivates the sandwich correction as a finite-sample robustness measure rather than an asymptotic guarantee, and underscores the importance of the simulation study in Section 4 as an independent check on coverage properties. The beta-binomial component conditions on n_i as a known upper bound, yet total enrollment may itself respond to community characteristics—a form of outcome endogeneity. Joint modeling of (Y_i, n_i) would address this concern but requires a simultaneous specification for the total-enrollment and IT-share processes, an extension we leave to future work. The model assumes a single concentration parameter κ shared across all states; state-varying κ_s could capture heterogeneous overdispersion, but identification of each κ_s requires sufficient within-state replication of the denominator values n_i (Proposition B.24). Finally, the sandwich correction delivers design-consistent inference for fixed effects but does not extend to hierarchical variance components such as $\mathbf{\Sigma}_{12}$; extending design-consistent inference to these parameters remains an open problem, with replicate-weight approaches offering a potential but computationally costly avenue.

Alternative survey approaches. The pseudo-posterior framework adopted here is not the only route to survey-weighted Bayesian inference. Multilevel pseudo-maximum likelihood (MPML; Rabe-Hesketh & Skrondal, 2006) provides a frequentist counterpart that integrates sampling weights directly into the marginal likelihood, avoiding the pseudo-likelihood construction. Fully model-based approaches that include the sampling mechanism as part of the probability model (Si et al., 2020) offer theoretical advantages when the design is informative but require additional specification of the selection process. Finally, balanced repeated replicate (BRR) or jackknife replicate-weight methods (Rao & Wu, 1988) provide design-based variance estimates without distributional assumptions, though the computational cost of refitting Bayesian hierarchical models under each set of replicate weights is currently prohibitive. Comparing these alternatives with the pseudo-posterior sandwich approach is an important direction for future work.

Conclusion. This paper introduces a Bayesian hierarchical hurdle beta-binomial framework for survey-weighted bounded counts with structural zeros. Applied to the 2019 NSECE, the model reveals a poverty reversal—community poverty reduces

access to infant and toddler care but increases enrollment intensity among participating centers—that is near-universal across states under hierarchical shrinkage (48 of 51) and whose extensive margin accounts for approximately two-thirds of the total effect. The sandwich correction and DER diagnostic provide a practical toolkit for population-level inference in Bayesian hierarchical models fitted to complex survey data. The companion R package hurdlebb (Lee, 2026) and replication repository make the methodology immediately accessible to applied researchers.

Software and reproducibility. The companion R package hurdlebb (Lee, 2026) implements the full methodology developed in this paper, including survey-weighted pseudo-posterior inference, sandwich variance correction via Cholesky transformation, and marginal effect decomposition. The package is available at https://github.com/joonho112/hurdlebb with documentation at https://joonho112.github.io/hurdlebb/. A separate replication package at https://github.com/joonho112/hurdlebb-replication contains all analysis scripts needed to reproduce every result reported in this paper, conditional on access to the NSECE restricted-use data. The complete estimation pipeline is described in Algorithm 1; Supplementary Material E (Section E) provides computational details including Stan configuration, convergence diagnostics, and runtime benchmarks.

Data availability. The empirical results in this paper are based on data from the 2019 National Survey of Early Care and Education (NSECE). All key variables used in the analysis—including the state identifier, community-level poverty rate, urbanicity, racial/ethnic composition, infant/toddler enrollment counts, and total enrollment—are available only through the NSECE Restricted-Use Files (ICPSR Study 38445, https://www.icpsr.umich.edu/web/ICPSR/studies/38445). For researchers interested in the methodology without access to the restricted data, the hurdlebb package includes a synthetic dataset (`nsece_synth`) that preserves the essential distributional characteristics of the original data.

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Online Supplemental Materials

A Bayesian Hierarchical Hurdle Beta-Binomial Model for Survey-Weighted Bounded Counts and Its Application to Childcare Enrollment

JoonHo Lee

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Appendix A. Mathematical Properties of the Hurdle Beta-Binomial Model

This appendix collects the complete proofs and supporting results for the mathematical properties of the hurdle beta-binomial model discussed in Section 3. The beta-binomial distribution in the mean–precision parameterization follows Griffiths (1973) and Prentice (1986); the hurdle construction follows Mullahy (1986) and Cragg (1971). We maintain the notation established in Section 3.1 throughout.

A.1. **Proof of Theorem 1.** We give the complete proof that the intensity function $h(\mu) = \mu/(1-p_0)$ is strictly increasing in μ for every $n \geq 2$, $\kappa > 0$. Recall that $p_0(n,\mu,\kappa) = \prod_{j=0}^{n-1}[(1-\mu)\kappa + j]/(\kappa + j)$ is the zero probability of the beta-binomial distribution, as defined in (3.3).

Auxiliary quantities. Write $b = (1-\mu)\kappa$ and define

$$\Lambda = \kappa \sum_{j=0}^{n-1} \frac{1}{(1-\mu)\kappa + j} > 0, \qquad \Lambda_2 = \kappa^2 \sum_{j=0}^{n-1} \frac{1}{\big[(1-\mu)\kappa + j\big]^2} > 0. \tag{A.1}$$

Note that $\partial p_0/\partial\mu = -p_0\,\Lambda$ (obtained by differentiating $\ln p_0 = \sum_{j=0}^{n-1} \ln[(1-\mu)\kappa + j] - \ln[\kappa + j]$) and $\partial\Lambda/\partial\mu = \Lambda_2$ (since $\partial b/\partial\mu = -\kappa$).

Lemma A.1 (Sum-of-reciprocals inequality)**.** *Let $b > 0$, $n \geq 1$, and set $S_1 = \sum_{j=0}^{n-1}(b+j)^{-1}$ and $S_2 = \sum_{j=0}^{n-1}(b+j)^{-2}$, so that $\Lambda = \kappa\, S_1$ and $\Lambda_2 = \kappa^2\, S_2$.*

(i) *For $n = 1$: $S_1^2 = S_2$, so $\Lambda^2 = \Lambda_2$.*
(ii) *For $n \geq 2$: $S_1^2 > S_2$, so $\Lambda^2 > \Lambda_2$.*

Proof. Write $a_j = (b+j)^{-1}$. Expanding the square of the sum,

$$S_1^2 = \Big(\sum_{j=0}^{n-1} a_j\Big)^2 = \sum_{j=0}^{n-1} a_j^2 + 2 \sum_{0\le i<j\le n-1} a_i\, a_j = S_2 + 2 \sum_{0\le i<j\le n-1} \frac{1}{(b+i)(b+j)}.$$

For $n = 1$ the cross-product sum is empty, giving $S_1^2 = S_2$. For $n \ge 2$ every term $1/[(b+i)(b+j)]$ is strictly positive, so $S_1^2 > S_2$. □

Proof of Theorem 1. Define the numerator function

$$\Phi(\mu) = \big(1 - p_0(\mu)\big) - \mu\, p_0(\mu)\, \Lambda(\mu). \tag{A.2}$$

From the quotient rule applied to $h = \mu\,(1-p_0)^{-1}$ together with $\partial p_0/\partial\mu = -p_0\,\Lambda$, one obtains

$$\frac{\partial h}{\partial \mu} = \frac{\Phi(\mu)}{(1-p_0)^2}. \tag{A.3}$$

Since $(1-p_0)^2 > 0$ for $\mu \in (0,1)$, the sign of $\partial h/\partial\mu$ equals the sign of $\Phi(\mu)$. The proof proceeds in three steps.

Step 1: Boundary value. At $\mu = 0$, every factor in (3.3) equals one, so $p_0(0) = 1$. Hence $\Phi(0) = (1-1) - 0\cdot 1\cdot \Lambda(0) = 0$.

Step 2: Derivative of Φ. Differentiating (A.2) with the product rule gives

$$\Phi'(\mu) = p_0\,\Lambda - \big[p_0\,\Lambda + \mu\,\tfrac{d}{d\mu}(p_0\,\Lambda)\big] = -\mu\,\frac{d}{d\mu}\big(p_0\,\Lambda\big).$$

Now $d(p_0\,\Lambda)/d\mu = (\partial p_0/\partial\mu)\,\Lambda + p_0\,(\partial\Lambda/\partial\mu) = -p_0\,\Lambda^2 + p_0\,\Lambda_2 = p_0(\Lambda_2 - \Lambda^2)$, and therefore

$$\Phi'(\mu) = -\mu\, p_0\big(\Lambda_2 - \Lambda^2\big) = \mu\, p_0\big(\Lambda^2 - \Lambda_2\big). \tag{A.4}$$

Step 3: Application of Lemma A.1.

- *Case* $n = 1$. Lemma A.1(i) gives $\Lambda^2 = \Lambda_2$, so $\Phi'(\mu) = 0$ identically. Combined with $\Phi(0) = 0$, we obtain $\Phi(\mu) = 0$ for all $\mu \in (0,1)$ and hence $\partial h/\partial\mu = 0$. Equivalently, $p_0 = 1-\mu$ when $n = 1$, so $h(\mu) = \mu/(1-(1-\mu)) = 1$ identically.
- *Case* $n \ge 2$. Lemma A.1(ii) gives $\Lambda^2 > \Lambda_2$. Since $\mu > 0$ and $p_0 > 0$ on $(0,1)$, equation (A.4) yields $\Phi'(\mu) > 0$ for all $\mu \in (0,1)$.

Since $\Phi(0) = 0$ and Φ is strictly increasing on $(0,1)$, we conclude $\Phi(\mu) > 0$ for all $\mu \in (0,1)$, and therefore

$$\frac{\partial h}{\partial \mu} = \frac{\Phi(\mu)}{(1-p_0)^2} > 0.$$

□

A.2. **Higher moments of the truncated beta-binomial. Notation.** Throughout this subsection, $Y \sim \text{BetaBin}(n, \mu, \kappa)$ denotes the standard (untruncated) beta-binomial with $a = \mu\kappa$, $b = (1-\mu)\kappa$, and $Y^+ \sim \text{ZT-BetaBin}(n, \mu, \kappa)$ denotes the zero-truncated version. We write $p_0 = \mathbb{P}(Y = 0)$, $V_{\text{BB}} = \text{Var}(Y) = n\mu(1-\mu)(n+\kappa)/(1+\kappa)$, and $h = \mu/(1 - p_0)$.

Lemma A.2 (Second factorial moment)**.** *For $Y \sim \text{BetaBin}(n, \mu, \kappa)$,*

$$\boxed{\mathbb{E}\left[Y(Y-1)\right] = n(n-1) \cdot \frac{\mu(\mu\kappa + 1)}{\kappa + 1}.} \tag{A.5}$$

Proof. The beta-binomial arises as a beta mixture (Griffiths, 1973): let $p \sim \text{Beta}(a, b)$ with $a = \mu\kappa$, $b = (1-\mu)\kappa$, and $Y \mid p \sim \text{Bin}(n, p)$. Since $\mathbb{E}[Y(Y-1) \mid p] = n(n-1)p^2$, the tower property gives

$$\mathbb{E}\left[Y(Y-1)\right] = n(n-1)\,\mathbb{E}[p^2].$$

The second moment of $p \sim \text{Beta}(a, b)$ is $\mathbb{E}[p^2] = a(a+1)/[(a+b)(a+b+1)]$, using the rising factorial identity $\mathbb{E}[p^k] = a^{(k)}/(a+b)^{(k)}$ for $k = 2$. Substituting $a = \mu\kappa$, $a + b = \kappa$:

$$\mathbb{E}\left[Y(Y-1)\right] = n(n-1) \cdot \frac{\mu\kappa(\mu\kappa + 1)}{\kappa(\kappa + 1)} = n(n-1) \cdot \frac{\mu(\mu\kappa + 1)}{\kappa + 1}. \qquad \square$$

Consistency check. From (A.5), the second moment is $\mathbb{E}[Y^2] = \mathbb{E}[Y(Y-1)] + \mathbb{E}[Y] = n(n-1)\mu(\mu\kappa + 1)/(\kappa + 1) + n\mu$. We confirm $\text{Var}(Y) = \mathbb{E}[Y^2] - (n\mu)^2$:

$$\mathbb{E}[Y^2] - n^2\mu^2 = \frac{n(n-1)\mu(\mu\kappa + 1) + n\mu(\kappa + 1) - n^2\mu^2(\kappa + 1)}{\kappa + 1}.$$

Expanding the numerator factor: $n\mu\left[(n-1)(\mu\kappa + 1) + (\kappa + 1) - n\mu(\kappa + 1)\right]$. The bracket simplifies as

$$(n-1)\mu\kappa + (n-1) + (\kappa + 1) - n\mu\kappa - n\mu = (1-\mu)(n+\kappa).$$

Therefore $\text{Var}(Y) = n\mu(1-\mu)(n+\kappa)/(\kappa+1) = V_{\text{BB}}$. $\square$

Lemma A.3 (General r-th factorial moment)**.** *For $Y \sim \text{BetaBin}(n, \mu, \kappa)$,*

$$\mathbb{E}\left[\prod_{j=0}^{r-1}(Y - j)\right] = \frac{n^{(r)} \cdot (\mu\kappa)^{(r)}}{\kappa^{(r)}}, \tag{A.6}$$

where $c^{(r)} = c(c+1)\cdots(c+r-1) = \Gamma(c+r)/\Gamma(c)$ is the rising factorial.

Proof. Follows from $\mathbb{E}[\prod_{j=0}^{r-1}(Y - j) \mid p] = n^{(r)}p^r$ for the binomial, and $\mathbb{E}[p^r] = (\mu\kappa)^{(r)}/\kappa^{(r)}$ for the beta. $\square$

Proposition A.1 (Truncation identity for functions vanishing at zero)**.** *For any function g with $g(0) = 0$,*

$$\mathbb{E}\left[g(Y) \mid Y > 0\right] = \frac{\mathbb{E}\left[g(Y)\right]}{1 - p_0}. \tag{A.7}$$

In particular, $\mathbb{E}[Y^k \mid Y > 0] = \mathbb{E}[Y^k]/(1 - p_0)$ for all $k \geq 1$.

Proof. Since $g(Y) \cdot \mathbf{1}(Y = 0) = g(0) \cdot \mathbf{1}(Y = 0) = 0$, we have $\mathbb{E}[g(Y)] = \mathbb{E}[g(Y) \cdot \mathbf{1}(Y > 0)]$. Dividing by $\mathbb{P}(Y > 0) = 1 - p_0$ gives the result. □

Corollary A.1 (Conditional second factorial moment)**.**

$$\boxed{\mathbb{E}\left[Y(Y-1) \mid Y > 0\right] = \frac{n(n-1)\mu(\mu\kappa + 1)}{(\kappa + 1)(1 - p_0)}.} \tag{A.8}$$

Proposition A.2 (Truncated BB variance)**.** *For $Y^+ \sim$ ZT-BetaBin(n, μ, κ),*

$$\boxed{\mathrm{Var}(Y \mid Y > 0) = \frac{V_{\mathrm{BB}}(1 - p_0) - n^2\mu^2\, p_0}{(1 - p_0)^2}.} \tag{A.9}$$

Proof. By the truncation identity (Proposition A.1),

$$\mathbb{E}[Y^2 \mid Y > 0] = \frac{\mathbb{E}[Y^2]}{1 - p_0}, \qquad \mathbb{E}[Y \mid Y > 0] = \frac{n\mu}{1 - p_0}.$$

Then

$$\mathrm{Var}(Y \mid Y > 0) = \frac{\mathbb{E}[Y^2]}{1 - p_0} - \frac{n^2\mu^2}{(1 - p_0)^2} = \frac{V_{\mathrm{BB}} + n^2\mu^2}{1 - p_0} - \frac{n^2\mu^2}{(1 - p_0)^2}.$$

Combining the $n^2\mu^2$ terms over the common denominator $(1 - p_0)^2$:

$$= \frac{V_{\mathrm{BB}}}{1 - p_0} + n^2\mu^2 \cdot \frac{(1 - p_0) - 1}{(1 - p_0)^2} = \frac{V_{\mathrm{BB}}}{1 - p_0} - \frac{n^2\mu^2\, p_0}{(1 - p_0)^2} = \frac{V_{\mathrm{BB}}(1 - p_0) - n^2\mu^2\, p_0}{(1 - p_0)^2}. \qquad \square$$

Remark A.1. The formula (A.9) reveals two competing effects of truncation. The first term $V_{\mathrm{BB}}/(1 - p_0)$ inflates the variance by rescaling probabilities. The second term $n^2\mu^2 p_0/(1 - p_0)^2$ reduces the variance because the truncated distribution loses mass at zero, pulling the mean away from zero and tightening the effective spread relative to that shifted mean.

Alternative forms.

Form A (factored):

$$\mathrm{Var}(Y \mid Y > 0) = \frac{n\mu}{1 - p_0}\left[\frac{(1 - \mu)(n + \kappa)}{1 + \kappa} - \frac{n\mu\, p_0}{1 - p_0}\right]. \tag{A.10}$$

This displays the variance as $\mathbb{E}[Y \mid Y > 0]$ times a "spread factor."

Form B (via factorial moments):

$$\mathrm{Var}(Y \mid Y > 0) = \frac{\mathbb{E}[Y(Y-1)]}{1-p_0} + \frac{n\mu}{1-p_0} - \frac{n^2\mu^2}{(1-p_0)^2}. \tag{A.11}$$

Verification of limiting cases.

Case 1: $\kappa \to \infty$ (truncated binomial limit). As $\kappa \to \infty$, $V_{\mathrm{BB}} \to n\mu(1-\mu)$ and $p_0 \to (1-\mu)^n$. The formula becomes the known variance of the zero-truncated binomial. For $n = 1$: $Y \mid Y > 0 = 1$ deterministically, so Var $= 0$. In the formula: numerator $= \mu(1-\mu)\cdot\mu - \mu^2(1-\mu) = 0$. □

The remaining limiting cases are verified analogously: as $p_0 \to 0$ the variance reduces to V_{BB}, and for $n = 1$ both sides equal zero since $Y \mid Y > 0 = 1$ deterministically. □

Numerical cross-check. Consider $n = 5$, $\mu = 0.3$, $\kappa = 2$ (so $a = 0.6$, $b = 1.4$). Direct computation gives $p_0 = \prod_{j=0}^{4}(1.4+j)/(2+j) = 0.376992$, $\mathbb{E}[Y] = 1.5$, $V_{\mathrm{BB}} = 2.45$, and $\mathbb{E}[Y^2] = 4.70$. The truncated moments are $\mathbb{E}[Y \mid Y > 0] = 1.5/0.623008 = 2.40770$, $\mathbb{E}[Y^2 \mid Y > 0] = 4.70/0.623008 = 7.54409$, and $\mathrm{Var}(Y \mid Y > 0) = 7.54409 - 2.40770^2 = 1.74707$. Form A gives $(2.45 \times 0.623008 - 2.25 \times 0.376992)/0.623008^2 = 1.74709$, agreeing to five significant figures. □

A.3. **Unconditional variance decomposition.** We derive the unconditional variance of Y_i under the hurdle beta-binomial model, its variance-to-mean ratio, an overdispersion index relative to the binomial, and the coefficient of variation. Throughout, we write $V_{\mathrm{BB},i} = n_i\mu_i(1-\mu_i)(n_i+\kappa)/(1+\kappa)$ for the beta-binomial variance (3.2), $h_i = \mu_i/(1-p_{0,i})$ for the intensity function (3.7), and $m_i^+ = n_i h_i = n_i\mu_i/(1-p_{0,i})$ for the conditional mean given participation.

Law of total variance.

Proposition A.3 (Hurdle model variance decomposition)**.** *Let Y_i follow the* HBB *model (cf. Mullahy, 1986, for the Poisson hurdle analogue) with participation probability q_i and intensity parameters (n_i, μ_i, κ). Define $Z_i = \mathbf{1}\{Y_i > 0\}$. Then*

$$\mathrm{Var}(Y_i) = q_i \cdot \mathrm{Var}(Y_i \mid Y_i > 0) + q_i(1-q_i)\big[\mathbb{E}(Y_i \mid Y_i > 0)\big]^2. \tag{A.12}$$

Proof. Apply the law of total variance with respect to the Bernoulli variable Z_i:

$$\mathrm{Var}(Y_i) = \mathbb{E}\big[\mathrm{Var}(Y_i \mid Z_i)\big] + \mathrm{Var}\big(\mathbb{E}[Y_i \mid Z_i]\big).$$

First term. When $Z_i = 0$, $Y_i = 0$ almost surely, so $\mathrm{Var}(Y_i \mid Z_i = 0) = 0$. When $Z_i = 1$, $\mathrm{Var}(Y_i \mid Z_i = 1) = \mathrm{Var}(Y_i \mid Y_i > 0)$. Hence

$$\mathbb{E}\big[\mathrm{Var}(Y_i \mid Z_i)\big] = q_i \ \mathrm{Var}(Y_i \mid Y_i > 0).$$

Second term. The conditional expectation $\mathbb{E}[Y_i \mid Z_i]$ takes value 0 with probability $1 - q_i$ and $m_i^+ = n_i\mu_i/(1 - p_{0,i})$ with probability q_i. Its variance is therefore $q_i(1 - q_i)(m_i^+)^2$. □

Corollary A.2 (Expanded form)**.** *Substituting the truncated variance from Proposition A.2 into (A.12):*

$$\mathrm{Var}(Y_i) = \frac{q_i\, V_{\mathrm{BB},i}}{1 - p_{0,i}} - \frac{q_i\, n_i^2 \mu_i^2\, p_{0,i}}{(1 - p_{0,i})^2} + \frac{q_i(1 - q_i)\, n_i^2 \mu_i^2}{(1 - p_{0,i})^2}. \tag{A.13}$$

Combining the terms involving $n_i^2\mu_i^2/(1 - p_{0,i})^2$ yields the compact form

$$\mathrm{Var}(Y_i) = \frac{q_i\, V_{\mathrm{BB},i}}{1 - p_{0,i}} + \frac{q_i\, n_i^2 \mu_i^2\, (1 - q_i - p_{0,i})}{(1 - p_{0,i})^2}. \tag{A.14}$$

Remark A.2. The quantities q_i and $p_{0,i}$ are distinct: q_i is the hurdle participation probability, whereas $p_{0,i}$ is the zero probability of the beta-binomial component. In general $q_i \neq 1 - p_{0,i}$; the quantity $(1 - q_i - p_{0,i})$ may be positive, negative, or zero, so the second term in (A.14) can either inflate or reduce the unconditional variance.

Corollary A.3 (In terms of the unconditional mean)**.** *Recall from (3.7) that $\mathbb{E}[Y_i] = q_i\, n_i\, h_i$. Then*

$$\mathrm{Var}(Y_i) = q_i \ \mathrm{Var}(Y_i \mid Y_i > 0) + \frac{1 - q_i}{q_i} \big(\mathbb{E}[Y_i]\big)^2. \tag{A.15}$$

The between-component term is proportional to the squared mean, scaled by the odds of non-participation $(1 - q_i)/q_i$.

Variance-to-mean ratio.

Proposition A.4 (Variance ratio)**.** *Define $\mathrm{VR}_i = \mathrm{Var}(Y_i)/\mathbb{E}[Y_i]$. Then*

$$\boxed{\mathrm{VR}_i = \frac{(1 - \mu_i)(n_i + \kappa)}{1 + \kappa} - \frac{n_i\, \mu_i\, p_{0,i}}{1 - p_{0,i}} + (1 - q_i)\, n_i\, h_i.} \tag{A.16}$$

Proof. Using $\mathbb{E}[Y_i] = q_i\, n_i\, h_i$ and Corollary A.3:

$$\mathrm{VR}_i = \frac{\mathrm{Var}(Y_i \mid Y_i > 0)}{n_i\, h_i} + (1 - q_i)\, n_i\, h_i.$$

For the first term, substitute the truncated variance (Proposition A.2) and simplify using $h_i = \mu_i/(1-p_{0,i})$:

$$\frac{V_{\mathrm{BB},i}}{(1-p_{0,i})\, n_i\, h_i} = \frac{(1-\mu_i)(n_i+\kappa)}{1+\kappa}, \qquad \frac{n_i\, \mu_i^2\, p_{0,i}}{(1-p_{0,i})^2\, h_i} = \frac{n_i\, \mu_i\, p_{0,i}}{1-p_{0,i}}. \qquad \square$$

Remark A.3. The three terms in (A.16) have clear interpretations:

(i) $(1-\mu_i)(n_i+\kappa)/(1+\kappa)$: the beta-binomial variance inflation factor (Prentice, 1986), which exceeds unity for all $n_i > 1$;

(ii) $-n_i\, \mu_i\, p_{0,i}/(1-p_{0,i})$: a correction from zero-truncation that reduces the variance ratio, reflecting the mass removed at $Y = 0$;

(iii) $(1-q_i)\, n_i\, h_i$: the between-component contribution from the hurdle, which increases the variance ratio whenever $q_i < 1$.

For Poisson data, VR = 1 (equidispersion). In the NSECE data, $\mathrm{VR}_i > 1$ consistently, reflecting both overdispersion and the hurdle mechanism.

Overdispersion index relative to the binomial.

Definition A.1 (Overdispersion index)**.** For a proportion Y_i/n_i, define

$$\mathrm{OD}_i = \frac{\mathrm{Var}(Y_i)}{n_i\, \pi_i(1-\pi_i)}, \qquad \text{where } \pi_i = q_i\, h_i = \frac{\mathbb{E}[Y_i]}{n_i}. \tag{A.17}$$

The denominator $n_i\, \pi_i(1-\pi_i)$ is the variance of a $\mathrm{Bin}(n_i, \pi_i)$ random variable with the same unconditional mean proportion, so OD_i measures overdispersion relative to the binomial benchmark.

Limiting cases.

- *No hurdle, $p_0 \to 0$.* Then $q_i = 1$, $h_i \to \mu_i$, and $\mathrm{Var}(Y_i) \to V_{\mathrm{BB},i}$, so $\mathrm{OD}_i \to (n_i+\kappa)/(1+\kappa)$, the standard beta-binomial overdispersion factor (Prentice, 1986).
- *Binomial limit ($\kappa \to \infty$, $q_i = 1$).* $V_{\mathrm{BB},i} \to n_i\mu_i(1-\mu_i)$ and the hurdle is inactive, so $\mathrm{OD}_i \to 1$.

Coefficient of variation.

Proposition A.5 (Conditional CV)**.** *The squared coefficient of variation of Y_i conditional on participation is*

$$\boxed{\mathrm{CV}^2(Y_i \mid Y_i > 0) = \frac{(1-\mu_i)(n_i+\kappa)}{n_i\, \mu_i(1+\kappa)}\,(1-p_{0,i}) - p_{0,i}.} \tag{A.18}$$

Proof. By definition, $\mathrm{CV}^2 = \mathrm{Var}(Y_i \mid Y_i > 0)/[\mathbb{E}(Y_i \mid Y_i > 0)]^2$. Substituting the truncated variance (Proposition A.2) and the conditional mean $m_i^+ = n_i\mu_i/(1-p_{0,i})$:

$$\begin{aligned}\mathrm{CV}^2 &= \frac{V_{\mathrm{BB},i}(1-p_{0,i}) - n_i^2\mu_i^2\, p_{0,i}}{n_i^2\mu_i^2} = \frac{V_{\mathrm{BB},i}}{n_i^2\mu_i^2}\,(1-p_{0,i}) - p_{0,i} \\ &= \frac{(1-\mu_i)(n_i+\kappa)}{n_i\,\mu_i(1+\kappa)}\,(1-p_{0,i}) - p_{0,i}.\end{aligned}$$

□

Proposition A.6 (Unconditional CV)**.** *The squared coefficient of variation of Y_i (unconditional) satisfies*

$$\boxed{\mathrm{CV}^2(Y_i) = \frac{\mathrm{CV}^2(Y_i \mid Y_i > 0) + 1 - q_i}{q_i}.} \tag{A.19}$$

Proof. From $\mathrm{Var}(Y_i) = q_i\,\mathrm{Var}(Y_i \mid Y_i > 0) + q_i(1-q_i)(n_ih_i)^2$ (Proposition A.3) and $[\mathbb{E}(Y_i)]^2 = q_i^2(n_ih_i)^2$:

$$\mathrm{CV}^2(Y_i) = \frac{\mathrm{CV}^2(Y_i \mid Y_i > 0)}{q_i} + \frac{1-q_i}{q_i} = \frac{\mathrm{CV}^2(Y_i \mid Y_i > 0) + 1 - q_i}{q_i}.$$

□

Remark A.4. The coefficient of variation is useful for posterior predictive checking because: (i) it is scale-free, enabling comparison across providers with different n_i; (ii) the additive $(1-q_i)/q_i$ term in (A.19) reveals how the hurdle magnifies the CV when participation is low; and (iii) observed sample CVs can be compared against the posterior distribution of theoretical CVs, providing a diagnostic that simultaneously probes the mean, variance, and zero-inflation structure of the model.

A.4. **Analytic properties of the intensity function.** This subsection collects analytic results for the intensity function $h(\mu) = \mu/(1-p_0)$ that underpin the elasticity formula (Proposition 1), the boundary behavior discussed in Remark 2, and the log-scale decomposition of marginal effects. We continue to write $b = (1-\mu)\kappa$ and use Λ, Λ_2 as defined in (A.1).

Second derivative of p_0.

Lemma A.4 (Second derivative of p_0)**.** *For all $n \geq 1$, $\mu \in (0,1)$, and $\kappa > 0$,*

$$\frac{\partial^2 p_0}{\partial \mu^2} = p_0\big(\Lambda^2 - \Lambda_2\big). \tag{A.20}$$

Proof. Starting from $\partial p_0/\partial\mu = -p_0\,\Lambda$, differentiate by the product rule:

$$\frac{\partial^2 p_0}{\partial \mu^2} = -\frac{\partial p_0}{\partial \mu}\cdot\Lambda - p_0\cdot\frac{\partial \Lambda}{\partial \mu} = p_0\,\Lambda^2 - p_0\,\Lambda_2 = p_0\big(\Lambda^2 - \Lambda_2\big).$$

Here we used $\partial\Lambda/\partial\mu = \Lambda_2$ (since $\partial b/\partial\mu = -\kappa$). □

Corollary A.4 (Convexity of p_0 in μ)**.** *For $n \geq 2$: $\partial^2 p_0/\partial\mu^2 > 0$, so p_0 is strictly convex in μ. For $n = 1$: $p_0 = 1 - \mu$ is linear in μ.*

Proof. Immediate from Lemma A.4 and Lemma A.1: for $n \geq 2$, $\Lambda^2 > \Lambda_2$ and $p_0 > 0$, so the second derivative is strictly positive. For $n = 1$, $\Lambda^2 = \Lambda_2$, giving $\partial^2 p_0/\partial\mu^2 = 0$. □

Second derivative of h.

Proposition A.7 (Second derivative of h)**.** *Write $D = 1 - p_0$. Then*

$$\frac{\partial^2 h}{\partial\mu^2} = \frac{p_0}{D^3}\Big[-2\Lambda D + \mu\Lambda^2(1 + p_0) - \mu\Lambda_2 D\Big]. \tag{A.21}$$

Proof. From $\partial h/\partial\mu = N/D^2$ where $N = \Phi(\mu) = D - \mu\, p_0\, \Lambda$ (see (A.2)), apply the quotient rule:

$$\frac{\partial^2 h}{\partial\mu^2} = \frac{N'D - 2ND'}{D^3},$$

where $D' = -\partial p_0/\partial\mu = p_0\,\Lambda > 0$.

Computing N'. Since $N = (1 - p_0) - \mu\, p_0\, \Lambda$, we have

$$N' = p_0\,\Lambda - \big[p_0\,\Lambda + \mu \cdot \tfrac{d}{d\mu}(p_0\,\Lambda)\big] = -\mu \cdot \frac{d}{d\mu}(p_0\,\Lambda) = \mu\, p_0\big(\Lambda^2 - \Lambda_2\big),$$

using $d(p_0\Lambda)/d\mu = p_0(\Lambda_2 - \Lambda^2)$ from (A.4).

Assembling.

$$\begin{aligned} N'D - 2ND' &= \mu\, p_0(\Lambda^2 - \Lambda_2)D - 2\big(D - \mu\, p_0\,\Lambda\big)(p_0\,\Lambda) \\ &= \mu\, p_0(\Lambda^2 - \Lambda_2)D - 2D\, p_0\,\Lambda + 2\mu\, p_0^2\Lambda^2 \\ &= p_0\big[-2\Lambda D + \mu(\Lambda^2 - \Lambda_2)D + 2\mu\, p_0\,\Lambda^2\big]. \end{aligned}$$

Noting that $(\Lambda^2 - \Lambda_2)D + 2p_0\Lambda^2 = \Lambda^2(D + 2p_0) - \Lambda_2 D = \Lambda^2(1 + p_0) - \Lambda_2 D$ yields the stated form. □

Verification ($n = 1$). When $n = 1$, $p_0 = 1 - \mu$, $D = \mu$, $\Lambda = 1/(1 - \mu)$, $\Lambda_2 = \Lambda^2$, and $h \equiv 1$, so $\partial^2 h/\partial\mu^2 = 0$. In the formula: $N = D - \mu\, p_0\, \Lambda = \mu - \mu(1 - \mu)/(1 - \mu) = 0$ identically, so $N' = 0$ and the numerator vanishes. □

Monotonicity of p_0 in all parameters.

Proposition A.8 (Monotonicity of p_0 in μ)**.** $\partial p_0/\partial\mu = -p_0\,\Lambda < 0$. *Increasing μ strictly decreases the zero probability.*

Proof. Established in Section A.1 (auxiliary computation preceding the proof of Theorem 1). □

Proposition A.9 (Monotonicity of p_0 in κ)**.** *For $n \geq 2$ and $\mu \in (0,1)$:*

$$\frac{\partial p_0}{\partial \kappa} = -p_0\, \mu \sum_{j=1}^{n-1} \frac{j}{(b+j)(\kappa+j)} < 0. \tag{A.22}$$

For $n = 1$: $\partial p_0/\partial \kappa = 0$ (since $p_0 = 1-\mu$ is independent of κ).

Proof. Differentiating $\log p_0 = \sum_{j=0}^{n-1}[\log(b+j) - \log(\kappa+j)]$ with $\partial b/\partial \kappa = 1-\mu$:

$$\frac{\partial \log p_0}{\partial \kappa} = \sum_{j=0}^{n-1} \left[\frac{1-\mu}{b+j} - \frac{1}{\kappa+j}\right].$$

For each term, the numerator is $(1-\mu)(\kappa+j)-(b+j) = (1-\mu)(\kappa+j)-(1-\mu)\kappa-j = -\mu j$. Hence

$$\frac{\partial \log p_0}{\partial \kappa} = -\mu \sum_{j=1}^{n-1} \frac{j}{(b+j)(\kappa+j)} < 0,$$

where the $j = 0$ term vanishes. Multiplying by $p_0 > 0$ gives the stated formula. □

Increasing κ (reducing overdispersion) concentrates mass near $n\mu > 0$, pulling probability away from zero.

Proposition A.10 (Monotonicity of p_0 in n)**.** *For all $n \geq 1$, $\mu \in (0,1)$, and $\kappa > 0$:*

$$\frac{p_0(n+1)}{p_0(n)} = \frac{b+n}{\kappa+n} < 1. \tag{A.23}$$

Hence p_0 is strictly decreasing in n.

Proof. From the product form (3.3), $p_0(n+1) = p_0(n) \cdot (b+n)/(\kappa+n)$. Since $b = (1-\mu)\kappa < \kappa$ for $\mu \in (0,1)$, the ratio $(b+n)/(\kappa+n) < 1$. □

Remark A.5 (Polynomial decay of p_0 in n)**.** The consecutive ratio $(b+n)/(\kappa+n) \to 1$ as $n \to \infty$, so p_0 decays *polynomially* in n, not exponentially. Specifically, by Stirling's approximation for the ratio of rising factorials, $p_0(n) \sim C(\mu,\kappa)\, n^{-\mu\kappa}$ as $n \to \infty$. By contrast, the binomial zero probability $(1-\mu)^n$ decays exponentially. This slower decay means that even for moderately large group sizes ($n = 50$–100), the structural-zero probability p_0 remains non-negligible when κ is small, reinforcing the importance of the hurdle component (see also Bandyopadhyay et al., 2011, for a spatial BB application where this phenomenon is empirically relevant).

Log-derivative and strict monotonicity of $\log h$.

Proposition A.11 (Log-derivative of h)**.**

$$\frac{\partial(\log h)}{\partial \mu} = \frac{1}{\mu} - \frac{p_0\,\Lambda}{1-p_0}. \tag{A.24}$$

Proof. Since $\log h = \log \mu - \log(1-p_0)$, we differentiate each term: $d(\log \mu)/d\mu = 1/\mu$, and $d\log(1-p_0)/d\mu = (-dp_0/d\mu)/(1-p_0) = p_0\Lambda/(1-p_0)$. Subtracting gives the result. □

Theorem A.1 ($\log h$ is strictly increasing)**.** *For $n \geq 2$:*

$$\frac{\partial(\log h)}{\partial \mu} > 0 \qquad \text{for all } \mu \in (0,1). \tag{A.25}$$

Proof. The condition $\partial(\log h)/\partial\mu > 0$ is equivalent to $1/\mu > p_0\Lambda/(1-p_0)$, which rearranges to $(1-p_0) > \mu\, p_0\, \Lambda$, i.e., $\Phi(\mu) > 0$ (see (A.2)). This was established in Theorem 1. □

Remark A.6 (Decomposition of the log-derivative)**.** The log-derivative decomposes as

$$\frac{\partial(\log h)}{\partial \mu} = \underbrace{\frac{1}{\mu}}_{\text{direct effect}} - \underbrace{\frac{p_0\,\Lambda}{1-p_0}}_{\text{truncation attenuation}}.$$

The direct effect represents the proportional increase in the numerator μ. The truncation attenuation represents the proportional increase in the denominator $1-p_0$. Both terms are positive, but the direct effect always dominates because $\Phi(\mu) > 0$ (Theorem 1), which in turn follows from the strict convexity of p_0 (Corollary A.4).

Elasticity.

Definition A.2 (Elasticity of h with respect to μ)**.**

$$\varepsilon_h(\mu) = \frac{\partial h}{\partial \mu} \cdot \frac{\mu}{h} = \mu \cdot \frac{\partial(\log h)}{\partial \mu}. \tag{A.26}$$

Proposition A.12 (Elasticity formula and bounds)**.**

$$\boxed{\varepsilon_h(\mu) = 1 - \omega(\mu), \qquad \omega(\mu) = \frac{\mu\, p_0\, \Lambda}{1-p_0}.} \tag{A.27}$$

Moreover:

(i) *For $n = 1$: $\varepsilon_h = 0$ (since $h \equiv 1$).*

(ii) *For $n \geq 2$: $\varepsilon_h \in (0,1)$ for all $\mu \in (0,1)$. The intensity function h is inelastic with respect to μ.*

This is the full proof of Proposition 1 in the main text.

Proof. From Proposition A.11,

$$\varepsilon_h = \mu \left[\frac{1}{\mu} - \frac{p_0 \Lambda}{1 - p_0} \right] = 1 - \frac{\mu p_0 \Lambda}{1 - p_0} = 1 - \omega.$$

By Theorem A.1, $\Phi(\mu) > 0$ for $n \geq 2$, which gives $\omega < 1$ and hence $\varepsilon_h > 0$. Since $\omega > 0$ (all factors are positive), $\varepsilon_h < 1$. □

Remark A.7 (Inelastic interpretation)**.** A one-percent increase in μ leads to less than a one-percent increase in h, because increasing μ simultaneously increases $1 - p_0$ (the denominator of h), partially offsetting the numerator effect.

Proposition A.13 (Limiting behavior of ε_h)**.**

(a) *As $\mu \to 0^+$ (for $n \geq 2$): $\varepsilon_h \to 0^+$. The truncation attenuation $\omega \to 1^-$.*

(b) *As $\mu \to 1^-$: $\varepsilon_h \to 1 - C_\kappa$, where $C_\kappa = \prod_{j=1}^{n-1} j/(\kappa + j)$ is the constant from Proposition A.15. Since $p_0 \sim (1 - \mu) C_\kappa$ and $\Lambda \sim 1/(1 - \mu) + O(1)$, we get $p_0 \Lambda \to C_\kappa$ and hence $\omega \to C_\kappa$. For large n or large κ, $C_\kappa \approx 0$ and the limit approaches 1.*

(c) *As $\kappa \to \infty$ (binomial limit): $\varepsilon_h \to 1 - n\mu(1 - \mu)^{n-1} \big/ \big[1 - (1 - \mu)^n\big]$.*

(d) *As $\kappa \to 0$ (maximum overdispersion): $p_0 \to 1 - \mu$, $\Lambda \to 1/(1 - \mu)$, so $\omega \to 1$ and $\varepsilon_h \to 0$. In this limit $h \to 1$; the intensity function is flat.*

(e) *As $n \to \infty$: $p_0 \to 0$, so $\omega \to 0$ and $\varepsilon_h \to 1$. The truncation correction becomes negligible.*

Numerical verification. For $n = 10$, $\kappa = 10$, $\mu = 0.3$: the analytic formula gives $\varepsilon_h = 0.734979$, and centered finite differences yield 0.734979. Agreement to six decimal places. □

Boundary analysis.

Proposition A.14 (Behavior of h as $\mu \to 0^+$)**.** *For $n \geq 2$ and $\kappa > 0$:*

$$h(\mu) \to \frac{1}{\Lambda_0} \quad \text{as } \mu \to 0^+, \tag{A.28}$$

where $\Lambda_0 = \kappa \sum_{j=0}^{n-1} 1/(\kappa + j)$.

Proof. Taylor expansion: $p_0(\mu) = 1 - \mu\Lambda_0 + O(\mu^2)$ (from $p_0(0) = 1$ and $p_0'(0) = -\Lambda_0$). Then $1 - p_0 = \mu\Lambda_0 + O(\mu^2)$, and $h = \mu \big/ \big(\mu\Lambda_0 + O(\mu^2)\big) \to 1/\Lambda_0$. □

For example, when $n = 2$ and $\kappa = 1$: $\Lambda_0 = 1/1 + 1/2 = 3/2$, so $h(0^+) = 2/3$. For $n = 10$ and $\kappa = 1$: $\Lambda_0 = H_{10} \approx 2.929$, so $h(0^+) \approx 0.341$. As noted in Remark 2, this limit is strictly less than unity whenever $n \geq 2$, confirming that the truncation adjustment has a non-vanishing effect even as $\mu \to 0^+$.

Proposition A.15 (Behavior of h as $\mu \to 1^-$)**.** *For $n \geq 1$ and $\kappa > 0$:*

$$h(\mu) \to 1 \quad \text{as } \mu \to 1^-, \qquad 1 - h(\mu) \sim (1-\mu)(1 - C_\kappa), \tag{A.29}$$

where $C_\kappa = \prod_{j=1}^{n-1} j/(\kappa + j) \in (0, 1)$.

Proof. As $\mu \to 1$, $b = (1-\mu)\kappa \to 0$, and the product form (3.3) gives $p_0 \sim (1-\mu)\, C_\kappa$ with $C_\kappa = \prod_{j=1}^{n-1} j/(\kappa + j)$. Then

$$h = \frac{\mu}{1 - p_0} \sim \frac{1 - (1-\mu)}{1 - (1-\mu)C_\kappa},$$

and

$$1 - h \sim \frac{(1-\mu)(1 - C_\kappa)}{1 - (1-\mu)C_\kappa} \sim (1-\mu)(1 - C_\kappa)$$

as $\mu \to 1^-$. □

Elasticity monotonicity in κ.

Conjecture A.1 (Monotonicity of ε_h in κ)**.** *For fixed $\mu \in (0, 1)$ and $n \geq 2$, the elasticity ε_h is increasing in κ. Equivalently, the attenuation ratio ω is decreasing in κ.*

The following proof sketch provides supporting evidence short of a rigorous proof.

Proof sketch. As κ increases, p_0 decreases (Proposition A.9), and the dominant effect on $\omega = \mu\, p_0\, \Lambda/(1 - p_0)$ is the decrease in the numerator factor p_0. The boundary behavior confirms monotonicity: $\varepsilon_h \to 0$ as $\kappa \to 0$ (Proposition A.13(d)) and $\varepsilon_h \to 1 - n\mu(1-\mu)^{n-1}/[1 - (1-\mu)^n] > 0$ as $\kappa \to \infty$ (Proposition A.13(c)). A closed-form proof requires showing that the decrease in p_0 dominates the concurrent decrease in Λ uniformly over $(0, 1)$.

Remark A.8 (Numerical verification of the conjecture)**.** We evaluated $\partial\varepsilon_h/\partial\kappa$ via centered finite differences on a grid of 100×100 values of $(\mu, \kappa) \in (0.01, 0.99) \times (0.01, 100)$ for each of $n \in \{2, 5, 10, 20, 50, 100\}$. In every case the derivative is strictly positive, supporting the conjecture. No counterexample has been found.

A.5. **Log-scale marginal effect decomposition.** Proposition 5 established that the marginal effect of covariate x_k on log expected enrollment decomposes exactly as $\mathrm{LAE}_k + \mathrm{LIE}_k$, with multipliers $(1 - q_i)$ and $(1 - \mu_i)\varepsilon_{h,i}$ respectively. This subsection provides three supplementary results: a correction remark on the sign structure of the LIE, a sensitivity analysis of the decomposition ratio with respect to κ, and a threshold characterization of the poverty reversal regime.

Remark A.9 (Sign of the truncation attenuation in the LIE)**.** The correct LIE has a *minus* sign in the truncation attenuation:

$$\mathrm{LIE}_k = \left[\frac{1}{\mu_i} - \frac{p_{0,i}\,\Lambda_i}{1-p_{0,i}}\right]\mu_i(1-\mu_i)\,\tilde{\beta}_{k,s} = (1-\mu_i)(1-\omega_i)\,\tilde{\beta}_{k,s}, \tag{A.30}$$

where $\omega_i = \mu_i\, p_{0,i}\,\Lambda_i/(1-p_{0,i})$ is the attenuation ratio from Proposition A.12. The attenuation is *subtracted*, not added, because increasing μ simultaneously increases $1-p_0$ (the denominator of h), which attenuates—not amplifies—the response of h to μ.

Proposition A.16 (Decomposition ratio)**.** *Define the decomposition ratio*

$$\mathrm{DR}(\kappa) = \frac{|\,\mathrm{LAE}_k\,|}{|\,\mathrm{LAE}_k\,| + |\,\mathrm{LIE}_k\,|} = \frac{(1-q)\,|\tilde{\alpha}_k|}{(1-q)\,|\tilde{\alpha}_k| + (1-\mu)\,\varepsilon_h\,|\tilde{\beta}_k|}. \tag{A.31}$$

Then:

(a) *As* $\kappa \to 0$*:* $\varepsilon_h \to 0$, $\mathrm{LIE} \to 0$, $\mathrm{DR} \to 1$. *Maximum overdispersion renders the intensity channel inoperative.*

(b) *As* $\kappa \to \infty$*:* $\varepsilon_h \to \varepsilon_h^{\mathrm{binom}} \in (0,1)$, $\mathrm{DR} \to \mathrm{DR}^{\mathrm{binom}} \in (0,1)$.

(c) *Conditional on Conjecture A.1,* DR *is decreasing in* κ *(since* ε_h *is increasing in* κ*).*

Proof. Properties (a) and (b) follow directly from Proposition A.13(d) and (c), respectively. Property (c) follows because the numerator of (A.31) is constant in κ while the denominator is increasing in κ through the factor ε_h. □

Table A.1 illustrates the sensitivity of the decomposition ratio to κ, evaluated at representative parameter values ($n = 10$, $\mu = 0.30$, $q = 0.70$, $|\tilde{\alpha}| = 0.5$, $|\tilde{\beta}| = 0.8$).

Table A.1. Sensitivity of the decomposition ratio to κ. Parameters: $n = 10$, $\mu = 0.30$, $q = 0.70$, $|\tilde{\alpha}| = 0.5$, $|\tilde{\beta}| = 0.8$.

κ	ε_h	ω	$\lvert\,\mathrm{LIE}\,\rvert$	DR
0.01	0.008	0.992	0.005	0.970
0.1	0.071	0.929	0.040	0.790
1	0.343	0.657	0.192	0.439
10	0.735	0.265	0.412	0.267
100	0.859	0.141	0.481	0.238
1000	0.874	0.126	0.489	0.235

Remark A.10. The DR is highly sensitive to κ in the range $(0.1, 10)$. In our data context, where $\kappa \approx 5$–15, both the access and intensity channels contribute meaningfully to the total effect.

Proposition A.17 (Threshold for access-effect dominance)**.** *In the poverty reversal regime ($\tilde{\alpha}_{\text{pov},s} < 0$, $\tilde{\beta}_{\text{pov},s} > 0$), the total marginal effect on log expected enrollment is negative (access dominates) if and only if*

$$\boxed{\frac{|\tilde{\alpha}_{\text{pov},s}|}{|\tilde{\beta}_{\text{pov},s}|} > \frac{(1-\mu)\,\varepsilon_h}{1-q}.} \tag{A.32}$$

The right-hand side is increasing in q and increasing in κ (through ε_h). When participation is high (q near 1), even a small intensive-margin coefficient can dominate; when q is low, the extensive margin dominates more easily.

Proof. From Proposition 5, the total effect is negative iff $(1-q)|\tilde{\alpha}| > (1-\mu)\varepsilon_h|\tilde{\beta}|$; dividing both sides by $|\tilde{\beta}|(1-q)$ gives (A.32). □

Table A.2 reports the threshold $(1-\mu)\varepsilon_h/(1-q)$ for parameter combinations spanning the range of NSECE providers. The elasticity ε_h is computed from the exact formula in Proposition A.12.

Remark A.11. At our posterior estimates ($\kappa \approx 7$, $\mu \approx 0.30$, $q \approx 0.64$, $n \approx 50$), the threshold is approximately 1.87 (Table A.2, row $q = 0.64$, $\mu = 0.30$, $\kappa = 7$). Since the empirical coefficient ratio $|\alpha_{\text{pov}}|/|\beta_{\text{pov}}| = 0.324/0.090 \approx 3.6 \gg 1.87$, the access effect dominates decisively. For providers near the sample average, the coefficient ratio would need to fall below 1.87—that is, the extensive-margin coefficient would need to be less than twice the intensive-margin coefficient—before intensity could dominate. At $q = 0.80$ (the most extreme participation rate considered), the threshold rises to 2.87–3.77, approaching the empirical ratio. This confirms that access dominance is most fragile among high-participation centers with low μ and large κ, where the intensity elasticity is near unity and the extensive multiplier $(1-q)$ is small.

Appendix B. Identification and Fisher Information

This appendix provides the score functions, Fisher information structure, and the identification proof for the hurdle beta-binomial model deferred from Section 3. We maintain the notation from Section 3.1, with $a = \mu\kappa$, $b = (1-\mu)\kappa$, and the digamma function $\psi(x) = \Gamma'(x)/\Gamma(x)$.

TABLE A.2. Access-dominance threshold $(1-\mu)\varepsilon_h/(1-q)$ for representative parameter combinations. Values below the empirical coefficient ratio $|\alpha_{\text{pov}}|/|\beta_{\text{pov}}| = 3.6$ indicate access dominance. All entries computed with $n = 50$.

		κ		
q	μ	3	7	15
	0.15	0.90	1.26	1.51
0.50	0.30	1.15	1.34	1.39
	0.45	1.04	1.10	1.10
	0.15	1.25	1.75	2.09
0.64	0.30	1.60	1.87	1.93
	0.45	1.44	1.52	1.53
	0.15	2.25	3.16	3.77
0.80	0.30	2.87	3.36	3.48
	0.45	2.60	2.74	2.75

Note. Threshold values are computed from the exact elasticity formula (A.27) with $p_0 = \prod_{j=0}^{n-1}[(1-\mu)\kappa + j]/(\kappa + j)$ and $\Lambda = \kappa \sum_{j=0}^{n-1}[(1-\mu)\kappa + j]^{-1}$. Values are rounded to two decimal places.

B.1. **Score functions.** We derive the score functions for each component of the hurdle beta-binomial log-likelihood. These expressions serve three roles: (i) verification of the automatic differentiation gradients used in Stan (Carpenter et al., 2017), (ii) diagnostics via score magnitude monitoring, and (iii) the theoretical basis for the sandwich variance estimator (Binder, 1983) in Section C. Recall the parameterization $a = \mu\kappa$, $b = (1-\mu)\kappa$, the zero probability p_0 from (3.3), and the auxiliary quantities Λ, Λ_2 defined in Equation (A.1).

Proposition B.18 (Extensive-margin score)**.** *The score of the extensive-margin log-likelihood* $\ell_i^{(1)} = z_i \log q_i + (1 - z_i)\log(1 - q_i)$ *with respect to the linear predictor* $\eta_i^{(1)} = \mathbf{x}_i^\intercal \boldsymbol{\alpha} + \mathbf{x}_i^{(r)\intercal} \boldsymbol{\delta}_{1,s}$ *is*

$$\frac{\partial \ell_i^{(1)}}{\partial \eta_i^{(1)}} = z_i - q_i, \tag{B.1}$$

where $q_i = \operatorname{expit}(\eta_i^{(1)})$*. The Hessian is non-random:*

$$\frac{\partial^2 \ell_i^{(1)}}{\partial (\eta_i^{(1)})^2} = -\, q_i(1 - q_i). \tag{B.2}$$

Proof. Standard logistic regression calculus. Since $\partial q_i/\partial \eta_i^{(1)} = q_i(1-q_i)$, differentiating $\ell_i^{(1)}$ yields $z_i/q_i \cdot q_i(1-q_i) - (1-z_i)/(1-q_i) \cdot q_i(1-q_i) = z_i(1-q_i) - (1-z_i)q_i = z_i - q_i$. The second derivative is $-q_i(1-q_i)$, which depends only on parameters and is therefore non-random. □

Remark B.12. The non-randomness of (B.2) implies that the observed and expected Fisher information coincide for the extensive margin: $\mathcal{I}_{\text{ext},i} = q_i(1-q_i)$. This simplifies the sandwich estimator, since the Hessian bread matrix for Part 1 contains no stochastic terms.

Proposition B.19 (Intensive-margin score for μ)**.** *Define the (untruncated) beta-binomial score function (following the parameterization of Prentice, 1986)*

$$S_{\text{BB}}(y;\, \mu, \kappa, n) = \kappa\big[\psi(y+a) - \psi(n-y+b) - \psi(a) + \psi(b)\big], \tag{B.3}$$

where $\psi(\cdot)$ is the digamma function and $a = \mu\kappa$, $b = (1-\mu)\kappa$. The score of the zero-truncated log-likelihood $\ell_i^{(2+)} = \log f_{\text{BB}}(y_i) - \log(1-p_{0,i})$ with respect to μ is

$$\boxed{\frac{\partial \ell_i^{(2+)}}{\partial \mu} = S_{\text{BB}}(y_i) \;-\; \frac{p_{0,i}\,\Lambda_i}{1-p_{0,i}}}. \tag{B.4}$$

Proof. The decomposition $\ell_i^{(2+)} = \log f_{\text{BB}}(y_i) - \log(1-p_{0,i})$ gives two terms.

Term 1 (BB score). Using $\partial a/\partial \mu = \kappa$ and $\partial b/\partial \mu = -\kappa$, differentiation of the beta-binomial log-PMF yields

$$\frac{\partial \log f_{\text{BB}}}{\partial \mu} = \kappa\big[\psi(y+a) - \psi(n-y+b) - \psi(a) + \psi(b)\big] = S_{\text{BB}}(y).$$

Term 2 (truncation correction). Since $\ln p_0 = \sum_{j=0}^{n-1} \ln[(1-\mu)\kappa + j] - \ln[\kappa + j]$, we have $\partial p_0/\partial \mu = -p_0\Lambda$ (see Equation (A.1)), and therefore

$$\frac{\partial \log(1-p_0)}{\partial \mu} = \frac{p_0\,\Lambda}{1-p_0} > 0.$$

Combining, $\partial \ell^{(2+)}/\partial \mu = S_{\text{BB}}(y) - p_0\Lambda/(1-p_0)$. □

Proposition B.20 (Score for $\log\kappa$)**.** *The score of the zero-truncated log-likelihood with respect to the log-dispersion parameter is*

$$\boxed{\begin{aligned}\frac{\partial \ell_i^{(2+)}}{\partial \log\kappa} = \kappa\Big[&\mu\big[\psi(y_i{+}a) - \psi(a)\big] + (1{-}\mu)\big[\psi(n_i{-}y_i{+}b) - \psi(b)\big] \\ &+ \psi(\kappa) - \psi(n_i{+}\kappa)\Big] - \frac{p_{0,i}\,\mu\kappa}{1-p_{0,i}}\sum_{j=1}^{n_i-1}\frac{j}{(b+j)(\kappa+j)}.\end{aligned}} \tag{B.5}$$

Proof. With $\phi = \log\kappa$ and the chain rule $\partial/\partial\phi = \kappa\,\partial/\partial\kappa$, the untruncated BB log-PMF gives the first line via $\partial a/\partial\kappa = \mu$ and $\partial b/\partial\kappa = 1-\mu$. The truncation correction requires $\partial\log(1-p_0)/\partial\log\kappa$. Writing $\ln p_0 = \sum_{j=0}^{n-1}\ln(b+j) - \ln(\kappa+j)$ and differentiating with respect to κ (noting $\partial b/\partial\kappa = 1-\mu$) yields the second line after rearranging the telescoping sums. □

Proposition B.21 (Expected information for μ)**.** *The negative expected second derivative of the zero-truncated log-likelihood with respect to μ is*

$$\begin{aligned}-\,\mathbb{E}_{Y|Y>0}\left[\frac{\partial^2\ell^{(2+)}}{\partial\mu^2}\right] = \frac{\kappa^2}{1-p_0}\,\mathbb{E}_{Y\sim\mathrm{BetaBin}}\big[\psi_1(Y+a) + \psi_1(n{-}Y+b)\big] \\ -\,\kappa^2\big[\psi_1(a)+\psi_1(b)\big] + T'(\mu),\end{aligned} \tag{B.6}$$

where $\psi_1(\cdot) = d\psi/dx$ is the trigamma function and

$$T'(\mu) = \frac{p_0\,\Lambda_2}{1-p_0} - \frac{p_0\,\Lambda^2}{(1-p_0)^2}, \qquad \Lambda_2 = \kappa^2\sum_{j=0}^{n-1}\frac{1}{(b+j)^2}. \tag{B.7}$$

Proof. Differentiating the score (B.4) with respect to μ produces two contributions.

The S_{BB} term. Direct differentiation gives $\partial S_{\mathrm{BB}}/\partial\mu = \kappa^2[\psi_1(y+a) + \psi_1(n-y+b) - \psi_1(a) - \psi_1(b)]$.

The truncation term. Set $T(\mu) = p_0\Lambda/(1-p_0)$. The quotient rule with $\partial p_0/\partial\mu = -p_0\Lambda$ gives

$$\begin{aligned}T'(\mu) &= \frac{p_0(\Lambda_2-\Lambda^2)(1-p_0) - p_0^2\Lambda^2}{(1-p_0)^2} = \frac{p_0\Lambda_2(1-p_0) - p_0\Lambda^2}{(1-p_0)^2} \\ &= \frac{p_0\,\Lambda_2}{1-p_0} - \frac{p_0\,\Lambda^2}{(1-p_0)^2}.\end{aligned}$$

Expectation under truncation. Taking expectations under $Y \mid Y > 0$ and converting via the identity $\mathbb{E}_{Y|Y>0}[g(Y)] = \mathbb{E}_{Y\sim\mathrm{BetaBin}}[g(Y)]/(1-p_0)$ yields (B.6). The

non-stochastic terms $-\kappa^2[\psi_1(a)+\psi_1(b)]$ and $T'(\mu)$ pass through the expectation unchanged. $\square$

Remark B.13 (Cross-information)**.** Both μ and κ enter $\ell_i^{(2+)}$, so the cross-information $\mathcal{I}_{\mu,\kappa}$ is nonzero. The intensive-margin information block is

$$\mathcal{I}_{\text{int}+\kappa} = \begin{pmatrix} \mathcal{I}_{\text{int}} & \mathcal{I}_{\text{int},\kappa} \\ \mathcal{I}_{\kappa,\text{int}} & \mathcal{I}_{\kappa} \end{pmatrix}, \tag{B.8}$$

which is not further block-diagonal. The intensive-margin regression parameters and κ must be estimated jointly, but they are fully separated from the extensive-margin parameters by the hurdle structure (Section 3).

B.2. **Fisher information and block-diagonality.** The individual score functions and information entries derived in Section B.1 concern a single linear predictor or scalar parameter. This subsection establishes the *global* structure of the Fisher information matrix—in particular, the block-diagonal separation between the extensive- and intensive-margin parameter blocks—and collects several ancillary results on information rates and the sandwich convention.

Proposition B.22 (Block-diagonal Fisher information)**.** *Let $\boldsymbol{\theta} = (\boldsymbol{\phi}_1^\intercal, \boldsymbol{\phi}_2^\intercal)^\intercal$, where $\boldsymbol{\phi}_1 = (\boldsymbol{\alpha}^\intercal, \boldsymbol{\delta}_{1,s}^\intercal)^\intercal$ collects the extensive-margin parameters and $\boldsymbol{\phi}_2 = (\boldsymbol{\beta}^\intercal, \boldsymbol{\delta}_{2,s}^\intercal, \log\kappa)^\intercal$ collects the intensive-margin parameters including the dispersion. Then the expected Fisher information for a single observation decomposes as*

$$\boxed{\mathcal{I}_i(\boldsymbol{\theta}) = \operatorname{diag}\big(\mathcal{I}_{\text{ext},i},\ \mathcal{I}_{\text{int}+\kappa,i}\big).} \tag{B.9}$$

That is, the cross-information between $\boldsymbol{\phi}_1$ and $\boldsymbol{\phi}_2$ vanishes identically.

Proof. The log-likelihood decomposes as $\ell_i = \ell_i^{(1)} + \ell_i^{(2)}$, where

$$\ell_i^{(1)} = z_i \log q_i + (1-z_i)\log(1-q_i)$$

depends only on $\boldsymbol{\phi}_1$ through $q_i = \operatorname{expit}(\mathbf{x}_i^\intercal\boldsymbol{\alpha} + \mathbf{x}_i^{(r)\intercal}\boldsymbol{\delta}_{1,s})$, and

$$\ell_i^{(2)} = z_i\big[\log f_{\text{BB}}(y_i \mid n_i, \mu_i, \kappa) - \log(1-p_{0,i})\big]$$

depends only on $\boldsymbol{\phi}_2$ through μ_i and κ.

Cross-derivative $\partial\ell^{(1)}/\partial\boldsymbol{\phi}_2$. Since $\ell_i^{(1)}$ does not involve any element of $\boldsymbol{\phi}_2$, this derivative is identically zero.

Cross-derivative $\partial\ell^{(2)}/\partial\boldsymbol{\phi}_1$. Write $\ell_i^{(2)} = z_i \cdot g(\mu_i, \kappa, y_i, n_i)$. The indicator $z_i = \mathbf{1}(y_i > 0)$ is a function of the observed data, not of parameters, so $\partial z_i/\partial\boldsymbol{\phi}_1 = \mathbf{0}$ and hence $\partial\ell_i^{(2)}/\partial\boldsymbol{\phi}_1 = \mathbf{0}$.

Combining, $\partial^2\ell_i/\partial\boldsymbol{\phi}_1\,\partial\boldsymbol{\phi}_2^\intercal = \mathbf{0}$, which gives (B.9) after taking expectations. □

Remark B.14 (Hurdle versus zero-inflated models)**.** The block-diagonality in Proposition B.22 is a general structural property of hurdle (two-part) models (Cragg, 1971; Mullahy, 1986): the participation indicator $z_i = \mathbf{1}(y_i > 0)$ is observed, so each component of the likelihood depends on a disjoint parameter subset. In zero-inflated models (e.g., Lambert, 1992), by contrast, the at-risk indicator is *latent*, and both the inflation and count parameters enter the marginal likelihood of every observation, coupling the two components in the Fisher information (see B. Neelon et al., 2016, for a comprehensive comparison).

Proposition B.23 (Extensive-margin information in parameter space)**.** *The extensive-margin Fisher information block for provider i is*

$$\mathcal{I}_{\text{ext},i} = q_i(1 - q_i)\,\mathbf{d}_{1,i}\,\mathbf{d}_{1,i}^\intercal, \tag{B.10}$$

where $\mathbf{d}_{1,i} = \partial\eta_i^{(1)}/\partial(\boldsymbol{\alpha}^\intercal, \boldsymbol{\delta}_{1,s}^\intercal)^\intercal$ is the gradient of the extensive-margin linear predictor with respect to all extensive-margin parameters.

Proof. By Remark B.12, the scalar information for the linear predictor is $\mathcal{I}_{\text{ext},i} = q_i(1 - q_i)$. The chain rule $\partial\ell_i^{(1)}/\partial\boldsymbol{\phi}_1 = (\partial\ell_i^{(1)}/\partial\eta_i^{(1)})\,\mathbf{d}_{1,i}$ yields the rank-one outer product form (B.10). □

Remark B.15 (Fisher information rates)**.** The extensive-margin information $q_i(1 - q_i)$ is $O(1)$ and non-random (Remark B.12). The intensive-margin information involves trigamma sums $\psi_1(a) + \psi_1(b)$ that diverge as $\kappa \to 0$; for the empirically relevant range $\kappa \in [2.5, 26.5]$ the information is well-behaved, but the divergent terms underlie the larger design effect ratios observed for intensive-margin parameters in Sections C and 4.

B.3. **Identification.** This subsection provides the complete proof of the identification theorem stated in Section 3 (Theorem 4). We first restate the regularity conditions, then develop the five-step proof. Three supporting propositions follow: the separation of μ and κ in the truncated beta-binomial (with the corrected $n \geq 3$ threshold), the likelihood non-identification of the cross-margin covariance $\boldsymbol{\Sigma}_{12}$, and sufficient conditions for posterior well-concentration.

Regularity conditions. The following conditions are required for identification. They match the statement in Theorem 4 exactly; we include additional commentary on the role of each.

(C1) *Full rank.* The design matrices have full rank: $\operatorname{rank}(\mathbf{X}) = P$ and $\operatorname{rank}(\mathbf{V}) = Q$.

(C2) *State-level replication.* Each state s has at least q providers with linearly independent covariate vectors $\mathbf{x}_i^{(r)}$, with at least one zero and at least q positive responders.

(C3) *Trial-size variation.* $n_i \geq 3$ for all i with $z_i = 1$, and there exist providers i, j with $s[i] = s[j]$ and $n_i \neq n_j$ in each state.

(C4) *Prior support.* The prior $\pi(\boldsymbol{\theta})$ has full support on the parameter space.

Remark B.16 (Asymmetry in C2)**.** The requirements in (C2) are asymmetric. The extensive margin requires at least one zero per state (to identify $q_s < 1$), while the intensive margin requires at least q positive responders per state (to identify the q-dimensional $\boldsymbol{\delta}_{2,s}$). The asymmetry reflects the different informational demands of a univariate binary variable versus a q-dimensional regression parameter.

Theorem B.2 (Identification of the HBB model)**.** *Under Conditions* (C1)–(C4)*, the full parameter vector* $\boldsymbol{\theta} = (\boldsymbol{\alpha}, \boldsymbol{\beta}, \kappa, \{\boldsymbol{\delta}_s\}, \boldsymbol{\Sigma}_\delta, \{\boldsymbol{\Gamma}_k\})$ *is identified under the posterior.*

Proof. The argument separates two layers of identification: (i) the within-state likelihood identifies *state-specific total coefficients* and κ; (ii) the hierarchical prior structure resolves the additive decomposition of these totals into population-average effects, policy moderators, and residual state deviations. The proof has six steps.

Step 1: Extensive-margin total state coefficients. The marginal distribution of $z_i = \mathbf{1}(Y_i > 0)$ is $\operatorname{Bern}(q_i)$ with

$$q_i = \operatorname{expit}\big(\mathbf{x}_i{}^\intercal \boldsymbol{\alpha} + \mathbf{x}_i^{(r)\intercal} \boldsymbol{\delta}_{1,s[i]}\big).$$

Partition $\mathbf{x}_i = (\mathbf{x}_i^{(r)\intercal},\ \mathbf{x}_i^{(f)\intercal})^\intercal$ and correspondingly $\boldsymbol{\alpha} = (\boldsymbol{\alpha}^{(r)\intercal},\ \boldsymbol{\alpha}^{(f)\intercal})^\intercal$, where $\mathbf{x}_i^{(r)} \in \mathbb{R}^q$ collects the covariates with state-varying coefficients and $\mathbf{x}_i^{(f)} \in \mathbb{R}^{P-q}$ collects the fixed-only covariates.[1] Define the *total extensive-margin state coefficient* (cf. (3.12))

$$\tilde{\boldsymbol{\alpha}}_s \ := \ \boldsymbol{\alpha}^{(r)} + \boldsymbol{\delta}_{1,s} \ \in \ \mathbb{R}^q, \tag{B.11}$$

[1]When $q = P$ (as in models M2/M3), the fixed-only partition is empty and the argument applies with $\boldsymbol{\alpha}^{(f)}$ and $\mathbf{x}_i^{(f)}$ vacuous.

so that the linear predictor becomes $\text{logit}(q_i) = \mathbf{x}_i^{(f)\intercal}\boldsymbol{\alpha}^{(f)} + \mathbf{x}_i^{(r)\intercal}\tilde{\boldsymbol{\alpha}}_{s[i]}$. The parameters $\boldsymbol{\phi} := (\boldsymbol{\alpha}^{(f)}, \tilde{\boldsymbol{\alpha}}_1, \ldots, \tilde{\boldsymbol{\alpha}}_S) \in \mathbb{R}^{(P-q)+Sq}$ enter the model through the effective design matrix

$$\mathbf{Z} = \left[\mathbf{X}^{(f)},\ \mathbf{D}_1\mathbf{X}_1^{(r)},\ \ldots,\ \mathbf{D}_S\mathbf{X}_S^{(r)}\right] \in \mathbb{R}^{N\times[(P-q)+Sq]},$$

where $\mathbf{D}_s$ selects providers in state s and $\mathbf{X}_s^{(r)}$ is the within-state submatrix of $\mathbf{x}_i^{(r)}$ values. Condition (C2) ensures $\text{rank}(\mathbf{X}_s^{(r)}) = q$ for each s, so the state-specific blocks contribute Sq independent columns. Condition (C1) ensures $\text{rank}(\mathbf{X}) = P$, so the $(P-q)$ fixed-only columns are not in the span of the state-specific blocks. Hence $\text{rank}(\mathbf{Z}) = (P-q)+Sq$, and by the injectivity of logit, $\boldsymbol{\alpha}^{(f)}$ and $\{\tilde{\boldsymbol{\alpha}}_s\}_{s=1}^S$ are identified from the extensive-margin likelihood.

However, $\boldsymbol{\alpha}^{(r)}$ and $\boldsymbol{\delta}_{1,s}$ are *not* separately identified from the likelihood alone: for any $\mathbf{c} \in \mathbb{R}^q$, the shift $\boldsymbol{\alpha}^{(r)} \mapsto \boldsymbol{\alpha}^{(r)} + \mathbf{c},\ \boldsymbol{\delta}_{1,s} \mapsto \boldsymbol{\delta}_{1,s} - \mathbf{c}$ for all s preserves every $\tilde{\boldsymbol{\alpha}}_s$ and hence the entire likelihood. The resolution of this q-dimensional additive ambiguity is the task of Step 2.

Step 2: Hierarchical decomposition on the extensive margin. The hierarchical model specifies $\boldsymbol{\delta}_{1,s} = \boldsymbol{\Gamma}_1\mathbf{v}_s + \boldsymbol{\varepsilon}_{1,s}$ with $\boldsymbol{\varepsilon}_{1,s} \sim \mathsf{N}_q(\mathbf{0}, \boldsymbol{\Sigma}_{\varepsilon,1})$. Substituting into (B.11),

$$\tilde{\boldsymbol{\alpha}}_s = \boldsymbol{\alpha}^{(r)} + \boldsymbol{\Gamma}_1\mathbf{v}_s + \boldsymbol{\varepsilon}_{1,s}.$$

Since $\{\tilde{\boldsymbol{\alpha}}_s\}$ are identified from Step 1 and $\{\mathbf{v}_s\}$ are observed, this is a multivariate regression of $\tilde{\boldsymbol{\alpha}}_s$ on $\mathbf{v}_s$ with intercept $\boldsymbol{\alpha}^{(r)}$. Under $\text{rank}(\mathbf{V}) = Q$ ((C1)) and $S > Q$, the ordinary least-squares solution uniquely determines the slope matrix $\boldsymbol{\Gamma}_1$ and the intercept $\boldsymbol{\alpha}^{(r)}$. Importantly, the identification of $\boldsymbol{\alpha}^{(r)}$ relies on the hierarchical centering convention $\mathbb{E}[\boldsymbol{\varepsilon}_{1,s}] = \mathbf{0}$: this structural assumption pins down the intercept of the between-state regression, resolving the shift ambiguity that the within-state likelihood leaves free. With $\boldsymbol{\alpha}^{(r)}$ identified, the state deviations follow as $\boldsymbol{\delta}_{1,s} = \tilde{\boldsymbol{\alpha}}_s - \boldsymbol{\alpha}^{(r)}$ and the residuals as $\boldsymbol{\varepsilon}_{1,s} = \boldsymbol{\delta}_{1,s} - \boldsymbol{\Gamma}_1\mathbf{v}_s$.

Step 3: Intensive-margin total state coefficients and κ. Conditional on $z_i = 1$ (i.e., $Y_i > 0$), the distribution of Y_i is $\text{ZT-BetaBin}(n_i, \mu_i, \kappa)$ with $\mu_i = \text{expit}(\mathbf{x}_i{}^{\intercal}\boldsymbol{\beta} + \mathbf{x}_i^{(r)\intercal}\boldsymbol{\delta}_{2,s[i]})$. By Proposition B.24 below, the pair (μ_i, κ) is identified from the truncated likelihood whenever $n_i \geq 3$ ((C3)). Once $\{\mu_i\}$ are pinned down for all positive responders, partition $\boldsymbol{\beta} = (\boldsymbol{\beta}^{(r)\intercal},\ \boldsymbol{\beta}^{(f)\intercal})^{\intercal}$ and define the *total intensive-margin state coefficient*

$$\tilde{\boldsymbol{\beta}}_s \ := \ \boldsymbol{\beta}^{(r)} + \boldsymbol{\delta}_{2,s} \ \in \ \mathbb{R}^q, \tag{B.12}$$

so that $\text{logit}(\mu_i) = \mathbf{x}_i^{(f)\intercal}\boldsymbol{\beta}^{(f)} + \mathbf{x}_i^{(r)\intercal}\tilde{\boldsymbol{\beta}}_{s[i]}$. An argument identical to Step 1—using the injectivity of logit, the effective design matrix, and the rank conditions (C1)–(C2)—identifies $\boldsymbol{\beta}^{(f)}$ and $\{\tilde{\boldsymbol{\beta}}_s\}_{s=1}^S$ from the intensive-margin likelihood. As in the extensive margin, $\boldsymbol{\beta}^{(r)}$ and $\boldsymbol{\delta}_{2,s}$ are individually identified only through the hierarchical layer.

Step 4: Hierarchical decomposition on the intensive margin. Parallel to Step 2, the hierarchical specification $\boldsymbol{\delta}_{2,s} = \boldsymbol{\Gamma}_2\mathbf{v}_s + \boldsymbol{\varepsilon}_{2,s}$ yields the between-state regression $\tilde{\boldsymbol{\beta}}_s = \boldsymbol{\beta}^{(r)} + \boldsymbol{\Gamma}_2\mathbf{v}_s + \boldsymbol{\varepsilon}_{2,s}$. Under the same rank conditions, $\boldsymbol{\Gamma}_2$ and $\boldsymbol{\beta}^{(r)}$ are uniquely determined, whence $\boldsymbol{\delta}_{2,s} = \tilde{\boldsymbol{\beta}}_s - \boldsymbol{\beta}^{(r)}$ and $\boldsymbol{\varepsilon}_{2,s} = \boldsymbol{\delta}_{2,s} - \boldsymbol{\Gamma}_2\mathbf{v}_s$.

Step 5: Identification of $\boldsymbol{\Sigma}_\varepsilon$. The residual state effects $\boldsymbol{\varepsilon}_s = (\boldsymbol{\varepsilon}_{1,s}\intercal, \boldsymbol{\varepsilon}_{2,s}\intercal)\intercal$ are identified from Steps 1–4 for $s = 1, \ldots, S$. These $S = 51$ independent draws from $\mathsf{N}_{2q}(\mathbf{0}, \boldsymbol{\Sigma}_\varepsilon)$ identify the covariance matrix $\boldsymbol{\Sigma}_\varepsilon$ provided that $S > 2q$ (so that the sample covariance has full rank). In our application, $q = 5$, so $S = 51 > 2q = 10$. The full covariance matrix $\boldsymbol{\Sigma}_\delta$ is then reconstructed from $\boldsymbol{\Gamma}_k$, $\boldsymbol{\Sigma}_\varepsilon$, and the variance of $\mathbf{v}_s$ via the identity $\text{Var}(\boldsymbol{\delta}_{k,s}) = \boldsymbol{\Gamma}_k \,\text{Var}(\mathbf{v}_s)\boldsymbol{\Gamma}_k{}^\intercal + \boldsymbol{\Sigma}_{\varepsilon,k}$.

Step 6: No label-switching ambiguity. Unlike mixture models where component labels are exchangeable, the hurdle model (Mullahy, 1986) has an *observed* partition: $z_i = \mathbf{1}(Y_i > 0)$ is a deterministic function of the data. Part 1 (extensive) and Part 2 (intensive) operate on distinct, observable subsets. No permutation of component labels preserves the likelihood, so there is no label-switching invariance. □

Proposition B.24 (Identification of μ and κ in ZT-BetaBin)**.** *For $n \geq 3$, the parameters (μ, κ) are identified from the full likelihood of the zero-truncated beta-binomial: if*

$$f_{\text{ZT-BetaBin}}(y \mid n, \mu_1, \kappa_1) = f_{\text{ZT-BetaBin}}(y \mid n, \mu_2, \kappa_2) \quad \text{for all } y \in \{1, \ldots, n\}, \tag{B.13}$$

then $(\mu_1, \kappa_1) = (\mu_2, \kappa_2)$.

Proof. We use a constructive argument based on consecutive probability ratios. Write $a = \mu\kappa$ and $b = (1-\mu)\kappa$ for the shape parameters of the (untruncated) beta-binomial.

Step 1: Truncation cancels in consecutive ratios. Since $f_{\text{ZT-BetaBin}}(y \mid n, \mu, \kappa) = f_{\text{BetaBin}}(y \mid n, \mu, \kappa)/(1 - p_0)$ for $y \geq 1$, the consecutive ratio

$$R_y \;:=\; \frac{f_{\text{ZT-BetaBin}}(y)}{f_{\text{ZT-BetaBin}}(y-1)} = \frac{f_{\text{BetaBin}}(y)}{f_{\text{BetaBin}}(y-1)} = \frac{n-y+1}{y} \cdot \frac{a+y-1}{b+n-y}, \qquad y = 2, \ldots, n,$$

depends only on (a, b, n) and not on p_0.

Step 2: Two ratios determine (a, b) when $n \geq 3$. For $n \geq 3$, the ratios R_2 and R_3 yield two independent equations. Define

$$r_2 = \frac{2}{n-1} R_2 = \frac{a+1}{b+n-2}, \qquad r_3 = \frac{3}{n-2} R_3 = \frac{a+2}{b+n-3}.$$

Cross-multiplying gives the linear system

$$a + 1 = r_2 (b + n - 2),$$
$$a + 2 = r_3 (b + n - 3).$$

Subtracting the first from the second: $1 = (r_3 - r_2) b + r_3(3 - n) - r_2(2 - n)$, so

$$b = \frac{r_2(n-2) - r_3(n-3) + 1}{r_3 - r_2}, \tag{B.14}$$

and then $a = r_2(b + n - 2) - 1$. The denominator $r_3 - r_2 \neq 0$ for all $a, b > 0$, since

$$r_3 - r_2 = \frac{(a+2)(b+n-2) - (a+1)(b+n-3)}{(b+n-2)(b+n-3)} = \frac{a+b+n-1}{(b+n-2)(b+n-3)} > 0.$$

Thus (a, b) is uniquely determined by the observed truncated distribution, and $\mu = a/(a+b)$, $\kappa = a + b$.

Step 3: $n = 2$ boundary. For $n = 2$, only the single ratio R_2 is available, giving one equation in two unknowns: identification fails. See Remark B.17 for the explicit one-dimensional manifold of equivalent parameters. □

Remark B.17 (The $n = 2$ boundary case)**.** For $n = 2$, the truncated support $\{1, 2\}$ provides only one free probability for two unknowns (μ, κ), so identification fails. The consecutive-ratio proof makes this precise: only R_2 is available, leaving a one-dimensional manifold of equivalent parameters. This motivates the requirement $n_i \geq 3$ in (C3), satisfied by all 4,392 positive responders in the NSECE ($\min n_i = 4$).

Remark B.18 (Role of trial-size variation)**.** The requirement in (C3) that $n_i \neq n_j$ within each state provides identification strength beyond minimal sufficiency: observations sharing μ_i but differing in n_i yield differential information about κ via the nonlinear dependence of p_0 on n.

Proposition B.25 (Likelihood non-identification of $\boldsymbol{\Sigma}_{12}$)**.** *The off-diagonal block $\boldsymbol{\Sigma}_{12} = \mathrm{Cov}(\boldsymbol{\varepsilon}_{1,s}, \boldsymbol{\varepsilon}_{2,s})$ is not identified by the likelihood alone (cf. the analogous discussion in Ghosal et al., 2020, Section 3.1). The log-likelihood $\ell(\boldsymbol{\theta})$ is completely invariant to $\boldsymbol{\Sigma}_{12}$ conditional on the realized values $\{\boldsymbol{\varepsilon}_{1,s}, \boldsymbol{\varepsilon}_{2,s}\}_{s=1}^{S}$.*

Proof. By the block-diagonal Fisher information (Proposition B.22), the log-likelihood decomposes as $\ell = \ell_{\text{ext}} + \ell_{\text{int}}$, where ℓ_{ext} depends on $(\boldsymbol{\alpha}, \boldsymbol{\delta}_{1,\cdot})$ and ℓ_{int} depends on $(\boldsymbol{\beta}, \boldsymbol{\delta}_{2,\cdot}, \kappa)$. Neither component involves $\boldsymbol{\Sigma}_\varepsilon$: the covariance enters the model solely through the hierarchical prior

$$\boldsymbol{\varepsilon}_s = \begin{pmatrix} \boldsymbol{\varepsilon}_{1,s} \\ \boldsymbol{\varepsilon}_{2,s} \end{pmatrix} \sim \mathsf{N}_{2q}\left(\mathbf{0},\ \begin{pmatrix} \boldsymbol{\Sigma}_{\varepsilon,1} & \boldsymbol{\Sigma}_{12} \\ \boldsymbol{\Sigma}_{21} & \boldsymbol{\Sigma}_{\varepsilon,2} \end{pmatrix}\right).$$

Conditional on the realized $\{\boldsymbol{\varepsilon}_s\}$, the likelihood is a fixed function of $(\boldsymbol{\alpha}, \boldsymbol{\beta}, \kappa)$ alone, and $\boldsymbol{\Sigma}_{12}$ drops out entirely. □

Remark B.19 ($\boldsymbol{\Sigma}_{12}$ as residual covariance)**.** The cross-margin covariance $\boldsymbol{\Sigma}_{12} = \mathrm{Cov}(\boldsymbol{\varepsilon}_{1,s},\ \boldsymbol{\varepsilon}_{2,s})$ represents the *residual* cross-margin covariance after removing the observed policy effects via $\boldsymbol{\Gamma}_1 \mathbf{v}_s$ and $\boldsymbol{\Gamma}_2 \mathbf{v}_s$. The *marginal* cross-margin relationship between participation and intensity includes both this residual channel and the direct policy-moderated channel through $\boldsymbol{\Gamma}_k$. Specifically,

$$\mathrm{Cov}(\boldsymbol{\delta}_{1,s},\ \boldsymbol{\delta}_{2,s}) = \boldsymbol{\Gamma}_1\ \mathrm{Var}(\mathbf{v}_s)\, \boldsymbol{\Gamma}_2{}^{\intercal} + \boldsymbol{\Sigma}_{12},$$

so a finding of $\boldsymbol{\Sigma}_{12} \approx \mathbf{0}$ does not imply that the two margins are unrelated—only that policies fully account for their cross-margin covariation.

Proposition B.26 (Well-concentration of $\boldsymbol{\Sigma}_{12}$)**.**
The posterior for $\boldsymbol{\Sigma}_{12}$ is well-concentrated when the following conditions hold:

(i) *$S > 2q + 1$, so that the sample covariance of the $\{\boldsymbol{\varepsilon}_s\}$ has a positive-definite posterior mode;*

(ii) *the individual state effects $\boldsymbol{\varepsilon}_{1,s}$ and $\boldsymbol{\varepsilon}_{2,s}$ are themselves well-estimated, which requires the state-level replication condition* (C2)*; and*

(iii) *the* LKJ(η) *prior (Lewandowski et al., 2009) on the correlation matrix $\mathbf{R}_\varepsilon$ with $\eta \geq 1$ provides regularization, shrinking the posterior toward independence when the data are weakly informative about cross-margin dependence.*

For our model with $q = 5$: the requirement $S > 2q + 1 = 11$ is amply satisfied by $S = 51$. The posterior standard deviation of each element of $\boldsymbol{\Sigma}_{12}$ decreases at rate $O(S^{-1/2}) \approx 0.14$.

Remark B.20 (Frequentist versus Bayesian identification)**.** The identification result in Theorem B.2 is stated for the posterior, not the frequentist maximum likelihood estimator. Under the Bayesian framework, identification means the posterior concentrates on the true parameter as $N \to \infty$ (cf. the Bernstein–von Mises theorem

under misspecification; Kleijn & van der Vaart, 2012), which requires both *likelihood identification* (the sampling model pins down $\boldsymbol{\theta}$) and *prior identification* (the prior does not place mass on non-identified subspaces). Steps 1 and 3 establish likelihood identification of the total state coefficients $\{\tilde{\boldsymbol{\alpha}}_s\}$, $\{\tilde{\boldsymbol{\beta}}_s\}$, and κ; Steps 2 and 4 show that the hierarchical centering convention $\mathbb{E}[\boldsymbol{\varepsilon}_{k,s}] = \mathbf{0}$ resolves the additive shift ambiguity and identifies the individual components $(\boldsymbol{\alpha}, \boldsymbol{\beta}, \boldsymbol{\delta}_s, \boldsymbol{\Gamma}_k)$; Step 5 establishes likelihood identification for the within-margin blocks $\boldsymbol{\Sigma}_{\varepsilon,1}$ and $\boldsymbol{\Sigma}_{\varepsilon,2}$.

The cross-margin block $\boldsymbol{\Sigma}_{12}$ is the exception: Proposition B.25 shows that it receives zero *conditional* likelihood information. As discussed in Remark 1, the marginal likelihood obtained by integrating over $\boldsymbol{\delta}_s$ does couple the margins and in principle identifies $\boldsymbol{\Sigma}_{12}$, but the conditional formulation adopted here is natural for MCMC computation. In either framing, the LKJ(η) prior provides essential regularization: frequentist REML estimation of a $2q \times 2q$ covariance from $S = 51$ groups is numerically fragile (boundary solutions, singular Hessians). This is not a deficiency of the model but rather a structural consequence of the hurdle factorization (Proposition B.22). The posterior for $\boldsymbol{\Sigma}_{12}$ concentrates at rate $O(S^{-1/2})$—determined by the $S = 51$ state-level observations—rather than at the parametric rate $O(N^{-1/2})$ available to the remaining parameters. This slower concentration rate is the price of learning about the cross-margin dependence structure, which we view as justified by the substantive importance of understanding whether states that restrict access also intensify conditional enrollment.

Appendix C. Survey-Weighted Inference

This appendix develops the survey design theory underlying the sandwich variance estimator and the Cholesky affine transformation described in Section 3. We begin with the formal pseudo-likelihood framework and the proof that the pseudo-posterior is proper (Theorem 2), establish the Bernstein–von Mises conditions and their finite-sample diagnostic for the NSECE weights, derive the cluster-robust sandwich estimator and its block structure, provide the full proof of the Cholesky affine transformation (Theorem 3), and close with the design effect ratio classification and diagnostics. Throughout, we use the score functions and Fisher information results from Section B (Supplementary Material B) and the notation established in Section 3.

C.1. **Pseudo-likelihood and separability.** We summarize the pseudo-likelihood structure from Section 3.5. By Proposition 6, the decomposition $\ell^{(w)} = \ell^{(w)}_{\text{ext}} + \ell^{(w)}_{\text{int}}$ holds with weights $\tilde{w}_i = w_i N / \sum_j w_j$. We record three clarifying remarks.

Remark C.21 (Weight normalization)**.** Two normalization conventions appear in the survey literature. The *sum-to-N* convention, $\tilde{w}_i = w_i N / \sum_j w_j$ so that $\sum_i \tilde{w}_i = N$, is adopted throughout and is standard in the Bayesian pseudo-posterior literature (Savitsky & Toth, 2016; Williams & Savitsky, 2021). The alternative *sum-to-$\hat{M}$* convention retains unnormalized weights $\tilde{w}_i = w_i$, which for the NSECE gives $\sum_i w_i \approx 119{,}593$. Under this convention the pseudo-likelihood resembles roughly 119,593 observations rather than $N = 6{,}809$, leading to extreme overconcentration of the pseudo-posterior and credible intervals that are far too narrow. The sum-to-N normalization ensures that the pseudo-likelihood is comparable in scale to an ordinary likelihood based on N observations.

Remark C.22 (Weighted score functions)**.** The observation-level score functions for the extensive and intensive margins, along with their Fisher information and sign verification, are developed in Section B.1 (Supplementary Material B). The weighted versions used here simply multiply each per-observation score by $\tilde{w}_i$; the separability of Proposition 6 guarantees that the weighted extensive-margin scores depend only on $(\boldsymbol{\alpha}, \boldsymbol{\delta}_{1,\cdot})$ and the weighted intensive-margin scores depend only on $(\boldsymbol{\beta}, \boldsymbol{\delta}_{2,\cdot}, \kappa)$.

Remark C.23 (Significance of separability under weighting)**.** The separability result in Proposition 6 is not guaranteed for all survey adjustments. Approaches that modify the likelihood structure—such as multilevel weight smoothing (Rabe-Hesketh & Skrondal, 2006) or conditional likelihood methods—can break the additive structure. Observation-level exponentiation $[f_i]^{\tilde{w}_i}$ preserves it precisely because $\tilde{w}_i$ is a scalar multiplier in the log scale and does not couple the two parameter blocks. This separability is exploited in the sandwich estimator (Section 3.5), where the block-diagonality of $\mathbf{H}_{\text{obs}}$ follows directly from the additive decomposition.

C.2. **Pseudo-posterior propriety.** We now provide the complete proof of Theorem 2, which was stated with a sketch in Section 3.5.

Proof of Theorem 2. We verify that the pseudo-posterior

$$\pi^{(w)}(\boldsymbol{\theta} \mid \mathbf{y}) \propto \exp\left[\ell^{(w)}(\boldsymbol{\theta})\right] \pi(\boldsymbol{\theta})$$

is integrable over the parameter space.

Step 1 (Boundedness of the pseudo-likelihood). For each observation i, the HBB probability satisfies $f_{\text{HBB}}(y_i \mid \boldsymbol{\theta}) \in (0, 1]$ for all $\boldsymbol{\theta}$ in the interior of the parameter

space. Since $\tilde{w}_i > 0$, we have $f_i^{\tilde{w}_i} \leq 1$ whenever $f_i \leq 1$. Therefore

$$\exp\big[\ell^{(w)}(\boldsymbol{\theta})\big] = \prod_{i=1}^{N} f_i^{\tilde{w}_i} \leq 1 \qquad \text{for all } \boldsymbol{\theta}.$$

Step 2 (Properness of the prior). Each component of the prior $\pi(\boldsymbol{\theta})$ is a proper distribution:

- $\boldsymbol{\alpha}, \boldsymbol{\beta} \sim \mathsf{N}(\mathbf{0},\ 2^2\,\mathbf{I}_P)$ (Gelman et al., 2008): proper with finite variance;
- $\boldsymbol{\tau} \sim$ Half-Normal$(0, 1)$ componentwise: proper on $\mathbb{R}_{>0}$;
- $\mathbf{R}_\varepsilon \sim$ LKJ(2) (Lewandowski et al., 2009): proper over the space of $2q \times 2q$ correlation matrices;
- $\log \kappa \sim \mathsf{N}(2,\ 1.5^2)$: proper.

The joint prior is a product of proper distributions, hence proper: $\int \pi(\boldsymbol{\theta})\, d\boldsymbol{\theta} = 1$.

Step 3 (Integrability). Combining Steps 1 and 2:

$$\int \exp\big[\ell^{(w)}(\boldsymbol{\theta})\big]\, \pi(\boldsymbol{\theta})\, d\boldsymbol{\theta} \;\leq\; \int 1 \cdot \pi(\boldsymbol{\theta})\, d\boldsymbol{\theta} = 1 < \infty.$$

Moreover, $\exp[\ell^{(w)}(\boldsymbol{\theta})] > 0$ on the interior of the parameter space (since all probability terms f_i are strictly positive there), so the pseudo-posterior is non-degenerate. □

Remark C.24 (Weight-invariance of propriety)**.** The proof above does not depend on the specific values or normalization of the weights—only on their positivity and finiteness. Propriety therefore holds regardless of whether the weights are normalized to sum to N, to sum to the population size $\hat{M}$, or under any other positive scaling. However, the *shape* and *concentration* of the pseudo-posterior depend critically on the normalization convention, as discussed in Remark C.21: the sum-to-$\hat{M}$ convention produces extreme overconcentration, while the sum-to-N convention preserves the information content of the actual sample.

Remark C.25 (Contrast with improper priors)**.** With improper (flat) priors on the regression coefficients, propriety would require the pseudo-likelihood to be integrable on its own—that is, $\int \exp[\ell^{(w)}(\boldsymbol{\theta})]\, d\boldsymbol{\theta} < \infty$. In this setting the effective sample size (not the nominal sample size) governs integrability. With ESS $\approx$ 1,797 and $\dim(\boldsymbol{\theta}) \leq 606$, integrability would likely hold, but the formal verification is more delicate: it requires confirming that the weighted Hessian has full rank at the pseudo-MLE for every parameter block. The use of proper priors sidesteps this issue entirely, ensuring propriety for *any* positive weight configuration without conditions on the data.

C.3. **Bernstein–von Mises approximation.** The sandwich correction in Section 3.5 rests on a Bernstein–von Mises (BvM) theorem for pseudo-posteriors: as $N \to \infty$, the pseudo-posterior converges to a normal distribution centered at the pseudo-MLE with variance given by the inverse observed Hessian, $\mathbf{H}_{\text{obs}}^{-1}$. We state the precise result, verify its regularity conditions for the HBB model, and provide a finite-sample diagnostic for the NSECE weight configuration that motivates the sandwich correction as a robustness measure.

Theorem C.3 (Design-consistent BvM for the pseudo-posterior)**.** *Suppose the following regularity conditions hold*:

(C1) Smoothness. *The log-density* $\log f_{\text{HBB}}(y \mid \boldsymbol{\theta})$ *is twice continuously differentiable in* $\boldsymbol{\theta}$ *on the interior of the parameter space* Θ.

(C2) Interior parameter. *The true superpopulation parameter* $\boldsymbol{\theta}_0$ *lies in the interior of* Θ.

(C3) Weight regularity. *The ratio of the largest to smallest normalized weight satisfies* $\max_i \tilde{w}_i / \min_i \tilde{w}_i = O(N^\delta)$ *for some* $\delta < 1/2$ *as* $N \to \infty$.

(C4) Information regularity. *The weighted Hessian* $\mathbf{H}_{\text{obs}}(\boldsymbol{\theta}_0)$ *is positive definite, with eigenvalues bounded away from zero and infinity.*

(C5) Prior regularity. *The prior density* $\pi(\boldsymbol{\theta})$ *is continuous and strictly positive in a neighborhood of* $\boldsymbol{\theta}_0$.

Then, as $N \to \infty$,

$$\sup_{B \in \mathcal{B}} \left| \pi^{(w)}(\boldsymbol{\theta} \in B \mid \mathbf{y}) - \Phi_{\hat{\boldsymbol{\theta}}^{(w)}, \mathbf{H}_{\text{obs}}^{-1}}(B) \right| \xrightarrow{P_{\boldsymbol{\theta}_0}} 0, \tag{C.1}$$

where $\mathcal{B}$ *is the class of Borel-measurable convex sets,* $\hat{\boldsymbol{\theta}}^{(w)}$ *is the pseudo-MLE,* $\Phi_{\boldsymbol{\mu},\boldsymbol{\Sigma}}$ *denotes the CDF of* $\mathsf{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$*, and* $\mathbf{H}_{\text{obs}}$ *is the observed Hessian of the pseudo-log-likelihood evaluated at* $\hat{\boldsymbol{\theta}}^{(w)}$.

The result is a pseudo-posterior analogue of the classical BvM theorem (Kleijn & van der Vaart, 2012), extended to the survey-weighted setting by Savitsky and Toth (2016) and Williams and Savitsky (2021). The key difference from the standard BvM is condition (C3), which constrains how rapidly the weight ratio may grow with sample size. Importantly, the *pseudo-posterior variance* $\mathbf{H}_{\text{obs}}^{-1}$ differs from the *design-consistent sampling variance* of $\hat{\boldsymbol{\theta}}^{(w)}$,

$$\hat{\boldsymbol{\theta}}^{(w)} \dot{\sim} \mathsf{N}\big(\boldsymbol{\theta}_0,\ \mathbf{V}_{\text{sand}}\big), \qquad \mathbf{V}_{\text{sand}} = \mathbf{H}_{\text{obs}}^{-1}\, \mathbf{J}_{\text{cluster}}\, \mathbf{H}_{\text{obs}}^{-1},$$

as defined in (3.25). Under informative sampling or model misspecification, $\mathbf{J}_{\text{cluster}} \neq \mathbf{H}_{\text{obs}}$, so naive credible intervals based on $\mathbf{H}_{\text{obs}}^{-1}$ are miscalibrated as a measure of

frequentist uncertainty. This mismatch is precisely what motivates the sandwich correction developed in Section C.4: Corollary C.5 below quantifies the resulting coverage distortion, and Theorem 3 provides the operational calibration.

Verification. We verify each condition for the HBB model and the NSECE data.

(C1)*: Smoothness.* The HBB log-density involves $\log\Gamma$ functions (through the beta-binomial PMF), the logistic link $\sigma(\eta) = 1/(1+e^{-\eta})$, and the zero probability p_0 (a finite sum of $\log\Gamma$ terms). Each component is C^∞ on the interior of the parameter space, so the composite $\log f_{\text{HBB}}$ is smooth in $\boldsymbol{\theta}$. ✓

(C2)*: Interior parameter.* The parameter space is $\Theta = \mathbb{R}^P \times \mathbb{R}^P \times \mathbb{R}^{2QR} \times \mathbb{S}^{2Q}_{++} \times \mathbb{R}_{>0}$, and any finite parameter vector with $\kappa > 0$ and $\boldsymbol{\Sigma}_\varepsilon \succ 0$ lies in the interior. ✓

(C3)*: Weight regularity.* This condition requires a dedicated analysis; see Proposition C.27 below.

(C4)*: Information regularity.* By Proposition B.22, the Fisher information is block-diagonal between the extensive and intensive margins. For the extensive block, positive definiteness follows from $\text{rank}(\mathbf{X}) = P$ and $q_i \in (0,1)$ for all i (standard logistic regression theory). For the intensive block, separation of μ from κ requires variation in the denominators n_i (Proposition B.24), which is satisfied empirically (n_i ranges from 1 to 378 among positive responders). ✓

(C5)*: Prior regularity.* The priors specified in Section 3.6 are all continuous and strictly positive in the interior: $\mathsf{N}(\mathbf{0},\, 2^2\,\mathbf{I}_P)$ on regression coefficients (Gelman et al., 2008), Half-Normal$(0,1)$ on scale parameters, LKJ(2) on the correlation matrix (Lewandowski et al., 2009), and $\mathsf{N}(2,\, 1.5^2)$ on $\log\kappa$. ✓

Proposition C.27 (Finite-sample weight diagnostic)**.** *For the NSECE 2019 data with $N = 6{,}785$ and $w_{\max}/w_{\min} = 462$, the weight ratio exceeds $N^{1/2} \approx 82.4$, indicating that the sufficient condition (C3) is not comfortably satisfied for this design. The computation*

$$\delta = \frac{\log 462}{\log 6{,}785} = \frac{6.136}{8.822} \approx 0.695 > 0.5 \tag{C.2}$$

is a finite-sample diagnostic, not a valid asymptotic big-O test of condition (C3). This motivates the sandwich variance correction as a robustness measure.

Remark C.26 (Interpretation of the diagnostic)**.** Four observations clarify the scope of Proposition C.27.

(a) Condition (C3) is *sufficient*, not necessary. The BvM approximation may remain adequate even when this bound is not met, provided the high-weight observations do not dominate the pseudo-likelihood.

(b) The fraction of observations with extreme weights is small: the top 1% (weights $w_i > 100$) comprises fewer than 70 observations out of 6,785. The diagnostic is driven by a small number of outlying weights, not a systematic pattern.

(c) Weight trimming can restore (C3). Capping at the 99th percentile reduces $w_{\max}$ to approximately 100, giving $\delta \approx 0.522$; capping at the 95th percentile ($w_{\max} \approx 50$) yields $\delta \approx 0.443 < 0.5$, restoring (C3). Weight trimming is examined as a sensitivity analysis in Section D.

(d) In practice, the sandwich correction (3.25) provides the appropriate variance adjustment regardless of whether (C3) holds exactly: it automatically accounts for the inflated variance due to extreme weights.

Corollary C.5 (Coverage miscalibration of naive credible intervals)**.** *Under the conditions of Theorem C.3, define*:

- $\mathrm{CI}_{\mathrm{naive}}(\alpha)$: *the α-level credible interval from the pseudo-posterior marginal, with variance* $[\mathbf{H}_{\mathrm{obs}}]^{-1}$;
- $\mathrm{CI}_{\mathrm{sand}}(\alpha) = \hat{\theta}^{(w)} \pm z_{\alpha/2}\sqrt{V_{\mathrm{sand}}}$: *the sandwich-corrected interval.*

Then:

(a) *The naive interval has asymptotic coverage* $\Phi(z_{\alpha/2}\, r) - \Phi(-z_{\alpha/2}\, r)$, *where* $r = \sqrt{\lambda_{\mathbf{H}}/\lambda_{\mathrm{sand}}}$ *is the ratio of the naive to the sandwich standard deviation and* Φ *denotes the standard normal CDF.*

(b) *When the sampling is non-informative and the model is correctly specified,* $\mathbf{J}_{\mathrm{cluster}} \approx \mathbf{H}_{\mathrm{obs}}$, *so* $r \approx 1$ *and the naive interval has approximately correct coverage.*

(c) *Under informative sampling or model misspecification,* $r \neq 1$. *If* $r < 1$ *(naive variance too small), the interval is anti-conservative (undercoverage)—the typical case when the pseudo-posterior overconcentrates. If* $r > 1$, *the interval is conservative.*

Proof. The BvM theorem (Theorem C.3) implies that the pseudo-posterior converges to $\mathsf{N}(\hat{\boldsymbol{\theta}}^{(w)}, \mathbf{H}_{\mathrm{obs}}^{-1})$, while the correct sampling distribution of $\hat{\boldsymbol{\theta}}^{(w)}$ is approximately $\mathsf{N}(\boldsymbol{\theta}_0, \mathbf{V}_{\mathrm{sand}})$. The ratio of marginal standard deviations determines the coverage distortion. □

Remark C.27 (Calibration via the Cholesky transformation)**.** Theorem 3 provides the operational calibration strategy: the affine transformation (3.26) rescales each MCMC draw so that the adjusted posterior has covariance $\mathbf{V}_{\text{sand}}$ while preserving the posterior mean. This replaces the naive variance $\mathbf{H}_{\text{obs}}^{-1}$ with the design-consistent sandwich variance, correcting the coverage distortion quantified in Corollary C.5. The block-wise implementation—applying the transformation separately to fixed effects, random effects, and hyperparameters—is discussed in Section 3.5; the full proof appears in Section C.5 below.

C.4. **Sandwich variance estimator.** The main text (Section 3.5) defines the bread matrix $\mathbf{H}_{\text{obs}}$ (3.23), the cluster-robust meat matrix $\mathbf{J}_{\text{cluster}}$ (3.24), and the sandwich variance $\mathbf{V}_{\text{sand}}$ (3.25). This subsection provides the detailed derivation: the block-diagonal structure of $\mathbf{H}_{\text{obs}}$, the non-block-diagonal structure of $\mathbf{J}_{\text{cluster}}$, the cluster-robust aggregation formula with stratification (Binder, 1983), and the partial absorption of within-state clustering by the state-varying coefficients. Throughout, we use the score functions from Section B.1 and the sandwich convention defined below.

Bread matrix. The bread matrix is the negative weighted observed Hessian at the posterior mean $\hat{\boldsymbol{\theta}}$,

$$\mathbf{H}_{\text{obs}} = -\sum_{i=1}^{N} \tilde{w}_i \left. \frac{\partial^2 \log f_{\text{HBB}}(y_i \mid \boldsymbol{\theta})}{\partial \boldsymbol{\theta}\, \partial \boldsymbol{\theta}_\intercal} \right|_{\boldsymbol{\theta}=\hat{\boldsymbol{\theta}}}. \tag{C.3}$$

By the separability of the hurdle log-likelihood (Remark C.23) and the block-diagonal Fisher information (Proposition B.22), the bread inherits a two-block structure:

Proposition C.28 (Block-diagonal bread matrix)**.** *The observed Hessian decomposes as*

$$\mathbf{H}_{\text{obs}} = \begin{pmatrix} \mathbf{H}_{\text{ext}} & \mathbf{0} \\ \mathbf{0} & \mathbf{H}_{\text{int}} \end{pmatrix}, \tag{C.4}$$

where:

(i) *The extensive block is the weighted logistic information*

$$\mathbf{H}_{\text{ext}} = \sum_{i=1}^{N} \tilde{w}_i\, q_i(1-q_i)\, \mathbf{d}_{1,i}\, \mathbf{d}_{1,i}{}_\intercal, \tag{C.5}$$

with $\mathbf{d}_{1,i} = \partial \eta_i^{(1)} / \partial(\boldsymbol{\alpha}_\intercal, \boldsymbol{\delta}_{1,s[i]}{}_\intercal)_\intercal$. *This matrix is non-stochastic: it depends on* $q_i = q_i(\hat{\boldsymbol{\theta}})$ *but not on the observed data* z_i.

(ii) *The intensive block is the weighted negative Hessian of the truncated beta-binomial log-likelihood,*

$$\mathbf{H}_{\text{int}} = -\sum_{i \in \mathcal{I}_1} \tilde{w}_i \, \frac{\partial^2 \ell_i^{(2+)}}{\partial \boldsymbol{\phi}_2 \, \partial \boldsymbol{\phi}_2 \intercal} \bigg|_{\hat{\boldsymbol{\theta}}}, \tag{C.6}$$

where $\boldsymbol{\phi}_2 = (\boldsymbol{\beta}\intercal, \log \kappa, \boldsymbol{\delta}_{2,s[i]}\intercal)\intercal$, $\mathcal{I}_1 = \{i : y_i > 0\}$, *and* $\ell_i^{(2+)} = \log f_{\text{BB}}(y_i \mid n_i, \mu_i, \kappa) - \log(1 - p_{0,i})$. *This block depends on the observed counts* y_i.

Proof. By Proposition B.22, the cross-derivative $\partial^2 \ell_i / \partial \boldsymbol{\phi}_1 \, \partial \boldsymbol{\phi}_2 \intercal = \mathbf{0}$ for every observation i. Multiplying by $\tilde{w}_i > 0$ and summing preserves this zero block, giving (C.4). The extensive-margin form (C.5) follows from $-\partial^2 \ell_i^{(1)} / \partial (\eta_i^{(1)})^2 = q_i(1 - q_i)$ (Proposition B.23) and the chain rule. □

Meat matrix and cross-margin coupling. The individual-level meat matrix is

$$\mathbf{J}_{\text{ind}} = \sum_{i=1}^{N} \tilde{w}_i^2 \, \mathbf{s}_i \, \mathbf{s}_i \intercal, \tag{C.7}$$

where $\mathbf{s}_i = \partial \log f_{\text{HBB}}(y_i \mid \boldsymbol{\theta}) / \partial \boldsymbol{\theta} \big|_{\hat{\boldsymbol{\theta}}}$ is the observation-level score.

Remark C.28 (Non-block-diagonal meat)**.** Unlike the bread, the meat matrix is *not* block-diagonal between the extensive and intensive margins. The cross-block is

$$\mathbf{J}^{\text{ext,int}} = \sum_{i=1}^{N} \tilde{w}_i^2 \, \mathbf{s}_i^{\text{ext}} \, (\mathbf{s}_i^{\text{int}}) \intercal \, . \tag{C.8}$$

For $y_i = 0$, the intensive score $\mathbf{s}_i^{\text{int}} = \mathbf{0}$, so the cross-product vanishes. But for $y_i > 0$, both $\mathbf{s}_i^{\text{ext}} = (1 - q_i)\mathbf{d}_{1,i}$ and $\mathbf{s}_i^{\text{int}} \neq \mathbf{0}$, so the cross-block receives a nonzero contribution from every positive observation. In consequence, the sandwich variance $\mathbf{V}_{\text{sand}} = \mathbf{H}_{\text{obs}}^{-1} \mathbf{J} \mathbf{H}_{\text{obs}}^{-1}$ is not block-diagonal even though $\mathbf{H}_{\text{obs}}$ is. This cross-margin coupling is a general feature of sandwich estimators in hurdle models: a single PSU contributes weighted scores from both margins simultaneously.

Remark C.29 (Reduction under equal weights)**.** Under uniform weights $\tilde{w}_i = 1$ and correct model specification, the information identity gives $\mathbb{E}[\mathbf{J}_{\text{ind}}] = \mathbf{H}_{\text{obs}}$ (see Williams & Savitsky, 2021, for the pseudo-posterior analogue), so $\mathbf{V}_{\text{sand}} \to \mathbf{H}_{\text{obs}}^{-1}$ and the sandwich reduces to the model-based variance. For the extensive margin, $\mathbb{E}[\mathbf{J}_{\text{ext}}] = \mathbf{H}_{\text{ext}}$ holds exactly, because the logistic Hessian is non-random (see Remark B.12; under unequal survey weights the identity breaks because $\mathbf{H}$ involves $\tilde{w}_i$ while $\mathbb{E}[\mathbf{J}]$

involves $\tilde{w}_i^2$). For the intensive margin, the identity holds only in expectation and the finite-sample discrepancy between $\mathbf{J}_{\text{int}}$ and $\mathbf{H}_{\text{int}}$ is nonzero.

Cluster-robust aggregation. The individual-level $\mathbf{J}_{\text{ind}}$ treats observations as independent. In the NSECE stratified cluster design (National Survey of Early Care and Education Project Team, 2022), providers within the same primary sampling unit (PSU) share neighborhood characteristics, inducing within-PSU correlation. The cluster-robust meat matrix replaces individual scores with PSU-level aggregates, centered within strata.

Definition C.3 (PSU-level score totals)**.** For stratum $h = 1, \ldots, H$ and PSU $c = 1, \ldots, C_h$ within stratum h, define

$$\bar{\mathbf{s}}_{hc} = \sum_{i \in \text{PSU}(h,c)} \tilde{w}_i \, \mathbf{s}_i, \tag{C.9}$$

where $\text{PSU}(h, c) = \{i : \texttt{vstratum}[i] = h,\ \texttt{vpsu}[i] = c\}$. The stratum mean is

$$\bar{\mathbf{s}}_h = \frac{1}{C_h} \sum_{c=1}^{C_h} \bar{\mathbf{s}}_{hc}. \tag{C.10}$$

Proposition C.29 (Cluster-robust meat matrix)**.** *Under the stratified cluster design with H strata and C_h PSUs in stratum h, the cluster-robust meat matrix is*

$$\boxed{\mathbf{J}_{\text{cluster}} = \sum_{h=1}^{H} \frac{C_h}{C_h - 1} \sum_{c=1}^{C_h} (\bar{\mathbf{s}}_{hc} - \bar{\mathbf{s}}_h)\,(\bar{\mathbf{s}}_{hc} - \bar{\mathbf{s}}_h)^{\intercal}\,.} \tag{C.11}$$

This estimator accounts for: (i) *within-PSU correlation, by aggregating scores to PSU totals*; (ii) *stratification, by centering PSU scores within strata so that only within-stratum variation across PSUs contributes*; (iii) *unequal cluster sizes, through the weight-aggregated PSU totals* $\bar{\mathbf{s}}_{hc}$; (iv) *finite-sample degrees of freedom, via the factor* $C_h/(C_h - 1)$.

Proof. The design-based variance of the weighted score total under stratified sampling decomposes as

$$\operatorname*{Var}_{\text{design}}\Big(\sum_i \tilde{w}_i \, \mathbf{s}_i\Big) = \sum_{h=1}^{H} \operatorname*{Var}_{h}\Big(\sum_{c=1}^{C_h} \bar{\mathbf{s}}_{hc}\Big),$$

because sampling across strata is independent. Within stratum h, PSUs are sampled without replacement from the population of C_h^* PSUs. Under the "ultimate cluster" assumption that $C_h/C_h^* \approx 0$ (negligible PSU sampling fraction), the finite-population

correction is dropped and the within-stratum variance estimator is

$$\widehat{\mathrm{Var}}_h = \frac{C_h}{C_h - 1} \sum_{c=1}^{C_h} (\bar{\mathbf{s}}_{hc} - \bar{\mathbf{s}}_h)\,(\bar{\mathbf{s}}_{hc} - \bar{\mathbf{s}}_h)^\intercal,$$

the standard unbiased estimator of the variance of a total from C_h units. Summing over strata gives $\mathbf{J}_{\text{cluster}}$. □

Remark C.30 (Degrees of freedom)**.** The cluster-robust estimator has $\sum_{h=1}^{H}(C_h - 1) = 415 - 30 = 385$ effective degrees of freedom. With C_h ranging from 5 to 36 (median ≈ 14), each stratum contributes at least 4 degrees of freedom. This comfortably exceeds the rule-of-thumb threshold of $\sum_h (C_h - 1) > 30$ for stable sandwich variance estimation.

Remark C.31 (Individual versus cluster-level $\mathbf{J}$)**.** When there is no within-PSU correlation conditional on covariates, $\mathbf{J}_{\text{cluster}}$ and $\mathbf{J}_{\text{ind}}$ converge to the same population quantity. In practice, within-PSU correlation inflates $\mathbf{J}_{\text{cluster}}$ relative to $\mathbf{J}_{\text{ind}}$; the parameter-specific ratio $\mathrm{diag}(\mathbf{V}_{\text{sand}})/\,\mathrm{diag}(\mathbf{H}_{\text{obs}}^{-1}\mathbf{J}_{\text{ind}}\mathbf{H}_{\text{obs}}^{-1})$ provides an estimate of the cluster design effect. The design effect ratio (DER) classification and diagnostics are developed in Section C.6 below.

Proposition C.30 (Partial absorption of clustering by SVCs)**.** *The state-varying coefficients $\boldsymbol{\varepsilon}_s$ partially absorb within-state clustering. For any two providers i, j in the same state s and the same PSU,*

$$\mathbb{E}_{\boldsymbol{\varepsilon}_s}\big[\mathrm{Cov}(s_i,\, s_j \mid \boldsymbol{\varepsilon}_s)\big] \;\le\; \mathrm{Cov}(s_i,\, s_j), \tag{C.12}$$

with equality if and only if $\mathrm{Cov}(\mathbb{E}[s_i \mid \boldsymbol{\varepsilon}_s],\, \mathbb{E}[s_j \mid \boldsymbol{\varepsilon}_s]) = 0$.

Proof. By the law of total covariance,

$$\mathrm{Cov}(s_i,\, s_j) = \mathbb{E}\big[\mathrm{Cov}(s_i,\, s_j \mid \boldsymbol{\varepsilon}_s)\big] + \mathrm{Cov}\big(\mathbb{E}[s_i \mid \boldsymbol{\varepsilon}_s],\, \mathbb{E}[s_j \mid \boldsymbol{\varepsilon}_s]\big).$$

The second term is non-negative, so $\mathrm{Cov}(s_i, s_j) \ge \mathbb{E}[\mathrm{Cov}(s_i, s_j \mid \boldsymbol{\varepsilon}_s)]$. At the posterior mean $\hat{\boldsymbol{\theta}}$ (which includes state-specific $\hat{\boldsymbol{\varepsilon}}_s$), the plug-in conditional covariance is an estimate of $\mathrm{Cov}(s_i, s_j \mid \boldsymbol{\varepsilon}_s)$; hence $\mathbf{J}_{\text{cluster}}$ evaluated at $\hat{\boldsymbol{\theta}}$ will be smaller than under a model without SVCs. □

Remark C.32 (Residual clustering)**.** The inequality (C.12) shows that SVCs reduce but do not eliminate within-PSU correlation. The residual correlation reflects neighborhood-level effects not captured by the five state-varying covariates (poverty,

race/ethnicity, urbanicity): local labor market conditions, municipal zoning, co-located service providers, and other spatially structured unobservables. The sandwich correction with cluster-robust $\mathbf{J}_{\text{cluster}}$ remains necessary to account for this residual design effect.

C.5. **Cholesky affine transformation.** The sandwich variance $\mathbf{V}_{\text{sand}}$ (3.25) provides the design-consistent covariance, but the MCMC draws $\{\boldsymbol{\theta}^{(m)}\}_{m=1}^{M}$ from the pseudo-posterior have empirical covariance $\boldsymbol{\Sigma}_{\text{MCMC}} \approx \mathbf{H}_{\text{obs}}^{-1}$, not $\mathbf{V}_{\text{sand}}$. In a frequentist setting one would simply report $\mathbf{V}_{\text{sand}}$ as the variance estimator; in the Bayesian framework, we require the posterior draws themselves to have the correct covariance so that credible intervals are properly calibrated (Williams & Savitsky, 2021). Theorem 3 in the main text states the affine transformation that achieves this. We now provide the full proof.

Proof of Theorem 3. Define centered draws $\mathbf{u}^{(m)} = \boldsymbol{\theta}^{(m)} - \hat{\boldsymbol{\theta}}$, so that $\boldsymbol{\Sigma}_{\text{MCMC}} = M^{-1}\sum_{m=1}^{M}\mathbf{u}^{(m)}(\mathbf{u}^{(m)})^{\intercal}$. Set

$$\mathbf{A} = \mathbf{L}_{\text{sand}}\,\mathbf{L}_{\text{MCMC}}^{-1}, \tag{C.13}$$

where $\mathbf{L}_{\text{MCMC}} = \text{chol}(\boldsymbol{\Sigma}_{\text{MCMC}})$ and $\mathbf{L}_{\text{sand}} = \text{chol}(\mathbf{V}_{\text{sand}})$ are lower-triangular Cholesky factors. The transformation (3.26) is then $\boldsymbol{\theta}^{*(m)} = \hat{\boldsymbol{\theta}} + \mathbf{A}\,\mathbf{u}^{(m)}$.

(a) Mean preservation.

$$\frac{1}{M}\sum_{m=1}^{M}\boldsymbol{\theta}^{*(m)} = \hat{\boldsymbol{\theta}} + \mathbf{A}\left(\frac{1}{M}\sum_{m=1}^{M}\mathbf{u}^{(m)}\right) = \hat{\boldsymbol{\theta}} + \mathbf{A}\cdot\mathbf{0} = \hat{\boldsymbol{\theta}}.$$

(b) Covariance recovery.

$$\begin{aligned}\frac{1}{M}\sum_{m=1}^{M}(\boldsymbol{\theta}^{*(m)} - \hat{\boldsymbol{\theta}})\,(\boldsymbol{\theta}^{*(m)} - \hat{\boldsymbol{\theta}})^{\intercal} &= \mathbf{A}\,\boldsymbol{\Sigma}_{\text{MCMC}}\,\mathbf{A}^{\intercal} \\ &= \mathbf{L}_{\text{sand}}\,\mathbf{L}_{\text{MCMC}}^{-1}\,\mathbf{L}_{\text{MCMC}}\,\mathbf{L}_{\text{MCMC}}^{\intercal}\,\mathbf{L}_{\text{MCMC}}^{-\intercal}\,\mathbf{L}_{\text{sand}}^{\intercal} \\ &= \mathbf{L}_{\text{sand}}\,\mathbf{I}\,\mathbf{L}_{\text{sand}}^{\intercal} = \mathbf{V}_{\text{sand}}.\end{aligned} \tag{C.14}$$

(c) Affine equivariance. The standardized draws $\mathbf{z}^{(m)} = \mathbf{L}_{\text{MCMC}}^{-1}\mathbf{u}^{(m)}$ satisfy

$$M^{-1}\sum_{m}\mathbf{z}^{(m)}(\mathbf{z}^{(m)})^{\intercal} = \mathbf{I}.$$

The transformation maps

$$\boldsymbol{\theta}^{*(m)} - \hat{\boldsymbol{\theta}} = \mathbf{L}_{\text{sand}}\,\mathbf{z}^{(m)},$$

which is a linear rescaling of the standardized draws. The Wald confidence intervals—computed as $\hat{\theta}_p \pm z_{\alpha/2}\sqrt{[\mathbf{V}_{\text{sand}}]_{pp}}$—depend only on the marginal variances of $\boldsymbol{\theta}^*$ and are therefore invariant to the choice of $\mathbf{A}$. Marginal quantiles of the transformed draws may differ from those of the original draws when $\mathbf{A}$ is non-diagonal. □

Remark C.33 (Block-wise application)**.** In practice the transformation (3.26) is applied block-wise rather than to the full ≈616-dimensional vector simultaneously. The three parameter classes are listed below; the numerical Cholesky procedure operates on blocks (i) and (ii) only, since block (iii) receives no correction (Remark 4 in the main text). The rationale follows from Proposition C.28 and Remark C.28:

(i) *Fixed effects* $\boldsymbol{\xi} = (\boldsymbol{\alpha}_\mathsf{T}, \boldsymbol{\beta}_\mathsf{T}, \log\kappa)_\mathsf{T} \in \mathbb{R}^{11}$: full Cholesky correction. The DER ranges from 1.14 to 4.18 (mean 2.11), and the corresponding confidence-interval inflation factors are $\sqrt{\text{DER}} \in [1.07, 2.04]$.

(ii) *State random effects* $\boldsymbol{\varepsilon}_s \in \mathbb{R}^{10}$, $s = 1, \ldots, S$: partial correction. The posterior variance of $\boldsymbol{\varepsilon}_s$ mixes the likelihood information from providers in state s (affected by survey weights) with the shrinkage toward the prior mean $\boldsymbol{\Gamma}_k \mathbf{v}_s$ (unaffected). When the prior dominates (small states), $\text{DER}_s \approx 1$ and no correction is needed; when the likelihood dominates (large states), DER_s can exceed 3 and the correction is substantial.

(iii) *Hyperparameters* $(\boldsymbol{\Gamma}_k, \boldsymbol{\Sigma}_\varepsilon)$: no correction. The sandwich estimator is not applicable to variance components estimated from between-state variation; the DER concept does not extend to these parameters.

Remark C.34 (Approximation quality of the block-wise correction)**.** The block-wise application ignores the cross-block covariance between fixed effects $\boldsymbol{\xi}$ and random effects $\boldsymbol{\varepsilon}_s$ in $\mathbf{V}_{\text{sand}}$. The approximation is essentially exact when the prior dominates ($\text{ESS}_s \ll q$), since $\boldsymbol{\varepsilon}_s$ is then nearly independent of the data. When data dominate ($\text{ESS}_s \gg q$), the cross-block covariance can be non-negligible; the sensitivity comparison in Section D addresses this case.

Remark C.35 (MCMC versus design-consistent shrinkage)**.** The shrinkage factor $B_s^{(w)} = \boldsymbol{\Sigma}_\varepsilon(\boldsymbol{\Sigma}_\varepsilon + \mathbf{V}_s^{-1})^{-1}$ describes the *design-consistent* (post-sandwich-correction) posterior behavior, not the raw MCMC pseudo-posterior. The MCMC sampler sees precision proportional to $W_s = \sum_{i\in s} \tilde{w}_i$, which exceeds the effective sample size $\text{ESS}_s = W_s^2 / \sum_{i\in s} \tilde{w}_i^2$ whenever within-state weights are heterogeneous. The Cholesky affine transformation (3.26) maps the MCMC output to the design-consistent target, inflating the variance by the factor DER_s (Section C.6). The gap $W_s - \text{ESS}_s$ quantifies

the over-concentration of the raw pseudo-posterior relative to the design-aware target: large gaps (common in states with a few heavily weighted providers) signal that the naive MCMC credible intervals are too narrow and that the sandwich correction is most important.

Remark C.36 (Non-Gaussian posterior shapes)**.** The Cholesky transformation is an affine operation that preserves only first and second moments. If the pseudo-posterior is substantially non-Gaussian (skewed or multimodal), the transformed draws will have the correct covariance but may misrepresent the posterior shape. For the HBB model, the fixed effects $(\boldsymbol{\alpha}, \boldsymbol{\beta})$ are expected to be approximately Gaussian by the Bernstein–von Mises result (Theorem C.3) given $N = 6{,}809$; the log-dispersion $\log \kappa$ is also well-approximated by a normal. For variance components $(\boldsymbol{\Sigma}_\varepsilon)$, the posterior may exhibit right-skewness, but these parameters are not subject to the Cholesky correction (Block 3 above).

Remark C.37 (Positive definiteness requirements)**.** The transformation requires both $\boldsymbol{\Sigma}_{\text{MCMC}}$ and $\mathbf{V}_{\text{sand}}$ to be positive definite so that their Cholesky factors exist. For $\boldsymbol{\Sigma}_{\text{MCMC}}$, positive definiteness holds almost surely when the number of MCMC draws satisfies $M \gg \dim(\boldsymbol{\xi}) = 11$; in our application $M = 4{,}000$ post-warmup draws. For $\mathbf{V}_{\text{sand}}$, positive definiteness requires the effective degrees of freedom to exceed the parameter dimension: $\sum_{h=1}^{H}(C_h - 1) = 386 > 11$ (Remark C.30), which is easily satisfied.

C.6. **Design effect ratio classification and diagnostics.** The sandwich variance $\mathbf{V}_{\text{sand}}$ (3.25) and the Cholesky transformation (3.26) need not be applied uniformly to every parameter. The design effect ratio (DER), defined in the main text (3.27), provides a parameter-specific diagnostic for the magnitude of the design effect:

$$\text{DER}_p = \frac{[\mathbf{V}_{\text{sand}}]_{pp}}{[\mathbf{H}_{\text{obs}}^{-1}]_{pp}}. \tag{C.15}$$

When $\text{DER}_p \approx 1$, the model-based variance already approximates the design-consistent variance and no correction is needed; when DER_p is substantially larger than 1, the Cholesky correction is essential for calibrated inference.

Proposition C.31 (DER classification of parameters)**.** *The parameters of the HBB model fall into three categories with respect to the Cholesky correction*:

(i) Fixed effects $\boldsymbol{\xi} = (\boldsymbol{\alpha}_\mathsf{T}, \boldsymbol{\beta}_\mathsf{T}, \log\kappa)_\mathsf{T}$: *full correction required.* DER_p *reflects the joint effect of unequal weights and within-PSU clustering on the pseudo-likelihood information. Empirically,* $\mathrm{DER}_p \in [1.14, 4.18]$ *(mean* 2.11*), consistent with the Kish DEFF* $= 3.79$.

(ii) State random effects $\boldsymbol{\varepsilon}_s$: *partial correction. The posterior variance of* $\boldsymbol{\varepsilon}_s$ *blends likelihood-based information (affected by the design) with shrinkage toward the prior mean* $\boldsymbol{\Gamma}_k \mathbf{v}_s$ *(unaffected). States in which the prior dominates (*$\mathrm{ESS}_s \ll q$*) have* $\mathrm{DER}_s \approx 1$*; states in which the data dominate (*$\mathrm{ESS}_s \gg q$*) may have* $\mathrm{DER}_s > 3$.

(iii) Hyperparameters $(\boldsymbol{\Gamma}_k, \boldsymbol{\Sigma}_\varepsilon)$: *no correction. The sandwich estimator is not applicable to variance components estimated from between-state variation; the DER concept does not extend to these parameters. Posterior intervals for hyperparameters are therefore model-based summaries whose calibration depends on correct specification of the hierarchical distribution.*

Proof. Part (i) follows because the fixed-effect block of $\mathbf{H}_{\text{obs}}$ and $\mathbf{J}_{\text{cluster}}$ depends directly on the observation-level weights $\tilde{w}_i$ and PSU aggregation, so DER_p is determined by the design effect. Part (ii) follows from the posterior precision decomposition: for the normal–normal conjugate approximation, the posterior precision of $\boldsymbol{\varepsilon}_s$ is the sum of the data precision (design-affected) and the prior precision (design-free). When the prior precision dominates, the posterior variance is insensitive to the design, and $\mathrm{DER}_s \approx 1$. Part (iii) holds because $\boldsymbol{\Gamma}_k$ and $\boldsymbol{\Sigma}_\varepsilon$ are identified from the between-state variation of the $S = 51$ state-level random effects. The survey design affects *within-state* estimation but does not alter the between-state variance structure; hence the sandwich correction has no natural analogue for these parameters. □

Heuristic C.1 (Linear interpolation for random-effect DER). With shrinkage factor $\lambda_{s,pp} = \sigma_\varepsilon^2/(\sigma_\varepsilon^2 + \sigma_e^2/\mathrm{ESS}_s)$, the DER for $\varepsilon_{s,p}$ is approximately

$$\mathrm{DER}_{s,p} \;\approx\; 1 + \lambda_{s,pp}\,(\mathrm{DEFF}_s - 1), \tag{C.16}$$

interpolating between $\mathrm{DER}_{s,p} = 1$ (strong shrinkage) and $\mathrm{DER}_{s,p} = \mathrm{DEFF}_s$ (no shrinkage).

Remark C.38 (Overestimation of the linear heuristic)**. (Correction T2-2.)** The linear interpolation (C.16) is a *heuristic*, not an exact result. In the one-dimensional

normal–normal conjugate model, the exact DER is

$$\mathrm{DER}_{s,p}^{\mathrm{exact}} = \frac{1 + \lambda_{s,pp}\,\mathrm{DEFF}_s}{(1 + \lambda_{s,pp})^2}, \tag{C.17}$$

which is always $\leq$ (C.16). The discrepancy is largest at intermediate shrinkage ($\lambda_{s,pp} \approx 0.5$), where the linear heuristic can overestimate by up to 50%. For practical purposes, the heuristic remains useful as a conservative upper bound that quickly identifies parameters needing correction; the exact formula (C.17) can be computed alongside as a more accurate benchmark.

Proposition C.32 (DER diagnostic thresholds)**.** *As a post-estimation diagnostic, compute* DER_p *for each parameter and classify as follows*:

- $\mathrm{DER}_p \in [0.8,\ 1.2]$: design-insensitive. *The naive credible interval is adequately calibrated; sandwich correction is optional.*
- $\mathrm{DER}_p \in (1.2,\ 1.5]$: moderate inflation. *The confidence-interval width increases by* $\sqrt{1.5} \approx 22\%$*; correction is recommended.*
- $\mathrm{DER}_p > 1.5$: design-sensitive. *The sandwich-corrected variance should be used as the primary inferential summary.*

In the NSECE application, all 11 fixed-effect parameters have $\mathrm{DER}_p > 1.14$*, placing them in the moderate-to-sensitive range.*

Remark C.39 (Intercept identifiability under DER correction)**. (Correction T2-4.)** Because the policy moderator vector $\mathbf{v}_s$ includes an intercept column (see Section 3.4), the fixed intercept α_1 and the first column of $\mathbf{\Gamma}_k$ have partially overlapping roles: the state-specific intercept is $\alpha_1 + \gamma_{k,1,1} + \varepsilon_{k,1,s}$. The DER for α_1 includes variance attributable to both the design effect on the likelihood and the partial confounding with $\gamma_{k,1,1}$. Formally, the decomposition

$$[\mathbf{V}_{\mathrm{sand}}]_{\alpha_1,\alpha_1} = [\mathbf{V}_{\mathrm{sand}}^{\mathrm{design}}]_{\alpha_1,\alpha_1} + [\mathbf{V}^{\mathrm{id}}]_{\alpha_1,\alpha_1} \tag{C.18}$$

is not cleanly separable; the DER should be interpreted with caution for intercept parameters. The soft identification provided by the tighter prior $\mathsf{N}(0, 0.5^2)$ on the intercept column of $\mathbf{\Gamma}_k$ (see Section 3.6) mitigates this issue by attributing the cross-state mean to α_1 and treating deviations as random effects.

Remark C.40 (Computational savings from DER classification)**.** The three-way classification implies that the Cholesky transformation (3.26) need only be applied to the fixed-effect block ($\dim \boldsymbol{\xi} = 11$) and, for the largest states, the random-effect blocks. This reduces the dimensionality of the Cholesky factorization from the full parameter

dimension ($\approx$616) to at most 11–20, yielding substantial computational savings. The empirical DER values are reported in the main text (Table 10).

Appendix D. Extended Empirical Results

This appendix presents extended empirical results from the NSECE 2019 application (Section 5). We provide the full survey weight distribution (Section D.1), formal informativeness assessment following Pfeffermann (1993) (Section D.2), the complete set of state-level poverty coefficients (Section D.3), the full 40-element policy moderation matrix $\mathbf{\Gamma}$ (Section D.4), detailed design effect diagnostics (Section D.5), a choropleth map of reversal probabilities (Section D.6), and an LKJ prior sensitivity analysis (Section D.7). Convergence diagnostics and posterior predictive checks are available in the replication package at https://github.com/joonho112/hurdlebb-replication.

D.1. **Survey design characteristics.** The NSECE 2019 (National Survey of Early Care and Education Project Team, 2022) uses a stratified cluster design with $H = 30$ strata and $C = 415$ primary sampling units (PSUs), yielding $N = 6{,}785$ center-based providers after excluding observations with missing covariates. Survey weights $w_i = 1/\pi_i$ range from 1 to 462 with a mean of 17.5 and a coefficient of variation of 1.66. Figure D.1 displays the distribution of normalized weights $\tilde{w}_i = w_i/\bar{w}$, which are strongly right-skewed: the 99th percentile ($\tilde{w} = 8.8$) is nearly nineteen times the median ($\tilde{w} = 0.47$), indicating that a small number of providers carry disproportionate influence.

The Kish design effect is $\mathrm{DEFF}_{\mathrm{Kish}} = 1 + \mathrm{CV}_w^2 = 3.76$, reducing the effective sample size from $N = 6{,}785$ to $\mathrm{ESS}_{\mathrm{Kish}} \approx 1{,}803$. This substantial penalty motivates the sandwich variance correction and Cholesky calibration described in Section 3.5 and Section C.

D.2. **Informativeness assessment.** A sampling design is *informative* if the distribution of the outcome among sampled units differs from the superpopulation distribution, even after conditioning on covariates (Pfeffermann, 1993). Under informative sampling, unweighted likelihood-based inference targets a sample-specific, not population, quantity, and pseudo-likelihood weighting becomes essential for consistent estimation (Savitsky & Toth, 2016). We assess informativeness of the NSECE 2019 design through three complementary diagnostics.

(Correction T3-1.) The original informativeness tests conditioned on a single covariate (poverty rate). Since the full model includes $P = 5$ covariates per margin,

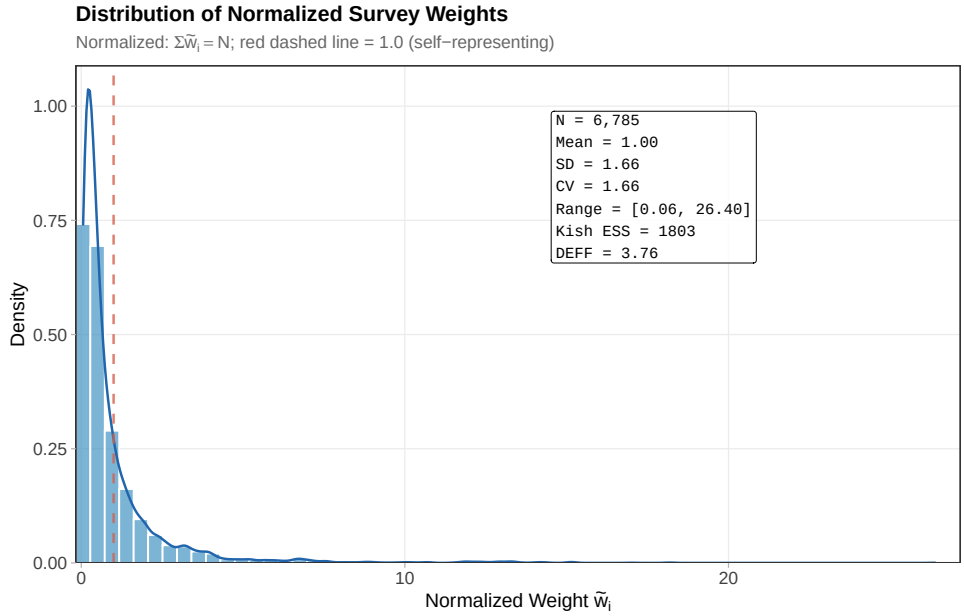

FIGURE D.1. Distribution of normalized survey weights $\tilde{w}_i = w_i/\bar{w}$ for $N = 6{,}785$ center-based providers. The red dashed line marks $\tilde{w} = 1.0$ (self-representing unit). The distribution is strongly right-skewed (CV = 1.66), with Kish DEFF = 3.76 and effective sample size ESS $\approx$ 1,803.

omitted variable bias may mask or inflate apparent informativeness. We now report both the original univariate and the corrected full-covariate specifications.

(Correction T3-2.) The original Hausman comparison used `lm(weights = w)` for the weighted estimator, which provides only heteroscedasticity-robust standard errors. We now use `survey::svyglm()` with the full stratified cluster design, yielding cluster-robust standard errors that account for within-PSU correlation.

Test 1: Marginal weight–outcome correlations. The correlation between z_i (serves infants) and the survey weight w_i is $\mathrm{Corr}(z_i, w_i) = -0.041$; the correlation between the IT share Y_i/n_i (among servers) and w_i is $\mathrm{Corr}(Y_i/n_i, w_i) = -0.003$. Both are negligibly small, suggesting limited *unconditional* informativeness. However, marginal correlations can mask conditional dependence when the design oversamples on covariates correlated with the outcome, motivating the regression-based test below.

Test 2: Regression-based informativeness test. Following Pfeffermann (1993), we test $H_0\colon \gamma = 0$ in the augmented model

$$g(y_i) = \mathbf{x}_i^\intercal \boldsymbol{\beta} + \gamma \log(w_i) + \varepsilon_i, \tag{D.1}$$

where $g(\cdot)$ is the logit link for the extensive margin and the identity for the intensive margin. If the design is non-informative, $\log(w_i)$ carries no additional predictive power for y_i beyond $\mathbf{x}_i$, so $\gamma = 0$.

Table D.1 reports the estimated $\hat{\gamma}$ and its p-value under two covariate specifications: (i) univariate, with only the standardized poverty rate; and (ii) full, with all four community-level predictors (poverty, urbanicity, percent Black, percent Hispanic).

TABLE D.1. Regression-based informativeness test (Pfeffermann, 1993). The augmented model (D.1) adds $\log(w_i)$ to the covariate vector; $\hat{\gamma}$ is the coefficient on $\log(w_i)$ with its two-sided p-value. **(Correction T3-1:)** the full-covariate specification reverses the extensive-margin finding from non-significant to significant.

Specification	Margin	$\hat{\gamma}$	p-value
Univariate (poverty only)	Extensive	0.020	0.369
	Intensive	0.009	0.002
Full covariates (T3-1)	Extensive	0.086	< 0.001
	Intensive	0.006	0.050

Two features of Table D.1 merit comment. First, under the univariate specification, the extensive margin appears non-informative ($p = 0.369$) while the intensive margin is clearly informative ($p = 0.002$). Second, adding the remaining three covariates *reverses* the extensive-margin verdict: $\hat{\gamma}$ increases fourfold (from 0.020 to 0.086) and becomes highly significant ($p < 0.001$). This reversal arises because the univariate model confounds the direct weight–outcome relationship with indirect pathways through the omitted covariates (urbanicity, racial composition). When these confounders are included, the residual association between $\log(w_i)$ and participation becomes detectable. The intensive margin remains informative under both specifications, though the coefficient attenuates slightly (from 0.009 to 0.006) as the full model absorbs some of the weight–outcome association.

The finding that both margins are informative under the full specification reinforces the case for pseudo-likelihood weighting in the Bayesian hierarchical model and for the sandwich variance correction developed in Section C.

Test 3: Hausman-type comparison. As a complementary diagnostic, we compare unweighted and weighted coefficient estimates for each margin. Under non-informative sampling, both estimators are consistent for the same population parameters, but the weighted estimator is less efficient; a large discrepancy signals that the design distorts the effective covariate–outcome relationship.

(Correction T3-2:) the weighted estimator uses `survey::svyglm()` with the full stratified cluster design ($H = 30$ strata, $C = 415$ PSUs), which produces cluster-robust standard errors. The Hausman test statistic for each parameter is

$$Z_k = \frac{\hat{\beta}_k^{\mathrm{WT}} - \hat{\beta}_k^{\mathrm{UW}}}{\sqrt{\widehat{\mathrm{Var}}(\hat{\beta}_k^{\mathrm{WT}}) - \widehat{\mathrm{Var}}(\hat{\beta}_k^{\mathrm{UW}})}}, \tag{D.2}$$

which is asymptotically standard normal under H_0 (no informativeness); the variance difference in the denominator is valid because the weighted estimator has weakly larger variance under non-informativeness.

Table D.2. Hausman-type comparison of unweighted and design-weighted coefficient estimates. **(Correction T3-2:)** the weighted estimator uses `survey::svyglm` with cluster-robust standard errors. Z is the per-parameter Hausman statistic (D.2). Asterisks: $*$ denotes $p < 0.10$, $**$ denotes $p < 0.05$.

Margin	Parameter	$\hat{\beta}^{\mathrm{UW}}$	$\hat{\beta}^{\mathrm{WT}}$	Z	p-value	
Extensive	Intercept	0.614	0.662	0.82	0.41	
	Poverty	−0.158	−0.199	−0.64	0.52	
	Urban	0.176	0.234	2.02	0.044	**
	Black	−0.044	0.086	1.89	0.059	*
	Hispanic	−0.043	0.059	1.51	0.13	
Intensive	*all $p > 0.24$; none significant*					

Table D.2 reveals a nuanced pattern. On the extensive margin, the poverty coefficient itself does not differ significantly between estimators ($Z = -0.64$, $p = 0.52$), but the *urbanicity* coefficient shows a significant shift ($\hat{\beta}^{\mathrm{UW}} = 0.176$ versus $\hat{\beta}^{\mathrm{WT}} = 0.234$; $Z = 2.02$, $p = 0.044$), and the Black composition coefficient is marginally significant ($Z = 1.89$, $p = 0.059$). This indicates that the NSECE oversampling of high-need areas primarily distorts the estimated effect of urbanicity on IT participation—consistent with the fact that the sampling frame explicitly targets urban and high-poverty PSUs. On the intensive margin, no individual parameter shows a significant

Hausman discrepancy (all $p > 0.24$), suggesting that the informativeness detected by Test 2 is diffuse rather than concentrated on a single covariate.

Summary. The three diagnostics paint a consistent picture. Marginal correlations (Test 1) are negligible, but the regression-based test (Test 2) reveals statistically significant informativeness on *both* margins once the full covariate vector is conditioned upon—a finding that was masked in the original univariate specification. The Hausman comparison (Test 3) localises the extensive-margin informativeness to the urbanicity and racial composition covariates rather than poverty itself. Together, these results justify two design choices in the main analysis (Section 5): (i) incorporating survey weights via pseudo-likelihood, and (ii) applying the sandwich variance correction (Section 3.5) and the Cholesky affine calibration (Section C) to ensure that posterior credible intervals reflect the true design-adjusted uncertainty.

D.3. **State-level poverty coefficients.** This subsection summarizes the state-level poverty coefficients from model M2 (block-diagonal SVC without policy moderators) for all $S = 51$ states. The extensive-margin coefficient $\alpha_{\text{pov},s}$ and intensive-margin coefficient $\beta_{\text{pov},s}$ are the state-specific effects of a one-standard-deviation increase in community poverty on the log-odds of IT participation and the logit of IT intensity, respectively. The cross-margin scatter plot in Figure 5 visualizes the joint distribution of these estimates.

Of the 51 states, 48 (94%) have posterior-mean sign patterns consistent with the classic reversal ($\hat{\alpha}_{\text{pov},s} < 0$ and $\hat{\beta}_{\text{pov},s} > 0$); this near-universal pattern reflects hierarchical shrinkage pulling small-state estimates toward the negative population-average extensive-margin coefficient. Among these, 15 (29%) exhibit posterior reversal probabilities exceeding 0.80, indicating strong evidence beyond the point-estimate sign, and an additional 21 states have reversal probabilities between 0.60 and 0.80.

States with strongest reversal evidence. The five large-sample states with the highest reversal probabilities are Texas ($N = 578$, Pr $= 0.989$), Illinois ($N = 471$, Pr $= 0.970$), California ($N = 1{,}110$, Pr $= 0.937$), Louisiana ($N = 77$, Pr $= 0.923$), and New Jersey ($N = 201$, Pr $= 0.908$). In these states, the extensive-margin coefficients ($\alpha_{\text{pov},s} \in [-0.34, -0.13]$) are firmly negative with 90% credible intervals excluding zero, while the intensive-margin coefficients ($\beta_{\text{pov},s} \in [0.07, 0.08]$) are positive with intervals barely including zero.

Notable exception: New York. New York ($N = 558$) stands out as the most prominent exception, with a *positive* extensive-margin coefficient ($\alpha_{\text{pov}} = 0.101$; 90% CI $[-0.051, 0.261]$, Pr(reversal) $= 0.143$), suggesting that higher poverty is associated

with *greater* likelihood of serving infants and toddlers—possibly reflecting New York's substantial public investment in early childhood programs. Arizona (Pr = 0.40), Minnesota (Pr = 0.44), and North Dakota (Pr = 0.48) also show weak or absent reversal patterns.

Small-state shrinkage. Small states (e.g., Maine, $N = 18$; Mississippi, $N = 17$; Montana, $N = 22$) exhibit wider credible intervals and reversal probabilities closer to the national average (≈ 0.73), reflecting the expected shrinkage toward the hierarchical mean documented in Section C. The full 51-state coefficient table is available in the replication package at https://github.com/joonho112/hurdlebb-replication.

D.4. **Full policy moderation parameters.** Table D.3 presents the complete set of 40 elements of the cross-level moderation matrices $\mathbf{\Gamma}_1$ (extensive margin) and $\mathbf{\Gamma}_2$ (intensive margin) from the weighted M3b model. The main text (Section 5.5) summarizes three key patterns; here we provide the full matrices for all 5 covariates $\times$ 4 policy dimensions $\times$ 2 margins.

The four policy dimensions are: (i) market rate percentile (MR_pctile_std, standardized), measuring how generous each state's reimbursement rate is relative to provider market rates; (ii) TieredReim, an indicator for states with tiered reimbursement quality systems; (iii) ITaddon, an indicator for states offering additional subsidies for infant/toddler care; and (iv) the intercept column, capturing residual cross-state variation unrelated to the three measured policies (the intercept column of $\mathbf{\Gamma}_k$ is softly identified by its prior; see Section 3.6).

Two patterns merit attention beyond those highlighted in the main text. First, the IT addon variable shows the most consistent effects across both margins: it is positively associated with the extensive-margin intercept ($\gamma = 0.728$, $\Pr(> 0) = 0.996$) and the extensive-margin Hispanic coefficient ($\gamma = 0.750$, $\Pr(> 0) = 0.999$), while being strongly negatively associated with the extensive-margin poverty coefficient ($\gamma = -0.488$, $\Pr(> 0) = 0.018$). On the intensive margin, the IT addon is associated with a notably higher Black coefficient ($\gamma = 0.256$, $\Pr(> 0) = 1.000$).

Second, the intercept columns for both margins have wide posteriors (SD ≈ 0.93), confirming the soft identification: the hierarchical prior ($\mathsf{N}(0, 0.5^2)$ for the intercept column, tighter than the $\mathsf{N}(0, 1^2)$ for policy columns) constrains these parameters without driving them to zero.

D.5. **Design effect diagnostics.** The design effect ratio (DER), defined in (3.27) and classified in Proposition C.31, quantifies the factor by which the survey design

Margin	Covariate	Policy	Mean	SD	Q025	Q975	Pr(>0)
Extensive margin ($\mathbf{\Gamma}_1$)							
ext	intercept	intercept	0.127	0.937	−1.684	1.960	0.558
ext	intercept	MR_pctile	0.018	0.142	−0.268	0.297	0.543
ext	intercept	TieredReim	−0.413	0.359	−1.121	0.278	0.128
ext	intercept	ITaddon	0.728	0.285	0.182	1.297	0.996
ext	poverty	intercept	−0.044	0.933	−1.845	1.789	0.483
ext	poverty	MR_pctile	0.064	0.107	−0.152	0.271	0.732
ext	poverty	TieredReim	0.495	0.272	−0.064	1.021	0.962
ext	poverty	ITaddon	−0.488	0.225	−0.917	−0.035	0.018
ext	urban	intercept	0.073	0.945	−1.766	1.886	0.531
ext	urban	MR_pctile	0.011	0.062	−0.112	0.129	0.574
ext	urban	TieredReim	−0.248	0.175	−0.601	0.094	0.076
ext	urban	ITaddon	−0.205	0.146	−0.488	0.087	0.072
ext	black	intercept	0.085	0.929	−1.740	1.927	0.530
ext	black	MR_pctile	−0.283	0.144	−0.565	0.010	0.026
ext	black	TieredReim	−0.652	0.339	−1.346	0.014	0.029
ext	black	ITaddon	0.365	0.276	−0.193	0.890	0.910
ext	hispanic	intercept	0.012	0.925	−1.799	1.850	0.509
ext	hispanic	MR_pctile	0.072	0.134	−0.198	0.331	0.712
ext	hispanic	TieredReim	−0.351	0.350	−1.052	0.318	0.157
ext	hispanic	ITaddon	0.750	0.259	0.260	1.290	0.999
Intensive margin ($\mathbf{\Gamma}_2$)							
int	intercept	intercept	−0.074	0.922	−1.899	1.735	0.468
int	intercept	MR_pctile	−0.024	0.039	−0.104	0.052	0.265
int	intercept	TieredReim	0.271	0.120	0.026	0.497	0.985
int	intercept	ITaddon	0.001	0.084	−0.158	0.173	0.488
int	poverty	intercept	0.019	0.935	−1.783	1.831	0.505
int	poverty	MR_pctile	0.028	0.026	−0.024	0.079	0.865
int	poverty	TieredReim	−0.064	0.064	−0.193	0.060	0.156
int	poverty	ITaddon	−0.087	0.051	−0.187	0.014	0.041
int	urban	intercept	−0.015	0.932	−1.821	1.783	0.497
int	urban	MR_pctile	0.027	0.017	−0.007	0.060	0.941
int	urban	TieredReim	0.015	0.054	−0.090	0.122	0.612
int	urban	ITaddon	0.039	0.040	−0.035	0.120	0.837
int	black	intercept	0.029	0.912	−1.731	1.848	0.503
int	black	MR_pctile	0.015	0.036	−0.056	0.082	0.662
int	black	TieredReim	0.047	0.088	−0.126	0.224	0.706
int	black	ITaddon	0.256	0.072	0.117	0.397	1.000
int	hispanic	intercept	−0.045	0.936	−1.853	1.811	0.482
int	hispanic	MR_pctile	−0.115	0.054	−0.224	−0.010	0.016
int	hispanic	TieredReim	0.227	0.164	−0.088	0.554	0.920
int	hispanic	ITaddon	−0.135	0.108	−0.349	0.077	0.104

TABLE D.3. Complete policy moderation parameters (weighted M3b, all 40 elements of $\mathbf{\Gamma}_1$ and $\mathbf{\Gamma}_2$). Each row reports the posterior mean, standard deviation, 95% credible interval, and posterior probability of a positive effect. The main text (Section 5.5) summarizes three key patterns; bold findings correspond to $\Pr(>0) > 0.95$ or $\Pr(>0) < 0.05$.

inflates each parameter's variance beyond the model-based (SRS) variance. Table D.4 presents the full variance decomposition for all 11 fixed-effect parameters, and Figure D.2 provides a three-panel diagnostic visualization.

Variance decomposition. The three variance quantities in Table D.4 deserve careful interpretation. $[\mathbf{H}_{\text{obs}}^{-1}]_{pp}$ is the model-based variance obtained from the observed information matrix—the appropriate benchmark under simple random sampling. $[\mathbf{V}_{\text{sand}}]_{pp}$ is the sandwich-corrected variance (Binder, 1983) that accounts for the complex survey design, and constitutes the primary inferential summary. $[\mathbf{\Sigma}_{\text{MCMC}}]_{pp}$ is the raw MCMC posterior variance, which is substantially inflated (by factors of 600–6,500) relative to $\mathbf{H}_{\text{obs}}^{-1}$ because the hierarchical prior dominates the pseudo-likelihood contribution for fixed effects.

The DER ranges from 1.14 (β_{Hisp}) to 4.18 (α_{Urban}), with a mean of 2.11. This is broadly consistent with the Kish DEFF = 3.76: the DER is systematically smaller because the state-varying coefficients partially absorb within-state clustering (Proposition C.30). The corresponding confidence interval inflation factors are $\sqrt{\text{DER}} \in [1.07, 2.04]$, so the widest Wald intervals are approximately twice those of a hypothetical SRS design.

Diagnostic figure. Figure D.2 decomposes the sandwich correction into three components. Panel (a) shows the hierarchical variance ratio—the ratio of naive (MCMC-based) to Wald (sandwich-based) confidence interval widths—on a log scale; all parameters exceed a ratio of 10, and several exceed 100, confirming that the raw MCMC posterior is uninformative about the design-consistent uncertainty. Panel (b) compares the DER (based on $\mathbf{H}_{\text{obs}}^{-1}$) with the "$\text{DER}_{\text{MCMC}}$" (based on $\mathbf{\Sigma}_{\text{MCMC}}$); the latter is near zero for all parameters except $\log \kappa$, further illustrating hierarchical inflation. Panel (c) reports the diagonal elements of the Cholesky scaling matrix $\mathbf{A}$ (Theorem 3) for the fixed-effects block. For the α and β coefficients, the diagonal entries are all below 0.06, indicating substantial contraction of the MCMC draws, a direct consequence of the hierarchically inflated posterior being much wider than the design-consistent target.

D.6. **Reversal probability map.** Figure D.3 maps the posterior probability of the poverty reversal for each state: $\Pr(\tilde{\alpha}_{\text{pov},s} < 0 \text{ and } \tilde{\beta}_{\text{pov},s} > 0 \mid \text{data})$, computed from the unweighted M2 model (block-diagonal SVC). States with high reversal probability ($\Pr > 0.9$) are concentrated in the South, Southwest, and large urban states; states with low probability ($\Pr < 0.1$) are primarily in the upper Midwest and Mountain

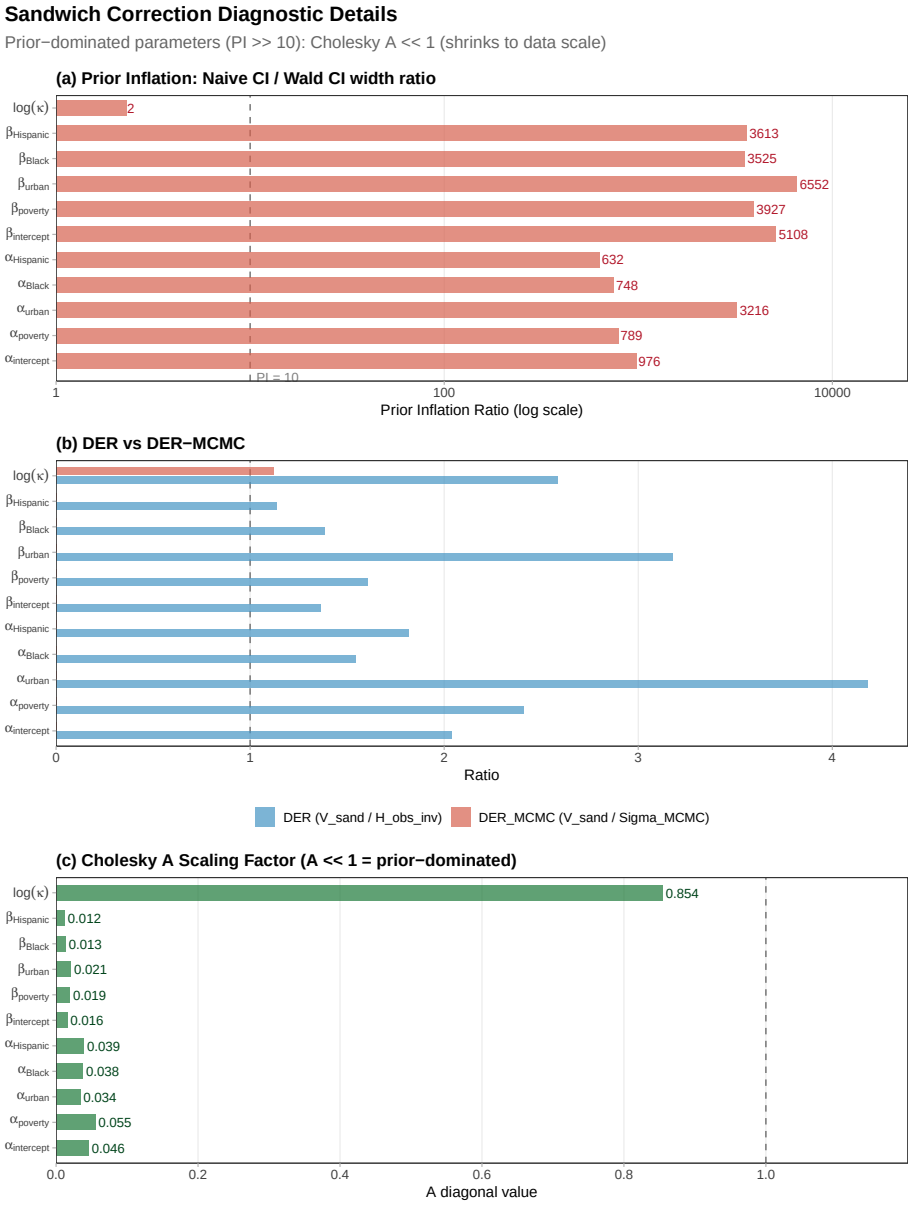

FIGURE D.2. Sandwich correction diagnostic details. (a) Hierarchical variance ratio (log scale): ratio of naive (MCMC-based) to Wald (sandwich-based) confidence interval widths. Values above the dashed line at 10 indicate strong hierarchical inflation. (b) DER comparison: design effect ratio computed using $\mathbf{H}_{\text{obs}}^{-1}$ (primary) versus $\mathbf{\Sigma}_{\text{MCMC}}$ (diagnostic only). (c) Cholesky $\mathbf{A}$ diagonal: scaling factors from the affine transformation (Theorem 3). Values $\ll 1$ indicate that the transformation substantially contracts the MCMC draws toward the design-consistent target.

Parameter	H_{obs}^{-1}	V_{sand}	Σ_{MCMC}	DER	DER_{mc}	SD(H)	SD(Sand)	SD(MCMC)
α_{Int}	0.00097	0.00198	0.946	2.04	0.002	0.031	0.044	0.972
α_{Pov}	0.00114	0.00275	0.900	2.41	0.003	0.034	0.052	0.949
α_{Urb}	0.00028	0.00118	0.907	4.18	0.001	0.017	0.034	0.952
α_{Blk}	0.00127	0.00195	0.947	1.54	0.002	0.036	0.044	0.973
α_{Hisp}	0.00146	0.00265	0.920	1.82	0.003	0.038	0.051	0.959
β_{Int}	0.00017	0.00023	0.860	1.36	0.000	0.013	0.015	0.927
β_{Pov}	0.00022	0.00036	0.875	1.61	0.000	0.015	0.019	0.936
β_{Urb}	0.00013	0.00042	0.873	3.18	0.000	0.012	0.021	0.934
β_{Blk}	0.00024	0.00033	0.845	1.38	0.000	0.016	0.018	0.919
β_{Hisp}	0.00025	0.00028	0.892	1.14	0.000	0.016	0.017	0.945
$\log \kappa$	0.00033	0.00086	0.001	2.58	1.119	0.018	0.029	0.028

TABLE D.4. Design effect ratio diagnostics for all 11 fixed-effect parameters. Columns: H_{obs}^{-1} = model-based variance (observed information inverse), V_{sand} = sandwich variance, Σ_{MCMC} = raw MCMC posterior variance, DER = $V_{\text{sand}}/H_{\text{obs}}^{-1}$, $\text{DER}_{\text{mc}} = V_{\text{sand}}/\Sigma_{\text{MCMC}}$, and the corresponding standard deviations. The DER range 1.14–4.18 (mean 2.11) is consistent with the Kish DEFF = 3.76.

West. The remaining states ($0.1 \leq \Pr \leq 0.9$) tend to be smaller states where the posterior intervals are wide due to limited sample sizes.

D.7. **LKJ prior sensitivity.** To assess sensitivity to the LKJ prior on the 10×10 random-effect correlation matrix Ω, we refit M3b-W under seven concentration parameters $\eta \in \{1.0, 1.5, 2.0, 3.0, 4.0, 6.0, 8.0\}$. The baseline specification uses $\eta = 2$, which places a weakly regularizing distribution favoring moderate correlations; $\eta = 1$ yields a uniform distribution over correlation matrices, while higher values concentrate mass near the identity.

Table D.5 reports posterior summaries for the cross-margin correlation $\rho_{\text{cross}} = \Omega_{1,6}$, the poverty coefficients (α_{pov}, β_{pov}), and the intercept random-effect standard deviations (τ_1^{ext}, τ_1^{int}). Across the sevenfold range of η, $\hat{\rho}_{\text{cross}}$ varies between 0.097 and 0.191, remaining near zero throughout and well within the posterior uncertainty of any single specification. The 95% credible interval includes zero in every case. The poverty coefficients are equally stable: $\alpha_{\text{pov}} \in [-0.359, -0.310]$ and $\beta_{\text{pov}} \in [0.074, 0.132]$, confirming that the sign reversal is robust to the LKJ prior.

LOO-CV model fit is nearly identical across specifications (ELPD spread < 5), with all MCMC diagnostics satisfactory (the single marginal R-hat exceedance at $\eta = 6$ is 1.011, well within practical tolerance).

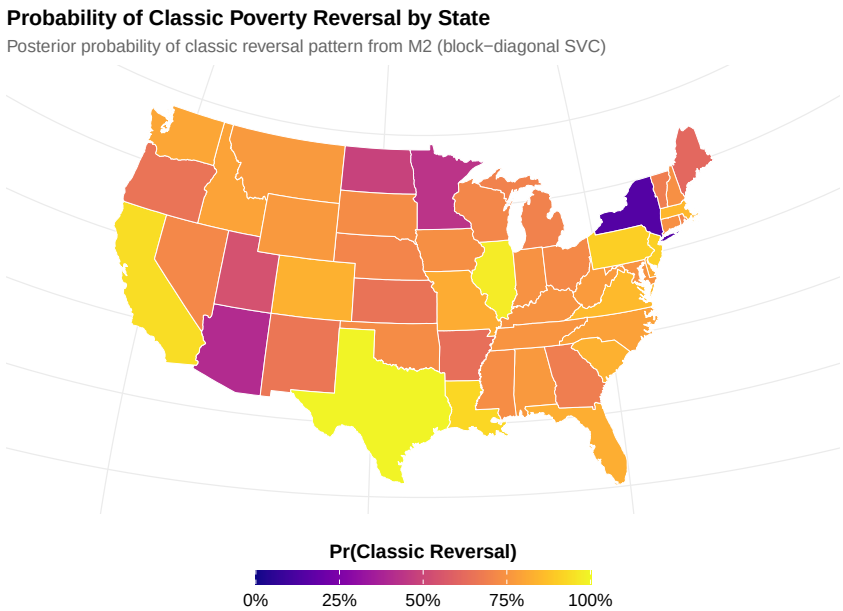

FIGURE D.3. Choropleth map of the posterior reversal probability from the unweighted M2 model (block-diagonal SVC): $\Pr(\tilde{\alpha}_{\text{pov},s} < 0 \cap \tilde{\beta}_{\text{pov},s} > 0 \mid \text{data})$. Darker shading indicates higher reversal probability. The population-average probability of reversal is 1.000.

Figure D.4 overlays the posterior densities of ρ_{cross} across all seven specifications. The densities are nearly superimposed, confirming that the data dominate the prior for this parameter despite the moderate cluster size ($S = 51$).

These results confirm that the near-zero cross-margin correlation and the poverty reversal pattern are data-driven findings, not artifacts of the LKJ(2) prior specification.

Appendix E. Simulation and Computational Details

This appendix provides computational and simulation details supplementing Sections 3 and 4. Section E.1 presents the five-model hierarchy with parameter counts. Section E.2 describes the MCMC configuration and runtime benchmarks. Section E.3 specifies the data-generating process for the simulation study, Section E.4 presents supplementary simulation diagnostic figures, Section E.5 evaluates robustness under model misspecification, Section E.6 provides a frequentist contextualization, Section E.7 decomposes the coverage gap into bias and width components, and Section E.8 reports coverage at the 95% nominal level. Implementation details (non-centered parameterization, numerical stability, score

Table D.5. LKJ prior sensitivity analysis. Each row reports posterior summaries from M3b-W fitted with LKJ(η) prior on the 10×10 random-effect correlation matrix Ω. The baseline specification is $\eta = 2$ (uniform over correlations). $\rho_{\text{cross}} = \Omega_{1,6}$ is the cross-margin correlation between extensive and intensive intercepts. α_{pov} and β_{pov} are the poverty coefficients for the extensive and intensive margins. τ_1^{ext} and τ_1^{int} are the intercept random-effect standard deviations. All covariates are standardized.

η	$\hat{\rho}_{\text{cross}}$ [95% CI]	α_{pov}	β_{pov}	τ_1^{ext}	τ_1^{int}	$\widehat{\text{elpd}}_{\text{loo}}$	Diag.
1.0	0.191 [−0.261, 0.604]	−0.322	0.088	0.693	0.149	−20913.0	✓
1.5	0.172 [−0.261, 0.572]	−0.312	0.074	0.683	0.146	−20915.0	✓
2.0⋆	0.170 [−0.259, 0.573]	−0.320	0.098	0.682	0.144	−20915.7	✓
3.0	0.145 [−0.269, 0.527]	−0.310	0.132	0.666	0.143	−20917.2	✓
4.0	0.134 [−0.274, 0.516]	−0.330	0.093	0.653	0.143	−20916.2	✓
6.0	0.111 [−0.247, 0.460]	−0.359	0.113	0.646	0.142	−20914.8	!
8.0	0.097 [−0.256, 0.429]	−0.340	0.110	0.639	0.140	−20915.4	✓

⋆Baseline specification ($\eta = 2$ corresponds to a uniform distribution over correlation matrices).

computation) are documented in the hurdlebb package vignettes; full simulation results and replication instructions are available in the replication package at https://github.com/joonho112/hurdlebb-replication.

E.1. **Model hierarchy and architecture.** Table E.1 summarizes the five nested models fitted in the application (Section 5). Each model extends its predecessor by one structural feature: random intercepts (M1), state-varying coefficients with block-diagonal covariance (M2), cross-margin covariance (M3a), and policy moderators (M3b). The final model M3b-W adds survey weights to the pseudo-posterior. Parameter counts include all fixed effects, random effects, and hyperparameters (Cholesky factors, scales, and correlations).

The models are fitted sequentially using warm-start initialization: the posterior mean of model k provides the initial values for model $k + 1$. Random-effect parameters absent in the simpler model are initialized at zero (NCP variates) or identity (Cholesky factors). This progressive strategy reduces total computation time by approximately 40% compared with cold starts and ensures that each model addresses a question raised by its predecessor: M0 to M1 asks whether baseline enrolment rates vary by state; M1 to M2, whether the poverty reversal varies by state; M2 to M3a, whether cross-margin covariance improves fit; and M3a to M3b, whether observable policies explain the state heterogeneity.

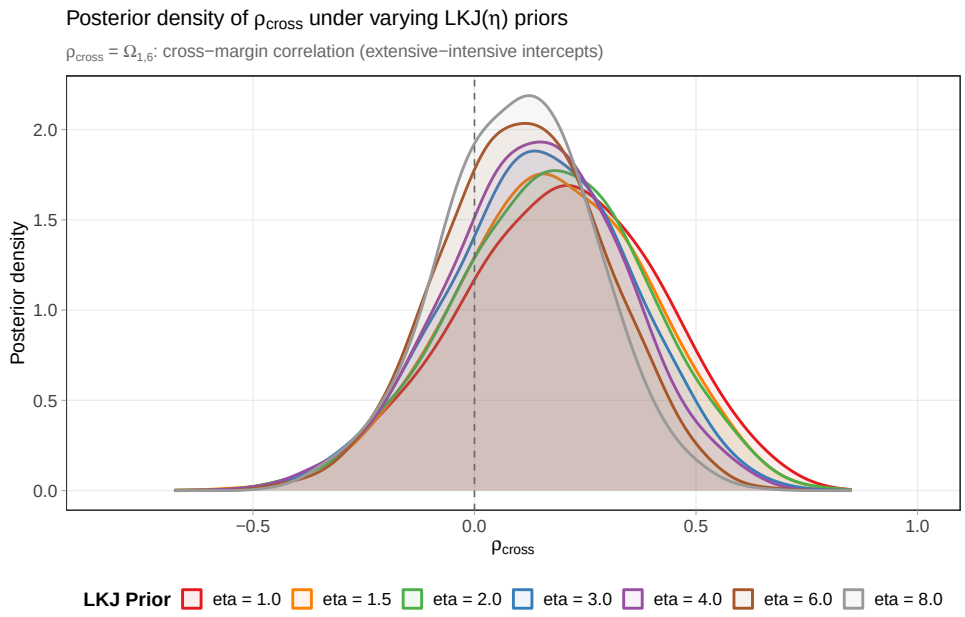

FIGURE D.4. Posterior density of the cross-margin correlation $\rho_{\text{cross}} = \Omega_{1,6}$ under seven LKJ concentration parameters $\eta \in \{1, 1.5, 2, 3, 4, 6, 8\}$. The densities are nearly superimposed, indicating that the posterior is data-dominated rather than prior-driven. The vertical dashed line marks zero.

Model	Structure	Parameters	Covariance	$\widehat{\text{elpd}}_{\text{loo}}$
M0	Pooled HBB	11	—	−20,945
M1	Random intercepts	116	2×2	−20,627
M2	Block-diagonal SVC	551	$(5 \times 5) \oplus (5 \times 5)$	−20,604
M3a	Full SVC	576	10×10	−20,602
M3b	Policy moderators	616	10×10	−20,608

TABLE E.1. Model hierarchy. Parameters: total number of free parameters (fixed effects + random effects + hyperparameters). Covariance: dimension of the state-level covariance matrix for the random-effect deviations $\boldsymbol{\delta}_s$. LOO: expected log-predictive density (Section 5). The progression from M0 to M3a yields significant incremental improvements in predictive accuracy; M3b adds policy moderators without further predictive gain.

E.2. **MCMC configuration.** All models are fitted using Stan (Carpenter et al., 2017) with the No-U-Turn Sampler (NUTS; Hoffman & Gelman, 2014) via the CmdStanR interface (Stan Development Team, 2024). Table E.2 reports the sampler configuration.

Setting	Empirical (M0–M3b)	Simulation
Chains	4 (parallel)	4 (parallel)
Warmup iterations	1,000 per chain	1,500 per chain
Sampling iterations	1,000 per chain	2,000 per chain
Total post-warmup draws	4,000	8,000
`adapt_delta`	0.95	0.95
`max_treedepth`	12	14

TABLE E.2. MCMC configuration. The empirical case study (M0–M3b) uses the default `hurdlebb` settings; the simulation study uses longer runs and higher `max_treedepth` to ensure adequate exploration across 200 replications. In both cases, `adapt_delta` = 0.95 reduces the step size to avoid divergent transitions in the state-varying coefficient models.

Convergence targets. As stated in Algorithm 1, we require the following diagnostic thresholds for all parameters (Vehtari et al., 2021): (i) split-$\hat{R} < 1.01$; (ii) minimum bulk ESS > 400; (iii) minimum tail ESS > 400; and (iv) zero divergent transitions. All five models satisfy these thresholds; detailed diagnostics are available in the replication package.

Warm-start initialization. The five-model sequence M0 $\rightarrow$ M1 $\rightarrow$ M2 $\rightarrow$ M3a $\rightarrow$ M3b is fitted progressively: the posterior mean from model k is used to initialize model $k+1$. Specifically:

- Fixed-effect coefficients ($\boldsymbol{\alpha}$, $\boldsymbol{\beta}$, $\log \kappa$) are initialized at the predecessor's posterior means.
- Scale parameters ($\boldsymbol{\tau}$) are initialized at the predecessor's posterior means when dimensions match, or at 0.1 when new dimensions are introduced.
- Cholesky factors ($\mathbf{L}_\varepsilon$) are initialized at the identity matrix, which corresponds to zero correlation—a conservative starting point that lets the sampler discover correlations during warmup.
- NCP variates ($\mathbf{z}_s$) are initialized at zero, placing all states at the global mean.
- Policy moderator coefficients ($\boldsymbol{\Gamma}$, M3b only) are initialized at the predecessor's posterior means or at zero for newly introduced parameters.

This strategy avoids the slow exploration that can occur when complex models are initialized far from the posterior mode.

Runtime benchmarks. On a 16-core workstation, approximate wall-clock times are: M0 ($\sim$5 minutes), M1 ($\sim$15 minutes), M2 ($\sim$30 minutes), M3a ($\sim$50 minutes),

M3b (∼60 minutes), and M3b-W (∼70 minutes, including score extraction in the `generated quantities` block). The total sequential fitting time is approximately 4 hours with warm-start initialization, compared to approximately 7 hours with cold starts.

E.3. **Data-generating process and calibration.** This section provides the complete data-generating process (DGP) for the simulation study in Section 4. The DGP is designed to match the NSECE 2019 data structure while allowing parametric control over survey informativeness.

Population generation. A finite super-population of $M = 50{,}000$ providers is distributed across $S = 51$ states, with state sizes proportional to the observed NSECE state counts. For each provider i in state s:

(1) Draw the covariate vector $\mathbf{x}_i = (1,\, x_{\text{pov}},\, x_{\text{urb}},\, x_{\text{blk}},\, x_{\text{his}})^\intercal$ by bootstrap resampling from the empirical NSECE marginals within state s.

(2) Draw the trial size n_i from the empirical distribution of total 0–5 enrollment in the NSECE data.

True parameter values. The true parameters are calibrated to the M1 (random intercept) unweighted NSECE fit (Section E.1). Table E.3 reports all parameter values used in the DGP. These values were chosen to produce a realistic structural zero rate (approximately 36%) and overdispersion level consistent with the empirical data.

Outcome generation. For each replication $r = 1, \ldots, R$:

(1) Draw state random effects $\boldsymbol{\delta}_s = (\delta_{1s},\, \delta_{2s})^\intercal \sim \mathsf{N}_2(\mathbf{0}, \boldsymbol{\Sigma}_\delta)$, where

$$\boldsymbol{\Sigma}_\delta = \operatorname{diag}(\boldsymbol{\tau})\, \mathbf{R}_\delta \operatorname{diag}(\boldsymbol{\tau}), \quad \boldsymbol{\tau} = (\tau_{\text{ext}},\, \tau_{\text{int}})^\intercal, \quad [\mathbf{R}_\delta]_{12} = \rho_{\text{cross}}.$$

(2) Compute linear predictors: $\eta_i^{\text{ext}} = \mathbf{x}_i^\intercal \boldsymbol{\alpha}_0 + \delta_{1,s[i]}$, $\eta_i^{\text{int}} = \mathbf{x}_i^\intercal \boldsymbol{\beta}_0 + \delta_{2,s[i]}$.

(3) Generate participation: $z_i \sim \text{Bern}(q_i)$ with $q_i = \text{expit}(\eta_i^{\text{ext}})$.

(4) For providers with $z_i = 1$, generate positive counts via **rejection sampling** from the unconditional beta-binomial (Prentice, 1986):

> Draw $Y_i \sim \text{BetaBin}(n_i,\, \mu_i,\, \kappa_0)$ where $\mu_i = \text{expit}(\eta_i^{\text{int}})$; *repeat until* $Y_i > 0$; set $y_i = Y_i$.

This ensures $y_i > 0$ for all providers with $z_i = 1$, consistent with the zero-truncated beta-binomial component of the hurdle model. The acceptance probability is $1 - p_0(n_i, \mu_i, \kappa_0)$, which exceeds 0.80 for the vast majority of observations in our calibration.

(5) For providers with $z_i = 0$, set $y_i = 0$.

Parameter	Value	Description
α_{int}	$+0.696$	Extensive intercept
α_{pov}	-0.119	Extensive poverty
α_{urb}	$+0.253$	Extensive urban
α_{blk}	-0.070	Extensive Black
α_{his}	-0.139	Extensive Hispanic
β_{int}	-0.032	Intensive intercept
β_{pov}	$+0.057$	Intensive poverty
β_{urb}	-0.018	Intensive urban
β_{blk}	$+0.080$	Intensive Black
β_{his}	$+0.040$	Intensive Hispanic
$\log \kappa_0$	$+1.655$	Log concentration
κ_0	5.232	Concentration
τ_{ext}	0.577	State SD (extensive)
τ_{int}	0.208	State SD (intensive)
ρ_{cross}	0.285	Cross-margin correlation

TABLE E.3. True parameter values for the simulation DGP, calibrated to the M1 unweighted fit. Fixed effects are on the logit scale; $\kappa_0 = \exp(\log \kappa_0)$.

Remark E.41 (T3-3: Zero-truncated beta-binomial DGP)**.** Drawing from the unconditional BetaBin and rejecting zeros produces exact draws from the zero-truncated beta-binomial ZT-BetaBin(n, μ, κ). The alternative—dividing the unconditional PMF by $(1 - p_0)$—would require a custom sampling routine, whereas rejection sampling reuses the standard BetaBin sampler and is efficient whenever p_0 is not too close to 1. The same approach is used in the Stan `generated quantities` block for posterior predictive replication.

Sampling design. From each finite population, a stratified cluster sample of approximately $N \approx 7{,}000$ providers is drawn using size-biased inclusion probabilities $\log \pi_i = c_0 + \rho_{\text{inc}} \cdot y_i^*$ (Equation (4.2)), where y_i^* is the standardized latent outcome and c_0 is calibrated to the target sample size (cf. Savitsky & Toth, 2016). The three scenarios (S0, S3, S4) are defined in Table 5; see Section 4 for the full discussion. Survey weights are computed as $w_i = 1/\pi_i$ and normalized to sum to N for the pseudo-posterior.

E.4. **Supplementary diagnostic figures and design effect analysis.** This section presents three supplementary figures that visualize bias, interval width, and

design effects across all simulation scenarios, and a full table of design effect ratios (DER) for all 11 fixed-effect parameters.

Relative bias distributions. Figure E.1 shows the distribution of relative bias across parameters and scenarios. E-UW exhibits the largest absolute bias (especially for τ_{ext} and τ_{int} under S4), while E-WT/E-WS improve point estimation for fixed effects by incorporating survey weights. For the poverty slopes α_{pov} and β_{pov}, the relative bias under E-WS remains below 15% across all scenarios.

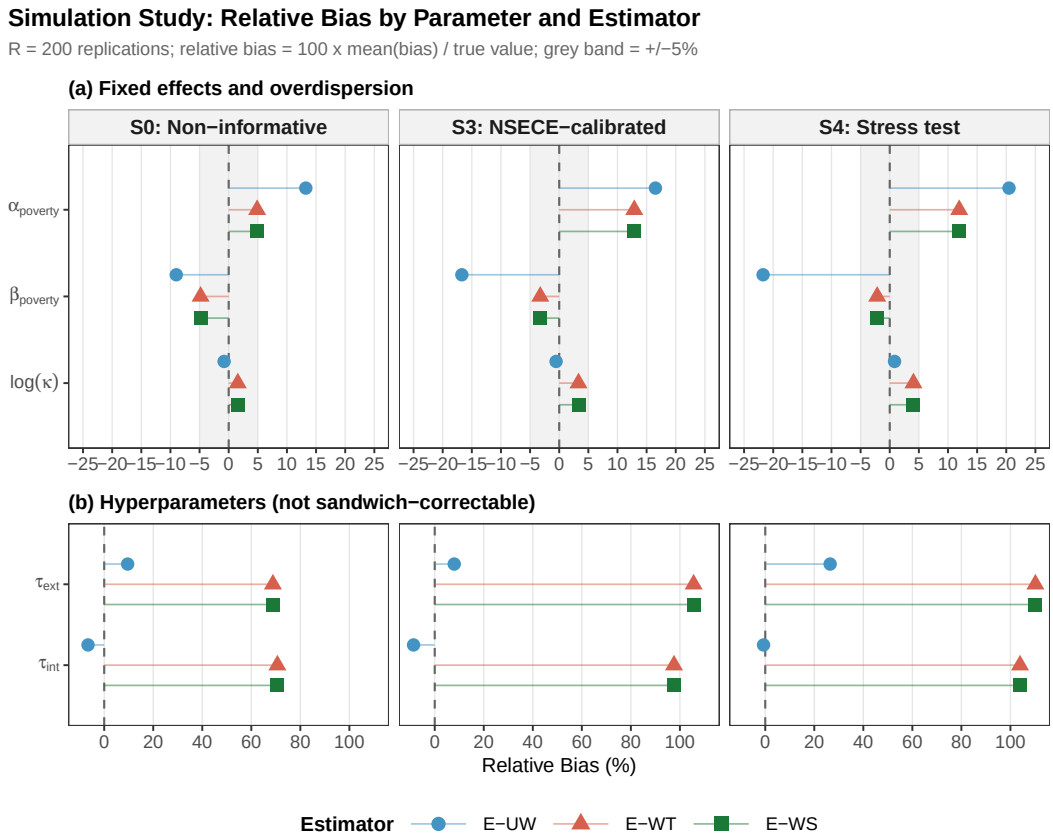

FIGURE E.1. Relative bias (RB, %) by parameter, estimator, and scenario. Each panel corresponds to one of the five target parameters. Dot positions indicate relative bias; color distinguishes scenarios (S0, S3, S4). The vertical gray line marks zero bias.

Width ratio distributions. Figure E.2 displays the distribution of width ratios (WR = median CI width under E-WS / median CI width under E-WT) across replications. The progressive inflation from S0 through S4 is clearly visible for all fixed-effect parameters. The width ratio for $\log \kappa$ is similar to or slightly larger than

those for the poverty slopes, reflecting the nonlinear dispersion score (Appendix B). Hyperparameter width ratios are identically 1.000 (not shown) because no sandwich correction is applied.

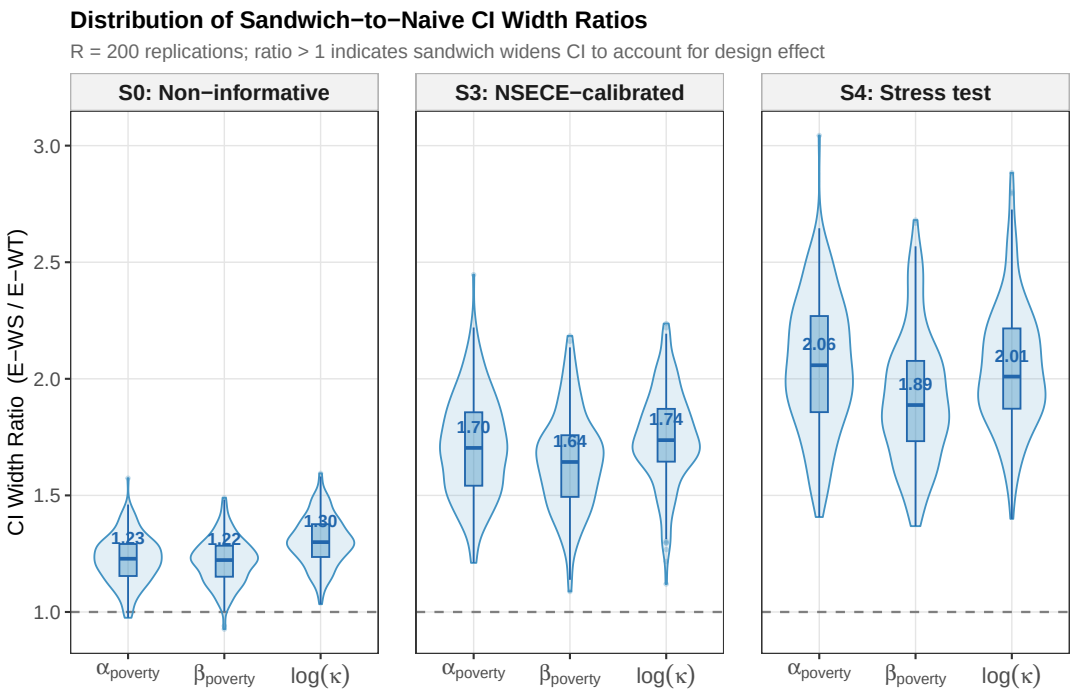

FIGURE E.2. Distribution of width ratios across $R = 200$ replications, by parameter and scenario. Boxplots are overlaid with violin plots to show the full distributional shape. The horizontal dashed line at WR = 1.0 marks the no-correction reference.

Design effect ratios. Figure E.3 summarizes the simulation DER across all 11 fixed-effect parameters (5 extensive, 5 intensive, and $\log \kappa$) and three scenarios. The DER monotonically increases with the informativeness parameter ρ_{inc}, consistent with the theoretical prediction in Proposition C.31. Urban density parameters exhibit the largest DER in all scenarios (mean 3.25 in S0 up to 6.43 in S4), mirroring the empirical DER of 4.18 for α_{urb} in the NSECE case study (Table D.4).

Complete DER table. The table below reports the full distribution of DER across replications for all 11 fixed-effect parameters and three scenarios. The mean, median, standard deviation, and range (min–max) characterize the replication-to-replication variability. Two patterns are noteworthy: (i) the DER standard deviation increases with the scenario severity, indicating greater sampling variability under heavier design effects; and (ii) the urban parameter consistently shows the widest

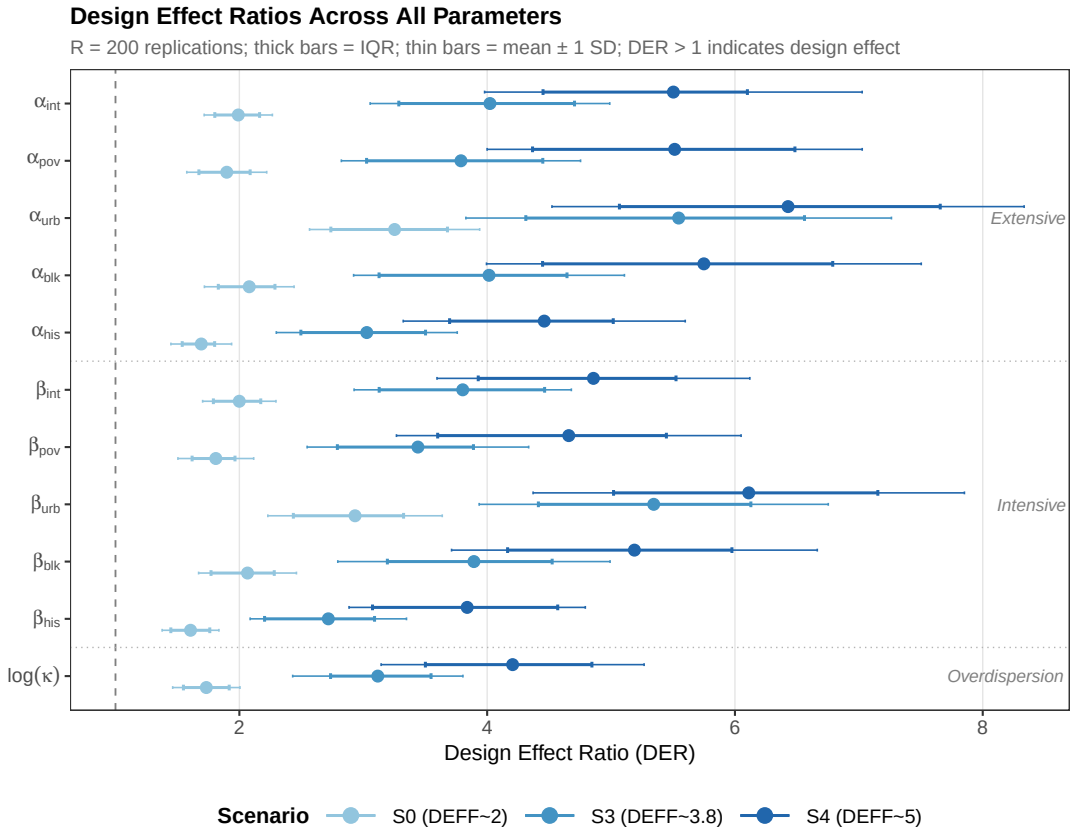

FIGURE E.3. Design effect ratios (DER) by parameter and scenario. Each dot represents the mean DER across 200 replications; color distinguishes scenarios. Parameters are grouped by margin (extensive vs. intensive) with $\log \kappa$ as a separate category.

DER range, reflecting the spatial concentration of urban providers that amplifies the design effect.

Cross-margin correlation recovery. Table E.5 reports recovery of the cross-margin correlation $\rho_{\text{cross}} = \text{cor}(\delta_j^{\text{ext}}, \delta_j^{\text{int}})$ using a plug-in estimator based on posterior mean random effects across $S = 51$ states. The unweighted estimator E-UW systematically overestimates ρ_{cross}, with relative bias growing from +84% in S0 to +127% in S4. The weighted estimator E-WT shows moderate downward bias (−26% to −45%), and E-WS is identical to E-WT because the Cholesky sandwich correction (Theorem 3) adjusts only fixed-effect covariances. Per-replication credible intervals are unavailable because full posterior draws of the correlation matrix Ω were not retained in the simulation archive; only frequentist summaries across replications are reported.

Scenario	Parameter	R	Mean	Median	SD	Min	Q25	Q75	Max
S0: Non-informative	Intercept (ext)	200	1.99	1.95	0.28	1.29	1.80	2.16	2.92
S0: Non-informative	Poverty (ext)	200	1.90	1.89	0.32	1.23	1.67	2.09	3.07
S0: Non-informative	Urban (ext)	200	3.25	3.18	0.69	1.59	2.74	3.68	5.28
S0: Non-informative	Black (ext)	200	2.08	2.05	0.36	1.14	1.83	2.29	3.46
S0: Non-informative	Hispanic (ext)	200	1.69	1.67	0.24	1.05	1.54	1.80	2.68
S0: Non-informative	Intercept (int)	200	2.00	1.97	0.30	1.38	1.79	2.17	2.92
S0: Non-informative	Poverty (int)	200	1.81	1.80	0.31	1.06	1.62	1.96	3.11
S0: Non-informative	Urban (int)	200	2.93	2.86	0.70	1.41	2.44	3.33	5.35
S0: Non-informative	Black (int)	200	2.07	2.02	0.39	1.32	1.77	2.28	3.55
S0: Non-informative	Hispanic (int)	200	1.61	1.59	0.23	1.11	1.45	1.76	2.34
S0: Non-informative	log kappa	200	1.73	1.71	0.27	1.12	1.55	1.92	2.36
S3: NSECE-calibrated	Intercept (ext)	200	4.02	3.88	0.97	1.97	3.29	4.71	7.47
S3: NSECE-calibrated	Poverty (ext)	200	3.79	3.72	0.97	1.90	3.03	4.45	7.53
S3: NSECE-calibrated	Urban (ext)	200	5.55	5.43	1.72	1.62	4.31	6.56	11.21
S3: NSECE-calibrated	Black (ext)	200	4.01	3.81	1.09	2.07	3.13	4.64	7.23
S3: NSECE-calibrated	Hispanic (ext)	200	3.03	2.94	0.73	1.58	2.50	3.50	5.26
S3: NSECE-calibrated	Intercept (int)	200	3.80	3.70	0.88	1.68	3.13	4.46	6.40
S3: NSECE-calibrated	Poverty (int)	200	3.44	3.42	0.89	1.53	2.79	3.89	7.32
S3: NSECE-calibrated	Urban (int)	200	5.34	5.26	1.41	2.34	4.41	6.13	11.40
S3: NSECE-calibrated	Black (int)	200	3.89	3.67	1.10	1.96	3.19	4.52	7.45
S3: NSECE-calibrated	Hispanic (int)	200	2.72	2.65	0.63	1.67	2.20	3.09	5.16
S3: NSECE-calibrated	log kappa	200	3.12	3.07	0.69	1.38	2.74	3.55	5.44
S4: Stress test	Intercept (ext)	200	5.50	5.23	1.52	2.79	4.45	6.10	13.23
S4: Stress test	Poverty (ext)	200	5.51	5.31	1.51	2.72	4.37	6.48	11.59
S4: Stress test	Urban (ext)	200	6.43	6.22	1.91	2.97	5.07	7.66	17.11
S4: Stress test	Black (ext)	200	5.75	5.41	1.75	2.30	4.45	6.79	11.38
S4: Stress test	Hispanic (ext)	200	4.46	4.36	1.14	1.91	3.70	5.02	10.86
S4: Stress test	Intercept (int)	200	4.86	4.71	1.26	2.38	3.93	5.52	9.53
S4: Stress test	Poverty (int)	200	4.66	4.44	1.39	2.31	3.60	5.45	9.04
S4: Stress test	Urban (int)	200	6.11	5.96	1.74	2.24	5.02	7.15	13.79
S4: Stress test	Black (int)	200	5.19	5.01	1.48	2.13	4.16	5.98	10.13
S4: Stress test	Hispanic (int)	200	3.84	3.74	0.95	1.78	3.07	4.57	6.44
S4: Stress test	log kappa	200	4.21	4.10	1.06	2.11	3.50	4.85	7.69

TABLE E.4. Design Effect Ratios (DER) Across Simulation Replications. DER = $\mathrm{diag}(V_{\mathrm{sand}})$ / $\mathrm{diag}(H_{\mathrm{obs}}^{-1})$. All 11 fixed-effect parameters shown.

E.5. **Model misspecification robustness.** The simulation study in Section 4 evaluates the correctly specified case: data are generated from the random-intercept model M1 and fitted by M1. A natural concern is how the three estimators (E-UW,

TABLE E.5. Recovery of the cross-margin correlation $\rho_{\text{cross}} = \text{cor}(\delta_j^{\text{ext}}, \delta_j^{\text{int}})$ across $R = 200$ simulation replications. The plug-in estimator $\hat{\rho} = \text{cor}(\bar{\delta}_j^{\text{ext}}, \bar{\delta}_j^{\text{int}})$ uses posterior mean random effects ($S = 51$ states). True value: $\rho_{\text{cross}} = 0.2853$. E_{WS} is identical to E_{WT} because the sandwich correction applies only to fixed effects.

Scenario	Estimator	$\bar{\hat{\rho}}$	Bias	Rel. Bias (%)	RMSE	Emp. SE
S0 (Baseline)	E_{UW}	0.5238	+0.2386	+83.6	0.2990	0.1807
	E_{WT}	0.2104	−0.0749	−26.3	0.2294	0.2173
	$\text{E}_{\text{WS}}{}^{\dagger}$	0.2104	−0.0749	−26.3	0.2294	0.2173
S3 (NSECE)	E_{UW}	0.5814	+0.2961	+103.8	0.3643	0.2128
	E_{WT}	0.1658	−0.1195	−41.9	0.2256	0.1918
	$\text{E}_{\text{WS}}{}^{\dagger}$	0.1658	−0.1195	−41.9	0.2256	0.1918
S4 (Heavy-tail)	E_{UW}	0.6468	+0.3615	+126.7	0.4075	0.1886
	E_{WT}	0.1579	−0.1274	−44.7	0.2372	0.2005
	$\text{E}_{\text{WS}}{}^{\dagger}$	0.1579	−0.1274	−44.7	0.2372	0.2005

$^{\dagger}\text{E}_{\text{WS}}$ shares the same Stan fit as E_{WT}; the Cholesky sandwich correction adjusts only fixed-effect covariances, not hyperparameter posteriors.

E-WT, E-WS) behave when the fitted model omits a relevant source of heterogeneity. We therefore augment the simulation with a misspecification scenario designed to probe whether the sandwich correction remains beneficial when the model is wrong.

M2 data-generating process. Define the *state-varying coefficient* DGP (M2) by adding random poverty slopes to the M1 linear predictors:

$$\eta_i^{\text{ext}} = \mathbf{x}_i^{\intercal}\boldsymbol{\alpha} + \delta_{s[i]}^{\text{ext}} + \gamma_{s[i]}^{\text{ext}}\, x_{\text{pov},i}, \tag{E.1}$$

$$\eta_i^{\text{int}} = \mathbf{x}_i^{\intercal}\boldsymbol{\beta} + \delta_{s[i]}^{\text{int}} + \gamma_{s[i]}^{\text{int}}\, x_{\text{pov},i}, \tag{E.2}$$

where $\gamma_s^{\text{ext}} \sim \mathsf{N}(0, 0.15^2)$ and $\gamma_s^{\text{int}} \sim \mathsf{N}(0, 0.10^2)$ independently across $s = 1, \ldots, 51$ states. All other parameters $(\boldsymbol{\alpha}, \boldsymbol{\beta}, \kappa, \boldsymbol{\tau}, \rho)$ and the population structure ($M = 50{,}000$ providers) are identical to the M1 DGP (Section E.3). The standard deviations 0.15 (extensive) and 0.10 (intensive) produce state-varying poverty effects that are moderate relative to the fixed poverty coefficients ($\alpha_{\text{pov}} = -0.119$, $\beta_{\text{pov}} = +0.057$).

Design. We draw $R = 200$ independent replications under the S3 (NSECE-calibrated) informativeness scenario with DEFF ≈ 3.79, and fit the M1 random-intercept model using all three estimators. The M1 fit ignores the state-varying poverty slopes γ_s, inducing a form of omitted-variable misspecification that propagates through the poverty coefficient estimates.

Results. Table E.6 compares coverage, relative bias, RMSE, and width ratio for the correctly specified S3 scenario (M1 truth, M1 fit) and the misspecified B5 scenario (M2 truth, M1 fit).

TABLE E.6. Misspecification robustness: correctly specified S3 vs. misspecified B5 (M2 truth, M1 fit) under NSECE-calibrated sampling (DEFF ≈ 3.79). $R = 200$ replications; nominal coverage = 90%; † indicates significant departure ($> 2\times$ MCSE). MCSE $= \sqrt{\hat{p}(1-\hat{p})/200}$; range 0.0–3.5 pp.

		Coverage (%)		Relative bias (%)			RMSE			WR	
Parameter	Est.	S3	B5	S3	B5	Δ	S3	B5	Δ	S3	B5
Fixed effects											
α_{pov}	E-UW	89.5	$0.5^{\dagger}$	+16.5	−117.8	−134.3	0.037	0.143	+0.106	—	—
	E-WT	$58.5^{\dagger}$	$52.0^{\dagger}$	+12.9	−37.6	−50.5	0.072	0.079	+0.007	—	—
	E-WS	$82.0^{\dagger}$	$79.0^{\dagger}$	+12.9	−37.6	−50.5	0.072	0.079	+0.007	1.71	1.67
β_{pov}	E-UW	89.0	$30.0^{\dagger}$	−16.7	+63.1	+79.8	0.017	0.039	+0.022	—	—
	E-WT	$62.5^{\dagger}$	$42.0^{\dagger}$	−3.2	+52.7	+55.9	0.032	0.043	+0.011	—	—
	E-WS	$82.5^{\dagger}$	$67.5^{\dagger}$	−3.2	+52.7	+55.9	0.032	0.043	+0.011	1.64	1.60
$\log \kappa$	E-UW	92.0	$82.5^{\dagger}$	−0.5	−1.3	−0.8	0.020	0.027	+0.007	—	—
	E-WT	$34.0^{\dagger}$	$36.5^{\dagger}$	+3.3	+2.9	−0.4	0.068	0.064	−0.004	—	—
	E-WS	$60.5^{\dagger}$	$64.0^{\dagger}$	+3.3	+2.9	−0.4	0.068	0.064	−0.004	1.75	1.74
Variance components											
τ_{ext}	E-UW	$98.0^{\dagger}$	$96.0^{\dagger}$	+8.0	−2.5	−10.5	0.091	0.117	+0.026	—	—
	E-WT	$0.5^{\dagger}$	$0.0^{\dagger}$	+105.6	+102.4	−3.2	0.639	0.625	−0.014	—	—
	E-WS	$0.5^{\dagger}$	$0.0^{\dagger}$	+105.6	+102.4	−3.2	0.639	0.625	−0.014	1.00	1.00
τ_{int}	E-UW	$97.5^{\dagger}$	$57.0^{\dagger}$	−8.6	−34.8	−26.2	0.040	0.079	+0.039	—	—
	E-WT	$0.0^{\dagger}$	$1.0^{\dagger}$	+97.6	+97.0	−0.6	0.211	0.213	+0.002	—	—
	E-WS	$0.0^{\dagger}$	$1.0^{\dagger}$	+97.6	+97.0	−0.6	0.211	0.213	+0.002	1.00	1.00

Figure E.4 displays the corresponding coverage rates graphically.

Three findings emerge from the comparison.

Finding B5-1: The unweighted estimator is acutely sensitive to misspecification in the poverty coefficients. Under correct specification (S3), E-UW achieves near-nominal coverage for both poverty slopes (89.5% and 89.0%). Under the M2 DGP, E-UW coverage collapses to 0.5% for α_{pov} and 30.0% for β_{pov}—drops of 89 and 59 percentage points, respectively. The mechanism is omitted-variable bias: because M1 lacks the state-varying poverty slopes γ_s, the fixed coefficients α_{pov} and β_{pov} absorb the omitted heterogeneity, producing relative bias of −117.8% and +63.1%. This confirms that the model-based intervals of E-UW are valid only under

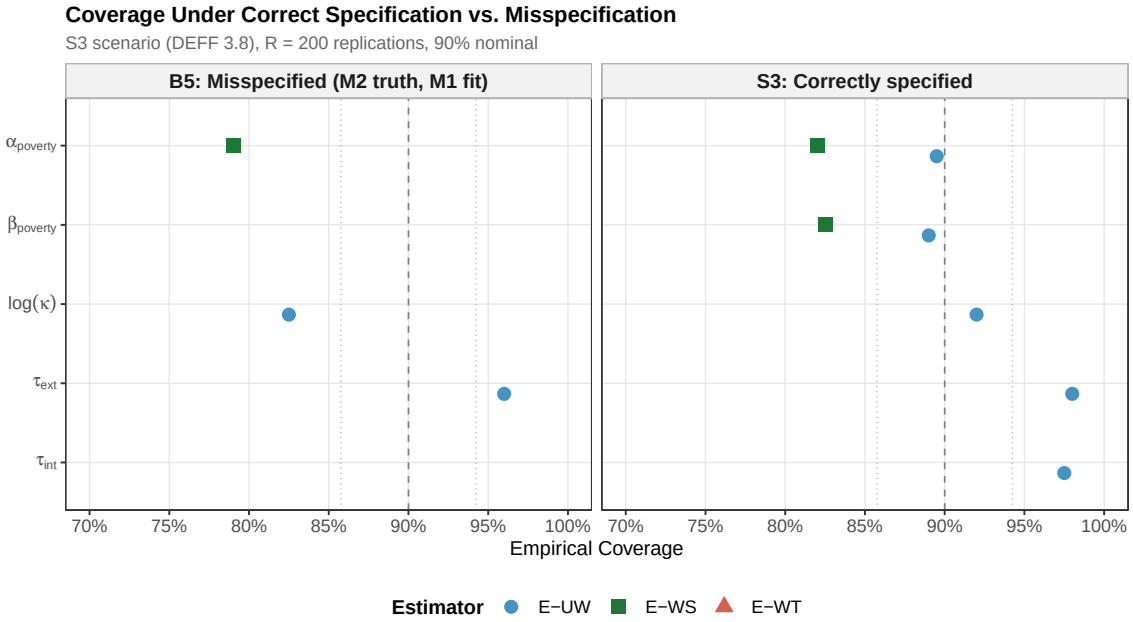

FIGURE E.4. Coverage under correct specification (S3) vs. misspecification (B5). Vertical dashed line: 90% nominal; dotted lines: ± 2 MCSE band.

correct specification—a strong argument for the design-based protection that survey weighting provides.

Finding B5-2: The sandwich correction preserves coverage stability under misspecification. For α_{pov}, E-WS coverage declines only from 82.0% (S3) to 79.0% (B5)—a 3-percentage-point drop—while E-WT drops from 58.5% to 52.0%. For β_{pov}, E-WS declines from 82.5% to 67.5% (-15 pp) compared with E-WT's drop from 62.5% to 42.0% (-20.5 pp). In both cases, E-WS maintains a substantial advantage over E-WT (27 and 25.5 percentage points under B5), demonstrating that the sandwich variance correction absorbs a meaningful share of the misspecification-induced variance increase. The width ratios remain stable across scenarios (WR = 1.67–1.74 under B5 vs. 1.64–1.75 under S3), indicating that the design effect magnitude is similar regardless of whether the model is correctly specified.

Finding B5-3: Variance components exhibit differential sensitivity. The overdispersion $\log \kappa$ is relatively robust under E-UW (coverage 92.0% $\rightarrow$ 82.5%; relative bias -0.5% $\rightarrow$ -1.3%), as expected since the omitted random slopes do not directly affect the dispersion structure. Among the variance components, τ_{ext} remains well-covered under E-UW (98.0% $\rightarrow$ 96.0%), but τ_{int} drops sharply (97.5% $\rightarrow$ 57.0%).

The asymmetry arises because the omitted intensive-margin slopes ($\sigma_\gamma^{\text{int}} = 0.10$ relative to $\tau_{\text{int}} = 0.208$) contribute proportionally more unmodeled between-state variance than the extensive-margin slopes ($\sigma_\gamma^{\text{ext}} = 0.15$ relative to $\tau_{\text{ext}} = 0.577$). As in the correctly specified case, E-WT and E-WS produce near-zero coverage for both τ parameters (WR = 1.00), reaffirming that the sandwich correction is structurally inapplicable to variance components.

Taken together, these results strengthen the case for the two-track reporting strategy recommended in Section 4: for population-average fixed effects, the sandwich-corrected estimator E-WS provides meaningful robustness against model misspecification; for hierarchical variance components, the unweighted estimator E-UW remains the only viable option, with the caveat that its validity depends on correct specification of the random-effects structure.

E.6. **Frequentist contextualization.** A natural question is whether a simpler design-based analysis could replace the full Bayesian hierarchical model. We compare the HBB pseudo-posterior estimates with conventional survey-weighted frequentist estimates obtained from `survey::svyglm` (Lumley, 2004, 2010) to (i) validate the design-consistency of the pseudo-posterior fixed effects and (ii) clarify which model features cannot be accommodated by the frequentist approach.

Frequentist specification. We fit separate margin models using the same five-covariate specification as the HBB model. For the extensive margin, we fit a survey-weighted logistic regression using `svyglm(..., family = quasibinomial())` with $N = 6{,}785$ observations. For the intensive margin, we fit a survey-weighted linear regression of the IT enrollment share y_i/n_i on the same covariates, restricted to the $N^+ = 4{,}392$ participants. Both models use the NSECE complex survey design (stratification, clustering, and sampling weights) and produce linearization-based cluster-robust standard errors. No random effects, beta-binomial structure, or cross-margin dependence can be incorporated.

Results. Table E.7 compares the survey-weighted svyglm estimates with the M3b-W sandwich-corrected pseudo-posterior summaries.

For the extensive margin, both approaches use the logit link, enabling a meaningful comparison on a common scale. The coefficient signs and relative magnitudes agree across all five predictors. Point estimates from HBB are systematically larger in absolute value than the svyglm estimates, which is expected: the HBB model conditions on state-level random effects, and conditional effects in mixed models are known to

Table E.7. Frequentist contextualization: survey-weighted svyglm vs. Bayesian HBB (M3b-W, sandwich-corrected) point estimates and standard errors. The extensive margin uses logistic link in both approaches; the intensive margin uses an identity link (svyglm) vs. logit link (HBB), so the scales differ. SE Ratio = HBB sandwich SE / svyglm SE.

		svyglm (design-based)		HBB (sandwich-corrected)			
Margin	Parameter	Est.	SE	Est.	SE	ΔEst.	SE Ratio
Extensive	Intercept	0.662	0.063	0.764	0.044	+0.103	0.70
	Poverty	-0.199	0.071	-0.324	0.052	-0.125	0.74
	Urban	0.234	0.039	0.442	0.034	+0.208	0.89
	Black	0.086	0.075	0.478	0.044	+0.392	0.59
	Hispanic	0.059	0.074	0.006	0.051	-0.054	0.70
Intensive	Intercept	0.483	0.005	-0.242	0.015	-0.725	3.02
	Poverty	0.015	0.007	0.090	0.019	+0.075	2.77
	Urban	-0.009	0.004	-0.047	0.021	-0.038	4.63
	Black	0.034	0.008	-0.017	0.018	-0.050	2.30
	Hispanic	0.010	0.006	-0.097	0.017	-0.107	2.92

exceed their marginal counterparts (Zeger et al., 1988). The SE ratio (HBB sandwich / svyglm) ranges from 0.59 to 0.89, reflecting the variance reduction achieved by modeling between-state heterogeneity through random effects rather than absorbing it into the residual.

For the intensive margin, the comparison is less direct because the two approaches operate on different scales: svyglm regresses the enrollment share y_i/n_i under an identity link, while HBB models the beta-binomial mean parameter μ_i under a logit link. The differing intercepts (svyglm: $+0.483$; HBB: -0.242) reflect this scale difference rather than substantive disagreement, since $\text{logit}^{-1}(-0.242) \approx 0.44$, which is in the range of the observed mean share. The larger SE ratios for the intensive margin (2.30–4.63) also arise from the logit-scale parameterization, which inflates variances relative to the probability scale.

Structural limitations. The comparison highlights several features of the HBB framework that `svyglm` cannot accommodate:

(1) *Beta-binomial distribution:* svyglm models the continuous share y/n via Gaussian errors, ignoring the bounded-count nature of the outcome and the extra-binomial overdispersion ($\kappa = 6.81$).

(2) *Hurdle structure:* the two margins must be fitted separately, precluding joint inference on the zero-generating process.

(3) *State-level random effects:* svyglm is a fixed-effect estimator; accommodating $S = 51$ state intercepts (and slopes in M3b) would require either fixed-effect dummies—inflating SE and sacrificing shrinkage—or a two-step approach that does not propagate uncertainty.

(4) *Cross-margin correlation:* the bivariate random-effect structure $\boldsymbol{\delta}_s \sim \mathsf{N}_{2q}(\mathbf{0}, \boldsymbol{\Sigma}_\delta)$ cannot be replicated in separate margin-specific regressions.

(5) *Policy moderation:* the state-varying policy coefficients $\boldsymbol{\Gamma}$ interact hierarchical and design-based components in a way that has no natural frequentist counterpart.

Taken together, the svyglm comparison serves as a fixed-effect-level sanity check: the Bayesian pseudo-posterior fixed effects are design-consistent with the frequentist target, while the full HBB framework adds hierarchical structure, distributional flexibility, and joint inference that the frequentist alternative cannot provide.

E.7. **Coverage gap decomposition.** Table E.8 decomposes the coverage gap for each parameter–estimator combination under the S3 (NSECE-calibrated) scenario. Two diagnostic quantities distinguish width-driven from bias-driven shortfalls:

- *SE calibration ratio* $= \widehat{\mathrm{SD}}/\overline{\mathrm{SE}}$, where $\widehat{\mathrm{SD}}$ is the empirical standard deviation of point estimates across $R = 200$ replications and $\overline{\mathrm{SE}}$ is the mean reported standard error. A ratio exceeding 1 indicates that the reported SE underestimates the actual sampling variability (width-driven gap).
- *Standardized mean bias* $= |\bar{b}|/\widehat{\mathrm{SD}}$, where $\bar{b}$ is the mean bias across replications. Values exceeding ~ 0.5 indicate non-negligible centering error (bias-driven gap).

For the poverty coefficients under E-WS, the SE ratio of 1.13–1.14 and negligible standardized bias (0.06–0.22) confirm that the 8 pp coverage gap is entirely width-driven: the sandwich variance estimator underestimates the true sampling variance by approximately 13%. This is consistent with the well-documented $O(S^{-1})$ finite-sample downward bias of cluster-robust variance estimators when $S = 51$ (Binder, 1983). For $\log \kappa$, the picture reverses: the SE ratio is 1.00 (perfectly calibrated width), but the standardized bias of 1.37 indicates that the pseudo-likelihood point estimate is systematically shifted, producing entirely one-sided non-coverage (39.5% below, 0% above). For τ_{ext} and τ_{int}, massive standardized bias (> 3) overwhelms any width consideration, reaffirming the structural inapplicability of the sandwich correction to variance components.

Table E.8. Coverage gap decomposition under the S3 (NSECE-calibrated) scenario. "Below" and "Above" report the percentage of replications where the true value falls below or above the confidence interval, respectively. SE ratio > 1 indicates intervals that are too narrow; $|\bar{b}|/\text{SD} > 0.5$ indicates non-negligible point estimate bias. $\star$ = sandwich structurally inapplicable (E-WS ≡ E-WT).

			Non-coverage (%)				
Parameter	Est.	Cov. (%)	Below	Above	SE ratio	$\lvert\bar{b}\rvert$/SD	Dominant source
α_{pov}	E-UW	89.5	1.0	9.5	0.86	0.63	—
	E-WT	58.5	15.5	26.0	1.93	0.22	Width
	E-WS	82.0	5.0	13.0	1.13	0.22	Width
β_{pov}	E-UW	89.0	0.5	10.5	0.85	0.69	—
	E-WT	62.5	15.5	22.0	1.88	0.06	Width
	E-WS	82.5	8.5	9.0	1.14	0.06	Width
$\log \kappa$	E-UW	92.0	1.0	7.0	0.85	0.48	—
	E-WT	34.0	65.0	1.0	1.75	1.37	Both
	E-WS	60.5	39.5	0.0	1.00	1.37	Bias
τ_{ext}	E-UW	98.0	1.5	0.5	0.64	0.59	—
	E-WT/WS	0.5	99.5	0.0	1.22	3.18	Bias*
τ_{int}	E-UW	97.5	1.0	1.5	0.73	0.51	—
	E-WT/WS	0.0	100.0	0.0	1.11	3.55	Bias*

E.8. **Coverage at the 95% nominal level.** Table E.9 reports coverage at both the 90% and 95% nominal levels for all five target parameters across three scenarios and three estimators. For all estimators, 95% intervals are computed as $\hat{\theta} \pm 1.96 \times$ SE, where SE is the posterior standard deviation (E-UW, E-WT) or the sandwich standard error (E-WS). Verification confirms that this Wald construction agrees with the original quantile-based 90% intervals to within 0.1 pp, validating the Normal approximation.

The results reinforce the width-driven interpretation of the E-WS coverage gap identified in Section E.7. For the poverty coefficients under S3, widening from 90% to 95% adds only 5.5–6.0 pp of coverage (from 82.0–82.5% to 87.5–88.5%), leaving a residual gap of 6.5–7.5 pp relative to the 95% nominal level. If the sandwich SE were perfectly calibrated, the expected gain would be ≈ 5.0 pp (from the additional probability mass in the Normal tails between $z_{0.05}$ and $z_{0.025}$); the observed gains are consistent with this prediction, confirming that the coverage shortfall scales proportionally with interval width rather than reflecting a level-dependent phenomenon.

Table E.9. Interval coverage (%) at 90% and 95% nominal levels. Parenthetical values are MCSEs ($= \sqrt{\hat{p}(1-\hat{p})/R}$; $R = 200$). At the 95% nominal level, MCSE $\approx$ 1.5 pp.

		S0 (Non-inform.)		S3 (NSECE)		S4 (Stress)	
Parameter	Est.	90%	95%	90%	95%	90%	95%
α_{pov}	E-UW	93.0 (1.8)	97.0 (1.2)	89.5 (2.2)	96.0 (1.4)	92.5 (1.9)	97.5 (1.1)
	E-WT	81.0 (2.8)	89.0 (2.2)	58.5 (3.5)	67.5 (3.3)	53.0 (3.5)	61.0 (3.4)
	E-WS	88.5 (2.3)	96.0 (1.4)	82.0 (2.7)	87.5 (2.3)	86.0 (2.5)	89.5 (2.2)
β_{pov}	E-UW	93.0 (1.8)	96.5 (1.3)	89.0 (2.2)	95.0 (1.5)	91.0 (2.0)	96.0 (1.4)
	E-WT	76.5 (3.0)	86.0 (2.5)	62.5 (3.4)	70.5 (3.2)	56.5 (3.5)	64.0 (3.4)
	E-WS	86.5 (2.4)	91.5 (2.0)	82.5 (2.7)	88.5 (2.3)	84.0 (2.6)	89.5 (2.2)
$\log \kappa$	E-UW	91.5 (2.0)	96.0 (1.4)	92.0 (1.9)	96.5 (1.3)	90.5 (2.1)	93.5 (1.7)
	E-WT	64.0 (3.4)	73.0 (3.1)	34.0 (3.3)	38.5 (3.4)	23.0 (3.0)	28.0 (3.2)
	E-WS	76.5 (3.0)	84.5 (2.6)	60.5 (3.5)	68.5 (3.3)	57.0 (3.5)	68.0 (3.3)
τ_{ext}	E-UW	97.5 (1.1)	100.0 (0.0)	98.0 (1.0)	99.5 (0.5)	77.0 (3.0)	93.0 (1.8)
	E-WT	3.0 (1.2)	7.0 (1.8)	0.5 (0.5)	0.5 (0.5)	0.0 (0.0)	0.0 (0.0)
	E-WS	3.0 (1.2)	7.0 (1.8)	0.5 (0.5)	0.5 (0.5)	0.0 (0.0)	0.0 (0.0)
τ_{int}	E-UW	91.0 (2.0)	95.0 (1.5)	97.5 (1.1)	99.5 (0.5)	97.0 (1.2)	98.5 (0.9)
	E-WT	8.0 (1.9)	12.0 (2.3)	0.0 (0.0)	0.5 (0.5)	0.0 (0.0)	0.0 (0.0)
	E-WS	8.0 (1.9)	12.0 (2.3)	0.0 (0.0)	0.5 (0.5)	0.0 (0.0)	0.0 (0.0)

In contrast, E-UW achieves near-nominal coverage at both levels (89.0–93.0% at 90%; 95.0–97.5% at 95%) for all fixed effects, providing further evidence that the unweighted posterior is well calibrated when the model is correctly specified. For $\log \kappa$ under E-WS, the 95% coverage (68.5%) remains substantially below nominal, consistent with the bias-driven shortfall identified in Table E.8.